

**INSTABILITIES AND CHAOS OF LASER BEAMS PROPAGATING
THROUGH NONLINEAR OPTICAL MEDIA**

by

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ABSTRACT

The interaction of two strong laser beams propagating through a nonlinear optical medium is studied theoretically and experimentally. It is shown that this interaction is unstable in that new, intense fields can be generated. An experiment has been performed to investigate the stability characteristics of near-resonant laser beams counterpropagating in a sodium vapor. It is found that the states of polarization of the transmitted waves change abruptly to a value different than the input polarizations when the intensities of the waves are above a well-defined threshold. This result demonstrates that multistability can occur in a cavity-less, nonlinear optical system.

When the intensities of the counterpropagating waves are far above the threshold for multistability there are no stable states of the system and the fields fluctuate chaotically in time. This result shows that an optical system can be dynamically unstable due solely to the effects of the nonlinear interaction and the effects of propagation. The attractors describing the dynamical evolution of the system are constructed in phase space and the fractal dimensions and entropies characterizing the attractors are determined. The boundary of the region of instability is measured and is found to depend on the detuning of the laser frequency from the sodium $3S_{1/2} \rightarrow 3P_{1/2}$ resonance, on the intensities and polarizations of the input beams, on the atomic number density, and on the buffer gas pressure.

The results are in good qualitative agreement with the predictions of a theory that models the nonlinear optical response in terms of the optical Kerr effect. Quantitative agreement is obtained through use of a more detailed model based of the solutions to the density-matrix equations of motion for the atomic system.

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CURRICULUM VITAE

The author was born on October 29, 1960 and raised in Peterborough, New Hampshire. In the fall of 1978 he entered the University of Rochester and graduated with distinction in the spring of 1982 with a Bachelors degree in Optics. His undergraduate education was partially funded by a Henry Lord Scholarship and a Naval Reserve Officer Training Corps Scholarship. Part of his undergraduate education included working as optical technician at the Laboratory for Laser Energetics from 1980 to 1982. In the fall of 1982 he began graduate work in Optics at the University of Rochester and in the spring of 1983 he received a Masters degree in Optics. His graduate research has been in the field of nonlinear optics and has been directed by Professor Robert W. Boyd. While at the University of Rochester he has held both research and teaching assistantships and a University Research Initiative fellowship. He is a member of the Optical Society of America and Tau Beta Pi.

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PRESENTATIONS

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"Competition Effects in Nonlinear Optics," Norwegian Defense Research Establishment Seminar, June 17, 1987, Oslo, Norway.

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CHAPTER I

INTRODUCTION

For many physical systems, it is possible to formulate models that allow us to predict the dynamical evolution of the system. However, seemingly simple phenomena, such as a dripping water faucet or the daily weather patterns in Rochester, can be very complicated and depend on small changes in the state of the system in an extremely sensitive manner. In this case it is impossible to predict the long-term behavior. However, studies¹⁻¹² have shown that nonlinear, deterministic models with a small number of degrees of freedom can describe such complex behavior and when the system displays sensitive dependence on initial conditions, the evolution is termed chaotic. It has also been shown that even when long-term predictability is lost, global order is still present, and that the 'form' of the order can provide useful information about the dynamical evolution. In many cases the form of the order is common to several chaotic systems (universal behavior);¹³ for instance, it has been shown¹⁴ that the model of a laser¹⁵ is the same as a model developed to explain convection in fluids.¹⁶

Universal behavior is an important characteristic of chaotic systems because controlled experiments can be performed for one system and the results can be used to determine the evolution of different systems for which it is difficult to conduct controlled experiments. The subject of this thesis is the study and characterization of a simple nonlinear optical system that displays chaotic dynamics. I will show that laser beams counterpropagating in a sodium vapor are unstable to the growth of temporal fluctuations due to the strong nonlinear coupling between the electromagnetic field and the atomic system. This study is of interest because complex

dynamical behavior is observed in a simple nonlinear optical interaction and a well-developed theoretical formulation is available to model the results. In addition, the mechanisms responsible for instabilities in this nonlinear interaction are also the source of instabilities in many nonlinear optical processes.

Unstable optical devices, such as the mode-locked laser, can be of practical value; however, instabilities can degrade the performance of nonlinear optical devices. We have shown¹⁷ that a phase-conjugate mirror^{18,19} based on internally self-pumped phase-conjugation in barium titanate²⁰ is dynamically unstable and that for large nonlinear coupling strengths the evolution is chaotic. When the phase-conjugate mirror is used under unstable operating conditions the performance of devices, such as phase-conjugate interferometers,²¹ is degraded.

An analysis of self-pumped phase conjugation in barium titanate is complicated because the mechanisms responsible for the phase-conjugation process are difficult to identify due to the geometry of the interacting waves. A model of self-pumped phase-conjugation²² assumes the geometry of the interacting waves shown in Fig. 1.1. The incident laser beam breaks up into new spatial components due to a self-defocusing process known as self-beam fanning.²³ The new beams undergo total internal reflection and are thereby redirected to intersect the incident beam a second time. At each of these interaction regions, counterpropagating waves are present and the phase-conjugate wave is generated by the four-wave mixing process. We have extended the model of self-pumped phase conjugation to include the temporal evolution of the optical nonlinearity and the contribution of all gratings formed by the interference of the interacting waves. It is found that competition among the various gratings leads to unstable behavior of the phase-conjugate wave. Figures 1.2 shows the theoretically predicted and experimentally observed chaotic evolution of the reflectivity of a self-pumped barium titanate phase-conjugate mirror. It is seen

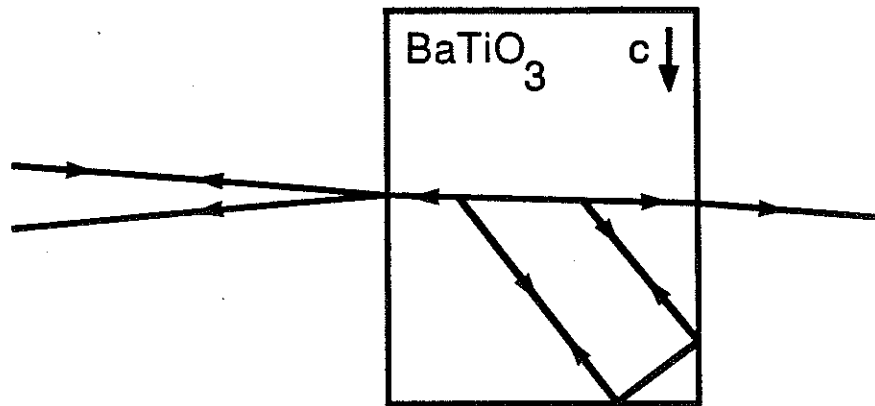


Figure 1.1 Geometry of the interacting waves responsible for internally self-pumped phase conjugation in barium titanate.

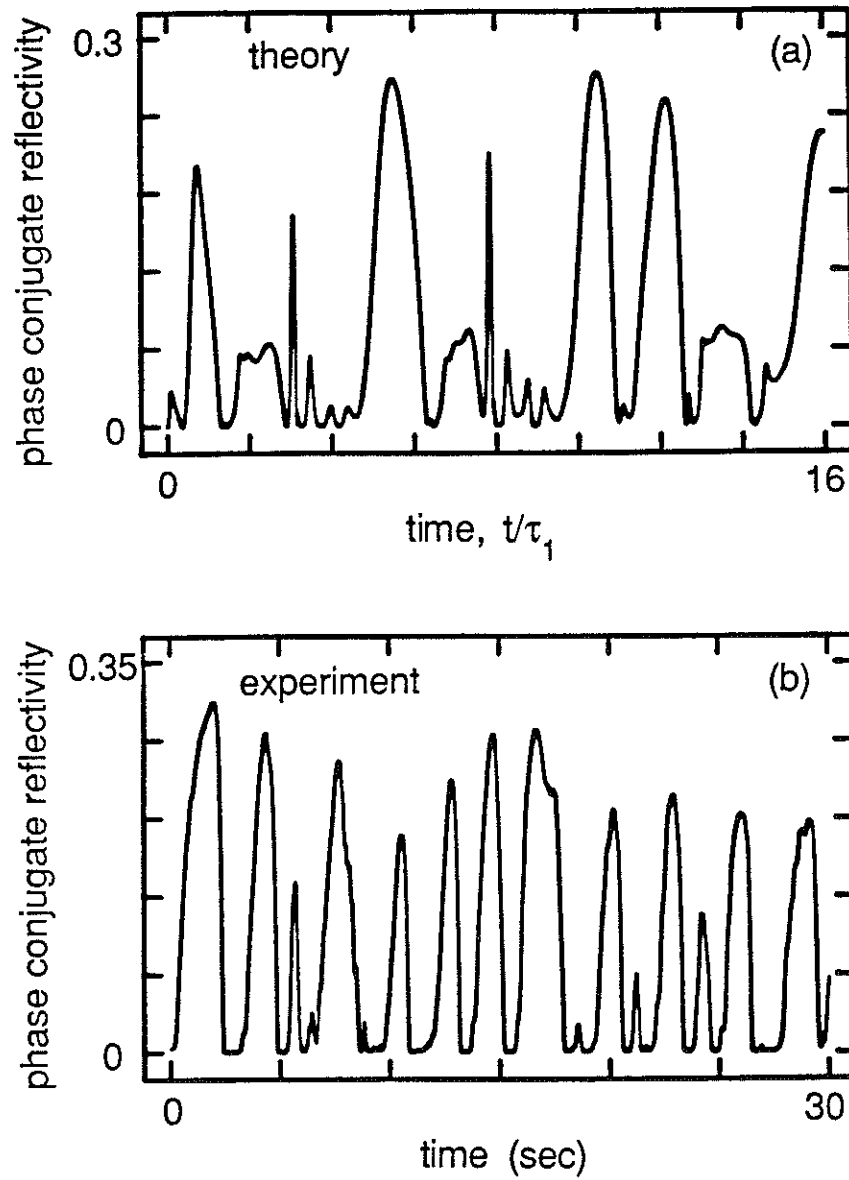


Figure 1.2 (a) Theoretically predicted and (b) experimentally observed chaotic evolution of the reflectivity of a self-pumped phase-conjugate mirror based on internally self-pumped phase conjugation in barium titanate. The time constant τ_1 is the formation time of one of the refractive index gratings.

that the reflectivity fluctuates wildly from zero to 30% and that the fluctuations are not periodic. It was determined that these fluctuations are chaotic from an analysis^{24,25} of the time series of the measured reflectivities.

The study of self-pumped phase conjugation in barium titanate showed that a cavity-less nonlinear optical system is unstable and that the temporal evolution can be chaotic. It is difficult to obtain a simple understanding of the reasons for unstable behavior because of the complicated geometry of the interaction. A more simple nonlinear optical process was analyzed by Silberberg and Bar-Joseph.^{26,27} It was predicted that laser beams counterpropagating in a Kerr medium are dynamically unstable. The conceptual simplicity of the geometry of this interaction and the potential usefulness of devices that use counterpropagating laser beams motivated the research presented in this thesis.

In the next chapter, a laser model for a single-mode laser for which solutions can be obtained easily will be reviewed to demonstrate that a system with a few degrees of freedom can display chaotic dynamics. It will be shown that as the laser pump rate is increased, the laser passes through a second threshold above which the laser output fluctuates periodically or chaotically. The nature of the fluctuations is characterized by analyzing the evolution of the trajectory in phase space.

In order to identify some of the possible mechanisms that cause instabilities in the mutual interaction of counterpropagating laser beams, the stability characteristics of laser beams interacting in a Kerr medium will be reviewed. A Kerr medium is studied because analytic solutions can be obtained and, under certain conditions, an atomic vapor can be approximated as a Kerr medium. It will be shown that laser fields propagating in a Kerr medium are unstable to the growth of new fields and that, for the case of counterpropagating laser beams, the state of polarization and the

amplitudes of the transmitted beams can be dynamically unstable for sufficiently high input intensities.

The first experimental observation²⁸ of instabilities and chaos in the state of polarization of counterpropagating laser beams in the absence of a cavity is described in Chapter IV. In this experiment, counterpropagating laser beams are derived from a continuous-wave dye laser whose frequency is tuned near to the sodium $3S_{1/2} \rightarrow 3P_{1/2}$ resonance and input to an optically dense sodium vapor. Measurements are performed to determine the states of polarization of the fields that are transmitted through the sodium cell. Data is presented which characterizes the nature of the instability and demonstrates that the evolution of the system can be chaotic. It is also shown that the polarization state of the transmitted fields can be a multistable function of the input intensities. In order to demonstrate the effects of this instability on a device that uses counterpropagating laser beams, the properties of phase conjugation by degenerate four-wave mixing are studied.

The results of this experiment can be qualitatively explained by the Kerr-medium model,²⁹ however, the effects of saturation modify the evolution of the system. In the fifth chapter, models which explain some of these observations are developed. The nonlinear interaction is treated by a semiclassical theory that treats the atomic sodium D_1 transition as a two-level atom. The gain experienced by a weak probe-wave in the presence of a single strong pump-wave is calculated in order to identify the gain mechanisms responsible for the observed instabilities. Numerical simulations are then used to study the stability characteristics of counterpropagating laser beams.

The equations in this thesis are written in the cgs system of units and in some cases, values of experimental parameters are given in mixed units as is customary in the field of nonlinear optics.

CHAPTER II

THE CHARACTERIZATION OF AN UNSTABLE OPTICAL SYSTEM

A. Introduction

The development of the laser brought about the possibility of studying nonlinear effects in optics because the radiation produced by the laser is coherent and very intense. For intense fields, the polarization is a nonlinear function of the field³⁰⁻³² and for low coupling strengths can be expanded in a Taylor series of the form

$$\mathbf{P} = \vec{\chi}^{(1)} \cdot \mathbf{E} + \vec{\chi}^{(2)} : \mathbf{E} \mathbf{E} + \vec{\chi}^{(3)} : \mathbf{E} \mathbf{E} \mathbf{E} + \dots \quad (2.1)$$

The polarization is the source term in the driven wave equation which governs electromagnetic phenomena. The wave equation is derived from Maxwell's equations and is given by

$$\left[\nabla \times \nabla \times + \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right] \mathbf{E} = - \frac{4\pi}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{P} \quad (2.2)$$

under the assumption that the medium through which the waves are propagating is nonconducting and nonmagnetic. In an effort to optimize practical nonlinear optical devices, the nonlinear susceptibilities $\chi^{(j)}$ can be increased. However, it has been found³³⁻³⁹ that devices can become unstable if the nonlinear coupling is too large. In

many instances, the performance of devices is degraded in the presence of instabilities. It is therefore important to determine the conditions under which nonlinear optical devices are unstable.

Interestingly, the laser itself is a highly nonlinear system and it was found that the radiation produced by the first laser (a ruby laser) consisted of random pulses of light⁴⁰ even under conditions of continuous-wave pumping.⁴¹ In this chapter, a simple model of a continuous-wave laser will be reviewed to show how an unstable optical system is characterized. I have chosen this example system because: (1) it is assumed that the slowly-varying amplitude of the electric field and polarization does not depend on the spatial coordinate (mean-field limit), (2) the evolution of the laser is governed by only three collective variables and (3) this model is isomorphic with the Lorenz model¹⁶ for fluid convection which has been studied extensively. It will be shown that it is useful to study the evolution of the laser in *phase space*⁴² and that even when the laser output fluctuates chaotically, there is a proper measure that uniquely characterizes the state of the system.

Chaotic evolution of the type predicted by the single-mode, homogeneously broadened ring-laser has been observed in recent experiments using far infra-red laser transitions.^{43,44} However, it has been³³⁻³⁹ found that many laser systems exhibit unstable behavior at a much lower threshold than that predicted by the single-mode ring-laser model.¹⁵ It has been determined⁴⁵ that additional complications present in real lasers, such as the transverse beam profile and deviations from the mean-field limit can change the instability threshold. It has also been found that spectral hole-burning in an inhomogeneously broadened transition⁴⁶ can cause the laser to operate at different frequencies, and the resulting fluctuations in the output are due to the beating of these new frequencies.

It has been observed that several lasers exhibit multimode instabilities when the gain bandwidth is greater than the longitudinal mode spacing. In particular, it has been found that spatial-hole burning in a standing-wave laser, in combination with multi-wave mixing, can give rise to very complicated fluctuations in the output intensity.⁴⁷ In addition, it has been found that gain at the Rabi frequency can give rise to a multimode instability in a ring dye-laser.⁴⁸

B. The Single-Mode Laser Model

The single-mode laser model has been discussed in a number of review articles^{33,37,38,39} and it has been found that the nature of instabilities in laser systems is strongly dependent on the decay rates of the cavity, atomic populations and atomic coherence. Other important parameters are the broadening mechanisms of the atomic system and the cavity-mode width in comparison to the gain width. The first models of the laser assumed that the decay rates of the atomic coherence were much larger than all other rates (the rate-equation limit);⁴⁹ however, it was found that the atomic coherence must be included in a model capable of predicting persistent pulsations, such as those observed in the study of the first laser systems.

Risken and Nummedal¹⁵ proposed a semi-classical model that predicts unstable behavior in the operation of a homogeneously broadened, unidirectional ring-laser. An important aspect of this model is the assumption that the spatial average of the variables is adequate to describe the dynamics of the laser (mean-field limit). In this case, the spatial derivatives of the slowly-varying amplitudes can be ignored. The collective variables of the model are the atomic coherence v , the population inversion w , and the electric field E . For line-center operation and in the plane-wave approximation, the equations of motion are given by

$$\frac{dv}{dt} = -\gamma_v(v - Ew),$$

$$\frac{dw}{dt} = -\gamma_w(w - r - 1 + rEv), \quad (2.3)$$

and

$$\frac{dE}{dt} = -\gamma_c(E - v),$$

where the atomic coherence, atomic inversion, and electric field are normalized by their 'continuous-wave' values $v' = \mu \gamma_v E' w' / \hbar$, $w' = w_p - \mu E' v' / \hbar \gamma_w$, and $E' = 2\pi N \mu v' \omega_0 / \gamma_c$, respectively. The cavity decay rate is γ_c ; the material parameters are: the dipole dephasing rate γ_v , the population decay rate γ_w , the dipole moment μ , the transition frequency ω_0 , the number density N , the inversion due to pumping in the absence of coupling w_p , and the normalized pump rate $r+1 = w_p/w'$. The case of the laser frequency detuned from the atomic resonance has been studied by several researchers.⁵¹

The behavior of the solutions to Eqs. (2.3) can be investigated by determining the fixed-points, denoted by $\mathbf{p}^{(j)} = (v^{(j)}, E^{(j)}, w^{(j)})$, of the set of equations and the stability characteristics of these fixed-points. The stability of these steady-state solutions is determined by perturbing the solutions and solving for the evolution of the perturbations. This procedure is accomplished by assuming that the variables are given by $x = x^{(j)} + \tilde{x}$ ($x = v, w, E$), linearizing the differential equations about the steady-state solutions $x^{(j)}$, and calculating the solutions to the linearized set of equations by assuming a trial solution of the form $\tilde{x} = \tilde{x}_0 e^{\lambda t}$. If all of the eigenvalues λ of the linearized flow are less than zero, the fixed-point is stable (except in the presence of a large amount of noise which could perturb the variables outside of the linear regime⁵²); for $\lambda = 0$, the fixed-point is marginally stable and will become

unstable in the presence of any amount of noise; and for any one eigenvalue $\lambda > 0$, the fixed-point is unstable.

The fixed-points of Eqs. (2.3) are equal to: $\mathbf{p}^{(1)} = (0, 0, r+1)$; $\mathbf{p}^{(2)} = (1, 1, 1)$; and $\mathbf{p}^{(3)} = (-1, -1, 1)$. The first fixed-point is the solution corresponding to the laser operating below the first laser threshold, i.e. no output. The characteristic equation that specifies the eigenvalues for the first fixed point is $(\lambda + \gamma_w)[\lambda^2 + \lambda(\gamma_v + \gamma_w) - r\gamma_v\gamma_c] = 0$. For a pump parameter r less than zero all the eigenvalues are less than zero and hence this fixed point is stable. For r less than but approximately equal to zero, one of the eigenvalues becomes very small. Therefore, perturbations will die out more slowly in this case. For $r = 0$, perturbations take an infinite amount of time to die out. This phenomena is known as *critical slowing down*,⁵³ and occurs in any system that is approaching a bifurcation point. The second and third fixed-points correspond to the two possible continuous-wave solutions. For a pump parameter r less than zero, the two continuous-wave solutions are unstable and, as mentioned above, the solution corresponding to the laser below the first laser threshold is stable. Since the set of Eqs. (2.3) is dissipative (this can be seen by considering how a volume element in phase space changes in time), all initial conditions $\mathbf{p}_0$ will eventually settle down to the point $\mathbf{p}^{(1)}$. The point $\mathbf{p}^{(1)}$ is an *attractor*,¹ defined as a bounded region of phase space to which all sufficiently close trajectories are attracted for sufficiently long times, and that the entire phase space is the *basin of attraction*, because all the initial points in phase space are 'attracted' to the point $\mathbf{p}^{(1)}$. For a pump parameter $r > 0$, the point $\mathbf{p}^{(1)}$ is always unstable and for $0 < r < r_c$ the continuous-wave fixed-points $\mathbf{p}^{(2)}$ and $\mathbf{p}^{(3)}$ are stable, where the critical pump parameter is

$$r_c = \frac{(\gamma_c + \gamma_v)(\gamma_c + \gamma_v + \gamma_w)}{\gamma_v(\gamma_c - \gamma_v - \gamma_w)}. \quad (2.4)$$

The basin of attraction for the point $\mathbf{p}^{(2)}$ is the half-space $v > 0$. Therefore, as the pump parameter is increased to a value greater than zero, an infinitely small source of noise (for the case of the laser, the noise is due to quantum fluctuations⁵⁴) will cause the point $\mathbf{p}^{(1)}$ to become unstable and the system will be attracted to one of the other fixed points, depending on the value of the initial perturbation. One of the eigenvalues of the linearized flow for each of the points $\mathbf{p}^{(2)}$ and $\mathbf{p}^{(3)}$ becomes greater than zero for values of the pump parameter larger than r_c and for $\gamma_c > \gamma_v + \gamma_w$ (bad cavity limit), implying that the system will be unstable. In this case, we must rely on a numerical integration of Eqs. (2.3) to determine the evolution of the system.

The dynamics of the laser in the unstable regime are found to be very rich. The minimum instability threshold is for the case of large collisional dephasing ($\gamma_w/\gamma_v \rightarrow 0$) and a 'bad' cavity ($\gamma_c/\gamma_v > 0$). For a pump parameter equal to the second laser threshold the evolution changes from a constant value to wildly fluctuating, noise-like behavior. However, for a limited set of parameters in the bad-cavity limit, periodic pulsations are predicted for the evolution of the laser. In particular, the laser undergoes regular pulsations for $r = 15$, $\gamma_c/\gamma_v = 4$ and $\gamma_v/\gamma_w = 0.14$, as shown in Fig. 2.1a. To demonstrate quantitatively that these fluctuations are periodic, it is useful to calculate the power spectrum and autocorrelation of the fluctuations. In Fig. 2.1b, it is seen that the power spectrum is sharply peaked at a few frequencies and their harmonics, and, from Fig. 2.1c, it is seen that the autocorrelation function is nearly one for long delay times. These results show that the evolution is indeed periodic. To visualize the evolution of the laser in phase space, a two-dimensional projection of the trajectory onto the v - E plane is shown in Fig. 2.2. It is seen that the trajectory forms a closed loop, and that the trajectory orbits about the three unstable fixed-points, which are indicated by the three dots in the figure. This attractor is referred to

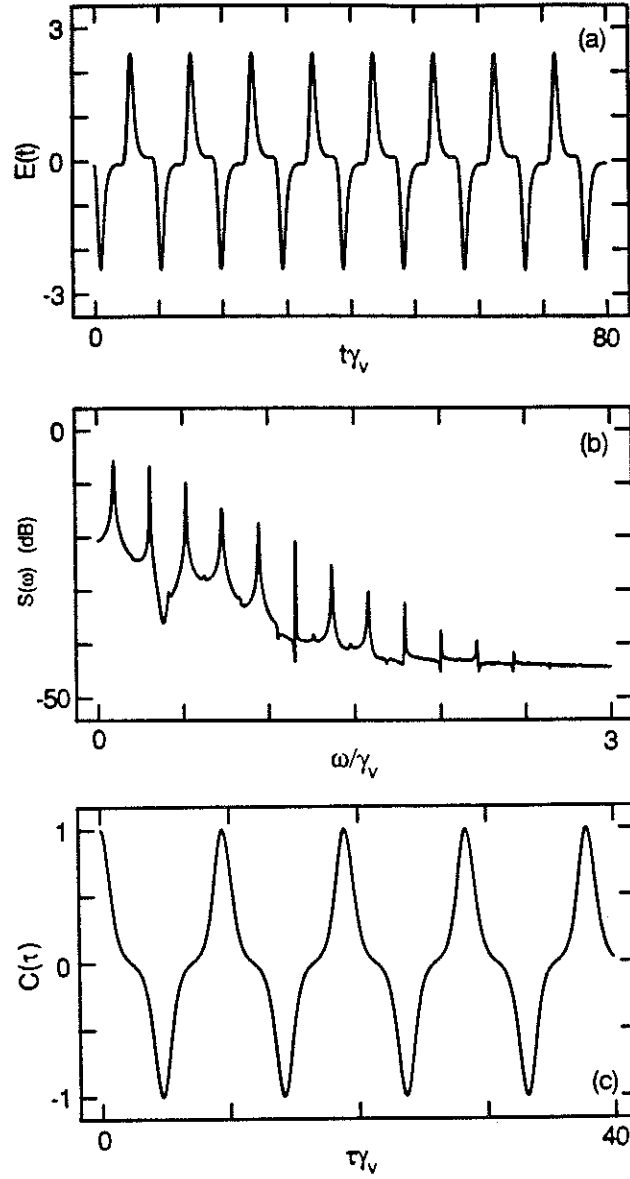


Figure 2.1 Predictions of the single-mode ring-laser model for $r = 15$, $\gamma_c/\gamma_v = 4$ and $\gamma_w/\gamma_v = 0.5$. (a) Temporal evolution of the electric field. (b) Power spectrum and (c) normalized autocorrelation function of the fluctuating time series shown in (a).

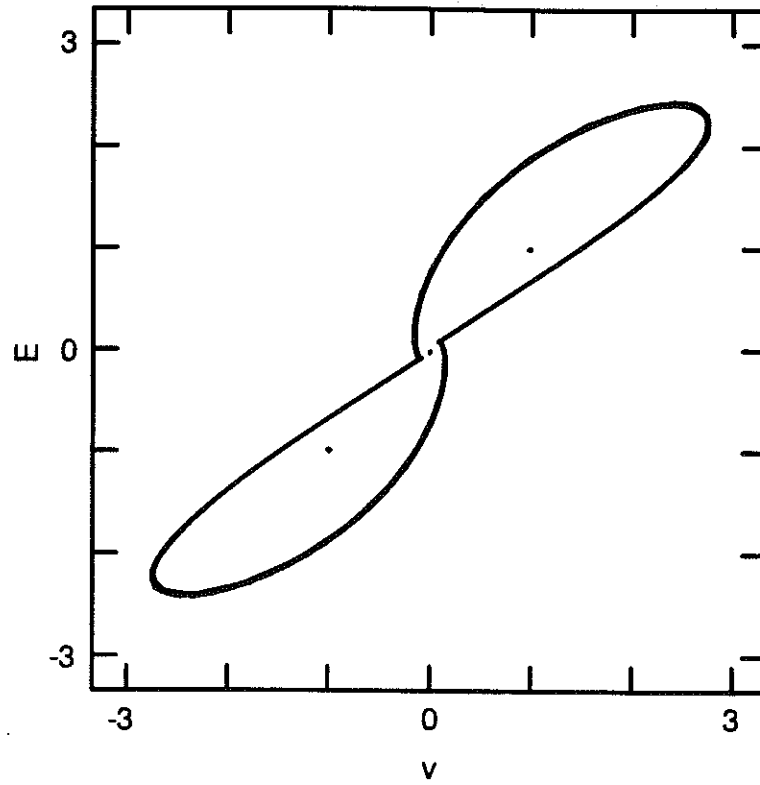


Figure 2.2 The projection of the trajectory onto the v - E plane. Note that the trajectory forms a closed loop and hence is characteristic of periodic motion. The three dots indicate the unstable fixed points of the system. The parameters used to generate this plot are the same as those used for Figs. 2.1.

as a *heteroclinic attractor*⁴⁴ since the trajectory evolves around more than one fixed-point. A *homoclinic attractor* is an attractor that orbits around a single fixed-point.

For different values of the population decay rate, dramatically different behavior is predicted. In particular, for $\gamma_c/\gamma_v = 4$ and $\gamma_w/\gamma_v = 0.5$, the critical pump parameter is equal to 11 and when the pump rate is greater than the critical pump rate, the electric field fluctuates wildly in time. Figure 2.3a shows the noise-like temporal evolution of the electric field for $r = 15$. In this case, the power spectrum contains no structure, as shown in Fig 2.3b, and the autocorrelation function rapidly drops to zero (Fig. 2.3c), leading one to the conclusion that these fluctuations are characteristic of random noise. However, by inspection of the evolution of the trajectory in phase space (Fig. 2.4), it is seen that the trajectory does not fill the phase space, and hence is not characteristic of random noise.

Landau and Lifshitz⁵⁵ postulated that the dynamics of a system change from regular to complex through a series of bifurcations. At each bifurcation point, a new set of incommensurate frequencies appear in the power spectrum of the resulting fluctuations. The state of the system is then characterized by the value of each (possibly infinite number) frequency. Ruelle and Takens¹ proposed a different scenario for the route to turbulence. They postulated that the dynamics could be complex just after a few bifurcations (perhaps none) and they referred to the evolution as *chaotic* evolution, which is characterized by a positive *Liapunov exponent*.⁵⁶ The Liapunov exponents of a set of equations are the rates of convergence (for an exponent less than zero) or divergence (greater than zero) of initially close trajectories. Therefore, a positive Liapunov exponent implies that the long-term evolution is extremely sensitive to initial conditions and the ability to predict the long-term behavior of the system is lost.

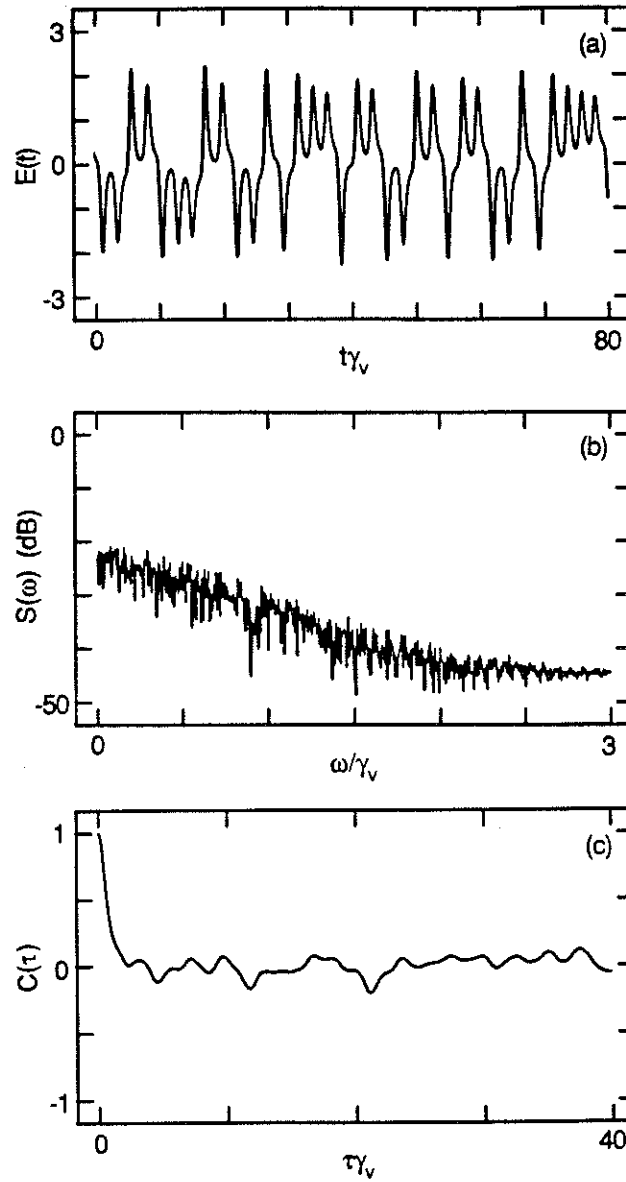


Figure 2.3 Chaotic evolution of the electric field (a) for $r = 15$, $\gamma_c/\gamma_v = 4$ and $\gamma_w/\gamma_v = 0.15$. The power spectrum (b) is broad and contains no peaks and the normalized autocorrelation function (c) drops rapidly to zero. The power spectrum and autocorrelation function indicate that the fluctuating time series is either that of random or chaotic motion.

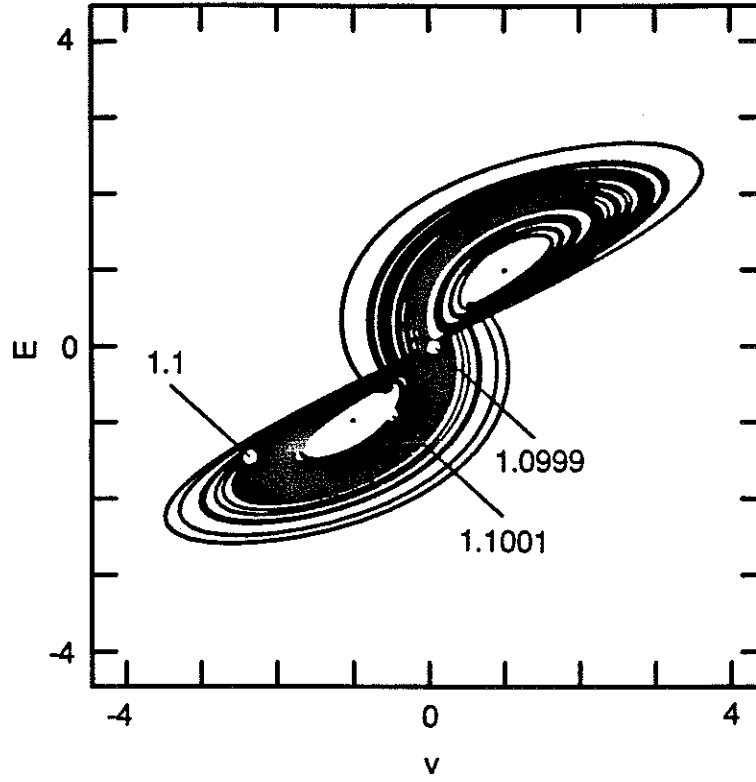


Figure 2.4 The projection of the trajectory onto the v - E plane. The three white dots indicate the positions of three trajectories at time t_f . These trajectories are determined by a numerical integration of Eqs. (2.3) for three different initial conditions. It is seen that the dots are at very different positions on the attractor. The evolution is chaotic and the attractor is a strange attractor since the trajectories do not fill phase-space.

$w(t=0)$	$v(t_f)$	$E(t_f)$	$w(t_f)$
1.1	-2.21	-1.44	3.64
1.1001	-0.513	-0.974	-0.795
1.0999	-0.0789	-0.0816	0.973

Table 2.1 Values of the variables describing the dynamics of the ring laser at time t_f determined from a numerical integration of Eqs. (2.3) for $\tau = 15$, $\gamma_c/\gamma_v = 4$, $\gamma_w/\gamma_v = 0.15$, $E(t=0) = v(t=0) = 1$ and three different values of the initial inversion $w(t=0)$.

Even when predictability is lost, useful information can still be obtained concerning the long-term behavior of the system because it is found that all trajectories eventually lie on the attractor. For example, consider the solutions to Eq. (2.3) for the initial conditions $E(t=0) = v(t=0) = 1$ and three different values for the initial condition of the inversion w . At a time t_f it is found that v , w and E take on very different values, which are listed in Table 2.1, and the positions of these points on the attractor are indicated by white dots in Fig. 2.4. We therefore see that it is impossible to specify the actual positions in phase space of the system at some later time because of the extreme sensitivity to initial conditions; however, we can say that the motion is bounded to some particular region (attractor) in phase space. For the case of a system characterized by a positive Liapunov exponent, the attractor is called a *strange attractor*.¹ In this case, different points on the attractor are dynamically uncorrelated because of the exponential divergence of nearby trajectories; however, points of the trajectory are spatially correlated because they are on the attractor. Therefore, a statistical measure that describes the geometry of the attractor is used to characterize the evolution of the chaotic trajectory.

C. Characterizing an Attractor by Dimensions and Entropies

It has been determined^{57,58} that a proper measure (i.e. a quantity that uniquely defines the attractor) for a chaotic system is a set of generalized dimensions and entropies of the associated attractor. The evolution of a chaotic system is well behaved, but the structure of the attractor is very 'rough'. It is found that as the structure of the attractor is examined in fine detail,⁴² the trajectories never follow a regular pattern. In most cases, however, it is found that the 'density of trajectories' is similar in the volume of space surrounding a point on the attractor as the volume

size is varied. The density of trajectories is defined formally through use of the concepts developed for information theory⁵⁹ and is directly related to the geometrical measure of the attractor.

The concept of density of trajectories is defined by considering the following construction: partition the n -dimensional phase space in which the attractor is embedded into n -cubes of dimension ϵ and count the number $N(\epsilon)$ of n -cubes that contain a portion of the attractor. The density of trajectories at a point on the attractor is given by the probability that a trajectory will pass through a cube centered on the point of interest. Formally, we assume that the trajectory is sampled at different times $t, t+\Delta t, \dots, t+(M-1)\Delta t$. The probability p_i of finding a point of the attractor in the i^{th} cube is given by

$$p_i = \lim_{M \rightarrow \infty} \frac{M_i}{M} \quad (2.5)$$

where M_i is the number of points in the i^{th} cube.

The dimensions D_q of the attractor are defined in terms of the order- q Renyi informations⁶⁰ $I_q(\epsilon)$ by the relationship

$$D_q = \lim_{\epsilon \rightarrow 0} \frac{I_q(\epsilon)}{\ln(1/\epsilon)} \quad (2.6)$$

where the informations are defined in terms of the probabilities as

$$I_q(\epsilon) = \frac{1}{(1-q)} \lim_{\epsilon \rightarrow 0} \frac{\ln \sum_{i=1}^{N(\epsilon)} p_i^q}{\ln \epsilon} \quad (q \geq 0, q \neq 1)$$

and

(2.7)

$$I_1(\epsilon) = \sum_{i=1}^{N(\epsilon)} p_i \ln p_i.$$

The sums in Eqs. (2.7) are to be taken only over those sampled points that are on the attractor; hence $I_1(\epsilon)$ is not strictly equal to the usual thermodynamic (or information) entropy. When $q = 0$, the dimension is given by

$$D_0 = - \lim_{\epsilon \rightarrow 0} \frac{\ln N(\epsilon)}{\ln \epsilon} \quad (2.8)$$

and is the scaling of the number of cubes required to cover the attractor as the cube size is varied. This dimension is called the *Hausdorff dimension*⁵⁷ (or capacity or similarity dimension) and in most cases is equal to the *fractal dimension*.⁶¹ For all chaotic attractors, it has been found that D_0 is fractional (hence the name fractal dimension), and that the nearest larger integer is the number of independent degrees of freedom of the system. The other dimensions D_q ($q \neq 0$) can be thought of as higher order moments describing the 'roughness' of the attractor. It has been shown⁵⁷ that the dimensions are bounded by the relationship $D_0 \geq D_1 \geq \dots \geq D_\infty$, and in many cases $D_0 \approx D_1 \approx D_2$. This last property of dimensions led to a great deal of confusion in the early studies of chaotic systems. It was thought that D_0 alone was a proper measure of the system,⁶² and that all other D_q were equal to D_0 . Therefore, in the early literature (and sometimes still in the present literature), D_2 was determined and its value was named the 'dimension' or fractal dimension of the attractor. It has

been pointed out by several researchers^{57,58} that D_2 can be significantly different than D_0 , and to make a comparison between theory and experiment, the entire spectrum of dimensions must be specified. However, a measurement of D_2 gives approximately the number of pertinent degrees of freedom of the system.

In the previous section, it was shown that a positive Liapunov exponent is a necessary and sufficient condition for a system to be chaotic. In many cases it is difficult to determine the Liapunov exponents, but it is easier to determine the spectrum of order- q entropies which are given by⁵⁷

$$K_q = \lim_{\Delta t \rightarrow 0} \lim_{\epsilon \rightarrow 0} \lim_{T \rightarrow \infty} \frac{1}{\Delta t T (1-q)} \ln \sum_{i_1, \dots, i_T} p_{\{i_1, \dots, i_T\}} \quad (q \geq 0) \quad (2.9)$$

where $p_{\{i_1, \dots, i_T\}}$ is the joint probability that the trajectory falls in cell i_1 at time t , cell i_2 at time $t+\Delta t$, ..., and cell i_T at time T . The sum in Eq. (2.9) is to be taken over all possible sequences $i_1, \dots, i_T$. In the limit that q approaches 1, the order-1 entropy is equal to the Kolmogorov entropy⁶³ which is, in most cases, equal to⁶⁴ the sum of the positive Liapunov exponents. The entropies are bounded by the relationship $K_0 \geq K_1 \geq \dots \geq K_\infty$ and for most attractors $K_0 \approx K_1 \approx K_2$. Therefore, a necessary and sufficient condition for a system to be chaotic is that K_1 is greater than zero and is finite. For a random system the Kolmogorov entropy tends to infinity.

Experimentally, it is difficult to measure the infinite spectrum of dimensions and entropies because noise in the measurement process obscures the structure of the attractor,⁶⁵ and a large number of points along the trajectory must be sampled. There is no well defined criterion for the total number of required points; however, it can be seen from Eq. (2.8) that the fractal dimension depends only on the number of cubes required to cover the attractor, irrespective of the rate of visitation of sections of the

attractor. Therefore, for a very nonuniform attractor (D_0 is very different from D_∞), there will be very sparse regions of the attractor, and a very long time series must be sampled so that the low probability regions will be taken into account. The total number of sampled points also depends on the sample interval Δt . There is no criterion for the sample time; however, a general rule is that there must be several samples during each 'characteristic' oscillation of the trajectory.

In all of the above discussions, it has been assumed that it is possible to construct an attractor that represents the evolving system. However, in an experiment, it is difficult to know the variables that form the basis of the phase space. Takens⁶⁶ showed that $2n+1$ measurements of a single variable is sufficient to specify a point of the trajectory in n -dimensional phase space. To construct the attractor, the evolution of a single variable x is sampled at times $t, t+\Delta t, \dots, t+(M-1)\Delta t$ (M very large). The set of sampled points $\{x_j | j=1, \dots, M\}$ are used to create d -dimensional time-delay vectors $\mathbf{x}_j = (x_j, x_{j+m}, \dots, x_{j+(d-1)m})$ where m is an integer. The dimensions D_q are determined for various values of the embedding dimension d , since the dimension of the true phase space is not known *a priori*, and for sufficiently large d , the D_q will converge to the correct value.

There are several criteria that can be used to determine the proper delay time $\tau = m\Delta t$ for the construction of the d -dimensional vectors. If τ is too short, the attractor will lie on a $(d-1)$ -dimensional plane and it is difficult to resolve the structure of the attractor. For τ very large, the coordinates of the vector will be uncorrelated due to the presence of experimental noise. It has been postulated⁶⁷ that the delay time corresponding to the first minimum in the autocorrelation function of the time series is a proper choice for τ . Fraser and Swinney⁶⁸ have developed a different algorithm that determines the delay time by ensuring that the coordinates of

the time-delay vector are mutually independent. However, this method is complicated and difficult to implement for $d > 2$.

I have developed a new algorithm based on simple arguments which has proven to be well suited for the construction of experimental attractors. It is based on the idea that the attractor must be 'opened up' to expose as much of the structure as possible.⁶⁵ Figures 2.5a-c show that attractor of the single-mode laser model constructed in a two-dimensional time-delay phase space by plotting $E(t+\tau)$ vs. $E(t)$ for various delay times τ . The parameters used when integrating Eqs. (2.3) are the same as those used to generate the attractor shown in Fig. 2.4 where $E(t)$ was plotted as a function of $v(t)$. For small delay times (Fig. 2.5a), the attractor is nearly a straight line and it is difficult to determine the structure of the attractor. For large delay times, the attractor is seen to unfold (Fig. 2.5b), and at the optimum delay time (Fig. 2.5c), the structure of the attractor is evident.

The optimum delay time τ_{opt} was determined by calculating the number of hyper-cubes required to cover the attractor $N(\epsilon)$ as the delay time is varied for ϵ small. The time-delay phase space is bounded because the values of the sampled variable x are bounded. The phase space is partitioned into d -dimensional hyper-cubes that have sides of length ϵ given by

$$\epsilon = \frac{x_{\text{max}} - x_{\text{min}}}{L}, \quad (2.10)$$

where L is an integer. It is convenient to normalize $N(\epsilon)$ by the maximum number of cubes in the phase space which is equal to

$$N(\epsilon)_{\text{max}} = L^d \propto \epsilon^{-d}. \quad (2.11)$$

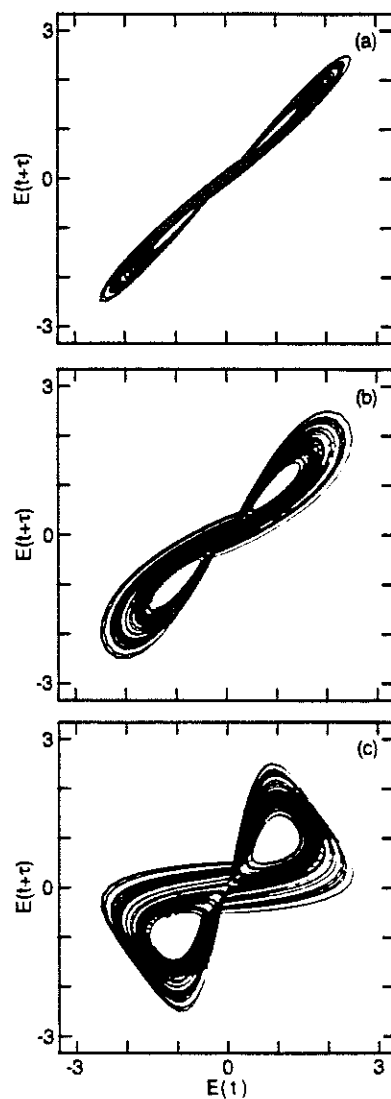


Figure 2.5 Attractors describing the chaotic evolution of the single-mode laser. The attractors are constructed by plotting the electric field at time $t+\tau$ vs. the electric field at time t for (a) $\tau\gamma_v = 1$, (b) $\tau\gamma_v = 5$ and (c) $\tau\gamma_v = 15$. At the optimum time delay (c), the attractor is opened up and the structure of the attractor is well resolved.

Figure 2.6 shows $N(\epsilon)/N(\epsilon)_{\max}$ as a function of the delay time for the attractors shown in Fig. 2.5. The optimum time delay corresponds to the first peak in the curve. This algorithm, applied to the case of a two-dimensional construction, gives a similar optimum time-delay to that of the algorithm of Fraser and Swinney. However, the present algorithm is simple and can easily be generalized to higher dimensions. It is found that even for high-dimensional attractors, the optimum time delay for a two-dimensional reconstruction is very close to the optimum delay for a higher dimensional reconstruction.

Two methods are commonly used to determine the dimensions and entropies of an attractor. Badii and Politi⁶⁹ have developed a method, called the fixed-mass method, to determine the infinite spectrum of dimensions and entropies. However, for most of the experimental attractors that I have studied, the convergence rate was slow. Therefore, I have primarily used the fixed-volume method developed by Grassberger and Procaccia¹² which can only determine D_2 and K_2 , but converges rapidly. The method involves measuring the correlation integral for a fixed value of the cube dimension ϵ . The correlation integral is defined as the number of pairs of points found in the i^{th} cube and average over all the cubes covering the attractor and is given by

$$C_d = \frac{1}{M(M-1)} \sum_{i=1}^{M-1} \sum_{j=i+1}^M \Theta(\epsilon - |\mathbf{x}_i - \mathbf{x}_j|), \quad (2.12)$$

where Θ is the Heavyside step function. The correlation dimension is determined from the correlation integral from the relation

$$D_2 = \lim_{\epsilon \rightarrow 0} \lim_{M \rightarrow \infty} \frac{\ln C_d(\epsilon)}{\ln \epsilon} \quad (d > 2n+1), \quad (2.13)$$

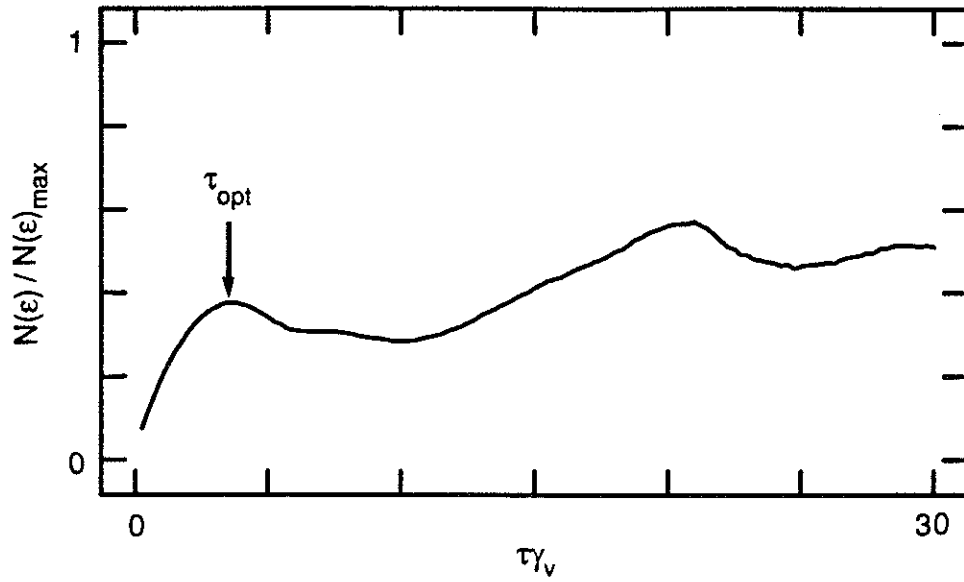


Figure 2.6 Fraction of the filled time-delay phase-space for a two-dimensional construction of the attractors describing the chaotic evolution of the single-mode laser as a function of the time delay. The first peak determines the optimum time delay τ_{opt} .

and the order-2 entropy is given by

$$K_2 = \lim_{d \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \lim_{M \rightarrow \infty} \frac{1}{\tau} \ln \frac{C_d(\epsilon)}{C_{d+1}(\epsilon)}. \quad (2.14)$$

For a system undergoing random evolution, the trajectory uniformly fills phase space. Hence, as can be seen from Eq. (2.11), $N(\epsilon) = N(\epsilon)_{\max}$ and by inspection of Eq. (2.7) it can be seen that $D_0 = d$. For a uniformly filled phase space D_0 is equal to D_∞ , which implies that $C_d(\epsilon) \propto \epsilon^d$. Hence, K_2 tends to infinity, as can be seen from Eq. (2.14). In practice, it is difficult to take the limits in Eqs. (2.6)-(2.8); the hypersphere radius ϵ cannot be taken too small nor can the embedding dimension d be taken too large because of experimental noise. However, there is a range of ϵ (the scaling region) where this algorithm is useful.

Figure 2.7 shows the correlation integral as a function of the cube dimension ϵ for various values of the embedding dimension d for the time series shown in Fig. 2.3a. It is seen that for $d > 2$, the curves are straight, parallel lines for a large range of ϵ . The slope of these lines is the correlation dimension D_2 and the order-2 entropy K_2 is determined from the spacing between successive lines for a fixed value of ϵ . From this plot it is found that $D_2 = 2.04$ and $K_2 = 32 \text{ bits } \gamma_v$. Therefore, the wildly fluctuating output of the laser shown in Fig. 2.3a is chaotic since K_2 is a lower bound to K_1 and is greater than zero.

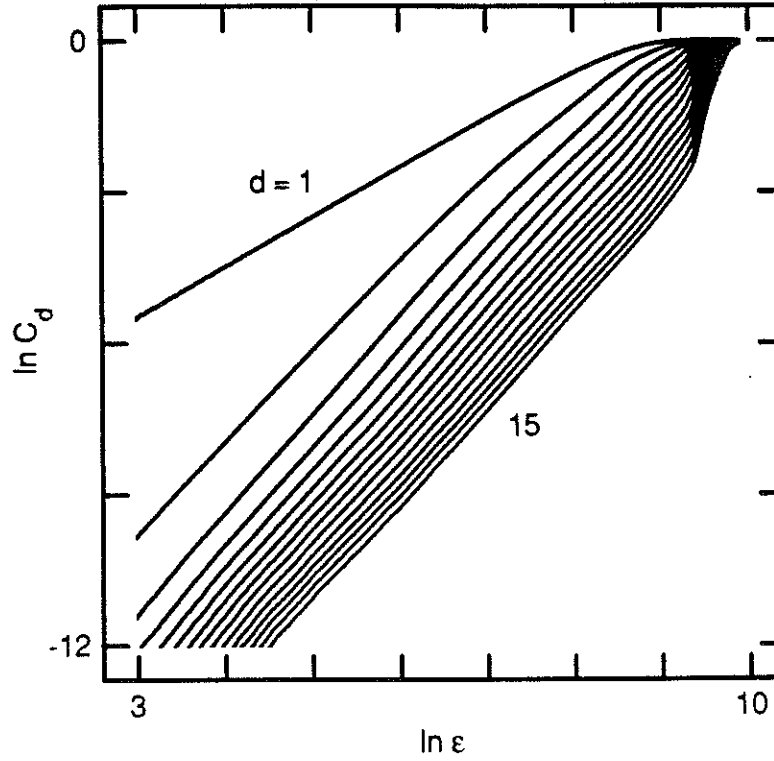


Figure 2.7 Logarithm of the correlation sum C_d as a function of the logarithm of the hypersphere radius ϵ for various embedding dimensions d . For $d > 2$, there is a range of ϵ over which the curves are straight, parallel lines. The slope of the lines is the correlation dimension D_2 , which is equal to 2.04. The spacing between successive curves is proportional to the order-2 entropy, which is determined to be 32 bits γ_v^{-1} . Note that the curves approach zero for large ϵ as required by the normalization.

CHAPTER III

REVIEW OF INSTABILITIES OF LASER BEAMS PROPAGATING IN A KERR MEDIUM

A. Introduction

In the early attempts to model the dynamics of the laser, it was difficult to obtain even qualitative agreement with experimental observations. It was conjectured that a possible source of the observed instabilities was noise in the pump radiation. In order to avoid some of these complexities, the interaction of monochromatic radiation with simple, uninverted material systems was studied theoretically. These models could easily be tested because the laser, under conditions of stable operation, provides a source of intense, monochromatic radiation which can induce nonlinearities in the response of optical materials, a necessary condition for the observation of chaotic instabilities. In this chapter, the instabilities that occur when a laser beam propagates in a nonlinear Kerr medium will be studied. It will be shown that there are several nonlinear optical interactions that can occur simultaneously and that competition among these processes can dramatically modify the threshold for instabilities. In particular, it will be shown that destructive interference between a process analogous to stimulated Rayleigh-wing scattering and parametric four-wave mixing leads to gain suppression. Gain suppression prevents the growth of instabilities when a single beam propagates in a Kerr medium and significantly raises the threshold for instabilities for the case of counterpropagating laser beams. This result demonstrates that propagation effects are an important aspect of the mechanisms for instabilities in optical systems.

McCall⁷⁰ found that a single laser beam propagating in a two-level atomic medium was unstable to the spatial growth of new frequency components. As a result of this study, it was predicted and later verified⁷¹ that the output field from a Fabry-Perot etalon filled with a collection of two-level atoms was bistable function of the input field intensities. There has been an extensive effort to understand optical bistability because of the potential usefulness of this effect for optical computing⁷² and because of the interesting quantum mechanical properties of this system.⁷³ Recently, an experiment was performed which quantitatively verified the predictions of the theoretical models of optical bistability.⁷⁴

Ikeda⁷⁵ showed that the output of a ring resonator filled with a nonlinear Kerr medium is temporally unstable when operated in the regime where optical bistability is expected. It was predicted that for sufficiently high nonlinearity, the output intensity would be chaotic and that the route to chaos would be through a period-doubling sequence.¹³ This prediction was verified in a hybrid optical system⁷⁶ and later in an all optical system.⁷⁷ Silberberg and Bar-Joseph²⁷ showed that the mechanism responsible for the instability predicted by Ikeda is a four-wave mixing process that produces new frequency components which are simultaneously resonant with the cavity and that the resulting fluctuations in the output intensity were due to the beating of the new frequencies components. There have been several other observations of chaotic behavior in passive nonlinear optical systems;^{78,79} however, all have required some form of external feedback.

Parallel to the study of instabilities in lasers and optical-bistable cavities was the study of phase conjugation by degenerate four-wave mixing. Early theoretical analysis⁸⁰ of the four-wave mixing interaction predicted that the phase-conjugate reflectivity would be infinite for particular values of the coupling strength. This form of parametric oscillation was not viewed as a deleterious instability; several novel

devices have been developed which require a phase-conjugate mirror with greater than unit reflectivity.⁸¹ However, it was found⁷⁹ that some of these devices displayed dynamical instabilities similar to those observed in lasers and other passive nonlinear optical devices. The property of infinite parametric gain in degenerate four-wave mixing will also be reviewed in this chapter and the threshold for this convective instability will be compared to other instabilities.

Since counterpropagating waves are present in many of the systems in which instabilities have previously been observed, several workers⁸² have studied the behavior of laser beams counterpropagating in a nonlinear material without any external feedback. Silberberg and Bar-Joseph²⁶ analyzed the case of a sluggish, scalar Kerr medium and found that the intensities of the transmitted beams are temporally unstable for sufficiently high input intensities. They determined that the mechanisms responsible for the instability are: a process analogous to stimulated Rayleigh-wing scattering,⁸³ which occurs at frequencies close to the input frequency, parametric four-wave mixing, and distributed feedback⁸⁴ due to the spatial modulation of the refractive index induced by the standing wave interference of the pump waves. It was also found that for a medium with an instantaneous response time, the threshold for instability is infinite.

Gaeta *et al.*²⁹ studied the case of a tensor Kerr medium and found that the states of polarization as well as the amplitudes of the transmitted waves could become temporally unstable. It was determined that the instability threshold for polarization fluctuations was lower than that for amplitude fluctuations and that instabilities exist even for the case of a medium with an instantaneous response. For the case of a medium with an instantaneous response, it was determined that the mechanism for instabilities was infinite gain due to parametric four-wave mixing.

The stability characteristics of laser beams counterpropagating in several different nonlinear optical materials has been studied by several researchers.⁸⁵⁻⁸⁷ The predictions of the simple Kerr-medium model are qualitatively similar to those predicted for different materials. In this chapter, the stability characteristics of laser beams counterpropagating in a sluggish, scalar Kerr medium will be reviewed in order to elucidate some of the mechanisms responsible for dynamical instabilities.

B. The Steady-State Behavior of a Single Wave Propagating in a Kerr Medium

In this section, the steady-state spatial variation of a linearly polarized, monochromatic plane-wave propagating in a lossless Kerr medium is studied. For a plane-wave field propagating in the z-direction, the complex electric field in the medium is given in Cartesian coordinates by $\mathbf{E} = (\tilde{E}_x \hat{x} + \tilde{E}_y \hat{y})\exp(-i\omega t)$, where $\tilde{E}_j = E_j \exp(ikz)$ ($j=x,y$), $k=n\omega/c$, and n is the linear refractive index of the medium at frequency ω . The complex polarization can be represented in a similar manner as $\mathbf{P} = (\tilde{P}_x \hat{x} + \tilde{P}_y \hat{y})\exp(-i\omega t)$. Each component of the polarization is given by $\tilde{P}_i = \sum \chi_{ij} \tilde{E}_j$, where χ_{ij} is the susceptibility tensor. The susceptibility tensor for a lossless Kerr medium is assumed to obey a Debye relaxation equation of the form²⁹

$$\tau \frac{d\chi_{ij}}{dt} + \chi_{ij} = (A-B/2)(\mathbf{E} \cdot \mathbf{E}^*)\delta_{ij} + (B/2)(\tilde{E}_i \tilde{E}_j^* + \tilde{E}_i^* \tilde{E}_j), \quad (3.1)$$

where δ_{ij} is the Kronecker-delta and A and B are real constants that characterize the process which gives rise to the Kerr effect.⁸⁸ The stationary solution for a single field

of the form $\tilde{E}_x = E_1 \exp(ik_1 z)$ and $E_y = 0$ is determined by solving Eq. (3.1) in steady-state and substituting the resulting polarization and the electric field into the wave equation (Eq. (2.2)). The equation for the slowly varying-amplitude is then given by

$$\frac{\partial E_1}{\partial z} + \frac{n_1}{c} \frac{\partial E_1}{\partial t} = i \frac{2\pi\omega_1}{n_1 c} (A+B/2) |E_1|^2 E_1. \quad (3.2)$$

It can be seen that the wave experiences only a phase-shift proportional to the intensity of the field since the nonlinear coupling constant is pure imaginary. The steady-state solution to Eq. (3.2) is

$$E_1(z) = E_1(0) e^{i n_{NL} I_1 z}, \quad (3.3)$$

where $n_{NL} = 2\pi\omega_1 (A+B/2)/(n_1 c)$ and $I_1 = |E_1|^2$. The stability of this stationary solution will be determined by investigating two different cases. In the first, the amplitude of the field will be perturbed and in the second, the polarization of the field will be perturbed.

B1. Stability of the Amplitude of the Stationary Solution

--- The Scalar Interaction

The stability of the amplitude of the single strong wave (pump wave) will be investigated by considering the interaction of the strong wave with a frequency shifted probe wave. The probe wave is linearly polarized in the x-direction (parallel to the polarization of the pump wave) and is taken to be of the form

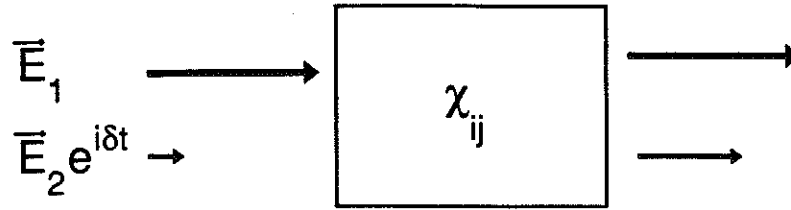
$$E_{\text{pert}} = E_2 e^{-i[(\omega - \delta)t - k_2 z]_{\Lambda}}, \quad (3.4)$$

where $k_2 = (\omega - \delta)n_2/c$. The geometry of the interaction is shown schematically in Fig. 3.1a. The evolution of the probe wave is determined by substituting the expressions for the fields into Eq. (3.1) and the resulting polarization is used as a source term in the wave equation (Eq. (2.2)). The coupled-amplitude equation is then found by making the slowly varying envelope approximation and is given by

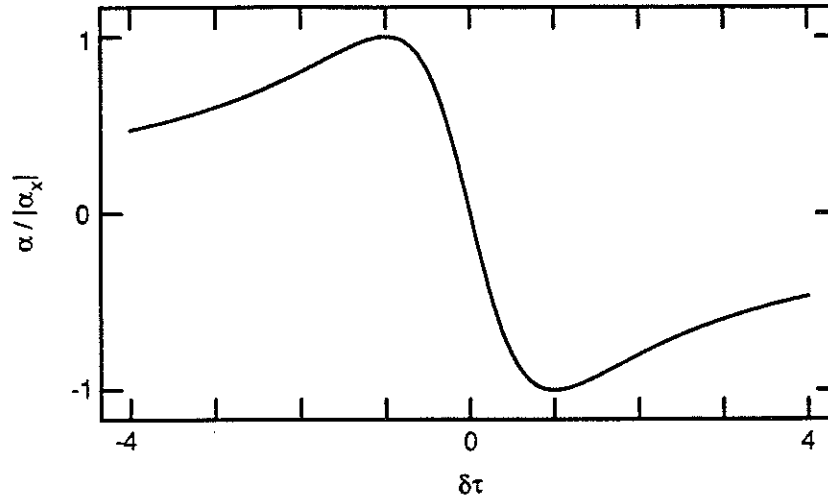
$$\frac{dE_2}{dz} = i \frac{2\pi\omega_2}{n_2 c} (A+B/2) \left(1 + \frac{1}{1+i\delta\tau}\right) |E_1|^2 E_2, \quad (3.5)$$

where $\omega_2 = \omega - \delta$. The first term of the coupling coefficient in Eq. (3.5) is purely imaginary and is the nonlinear phase-shift imparted to the probe wave by the pump wave. The second term in the coupling coefficient has a real part when $\delta\tau \neq 0$ and hence can give rise to absorption or gain of the probe wave due to the presence of the pump wave. This gain term is analogous to that which leads to stimulated Rayleigh-wing scattering.

The spatial variation of intensity of the probe wave is found from the solution to Eq. (3.5) and is given by $I_2(z) = I_2(0)e^{-\alpha z}$, where the absorption coefficient is $\alpha = -2|\alpha_x| \delta\tau / [1 + (\delta\tau)^2]$, where $|\alpha_x| = 2\pi\omega_2(A+B/2)I_1/n_2c$ is the maximum value of the absorption coefficient. The functional form of the absorption coefficient is shown in Fig. 3.1b for the case $(A+B/2) > 0$, where it is seen that for $\delta\tau > 0$ the absorption is negative implying that the probe-wave experiences gain. Note that the gain peaks when the frequency difference is equal to the response time τ of the medium. Therefore, the strong wave E_1 is unstable to the growth of a new field when the frequency of the probe wave is to the low-frequency (Stokes) side of the strong input



(a)



(b)

Figure 3.1 (a) Geometry of the interaction of a strong pump wave E_1 with a weak probe wave E_2 in a Kerr medium characterized by a susceptibility tensor χ_{ij} and a response time τ . Both waves have the same linear polarization. (b) Absorption experienced by the probe wave as a function of the normalized pump-probe detuning $\delta\tau$. For δ greater than zero, the probe wave experiences gain and hence the pump wave is unstable to the growth of new fields.

wave and for a medium with a finite response time. The threshold for this instability corresponds to $E_1 = 0$; however, in practice, the threshold corresponds to a field strength that gives an exponential gain of $\alpha L \sim -30$, where L is the length of the medium. A gain of $\sim e^{30}$ is necessary because a minimum probe wave intensity is required before it can be detected over the sources of experimental noise, such as scattered light, given that the input field strength corresponds to approximately a single input photon (quantum noise input). Note that these results are independent of the relative phase of the two fields. If the above calculation was repeated for an arbitrary propagation direction of the probe-wave, it would be found that the results are unchanged; however, a finite pump beam diameter is used in experiments, and hence only the nearly copropagating or counterpropagating probe-wave propagation directions will experience significant gain when the pump-beam diameter is small in comparison to the interaction length. The stimulated Rayleigh-wing gain process, which is independent of the relative phase and propagation directions of the interacting fields will be referred to as a 'gain' process. Hereafter, the E_2 wave will be referred to as the Stokes wave and it will be assumed that $\delta > 0$.

In the above analysis, we have ignored the response of the medium at the anti-Stokes sideband (at frequency $\omega + \delta$), which is due to the beating of the Stokes field at frequency $\omega + \delta$ with the strong field at frequency ω . To investigate the effect of the coupling to this sideband, the stability of the strong field in the presence of a perturbing field of the form

$$E_{\text{pert}} = (E_2 e^{-i[(\omega - \delta)t + k_2 z]} + E_3 e^{-i[(\omega + \delta)t + k_3 z]}) \hat{x} \quad (3.6)$$

will be determined, where $k_3 = n_3(\omega + \delta)/c$. The geometry of this interaction is shown in Fig. 3.2a. For this case, the spatial evolution of the perturbing field obeys the equations

$$\frac{\partial E_2}{\partial z} = i \frac{2\pi\omega_2}{n_2 c} (A+B/2) \left[\left(1 + \frac{1}{1+i\delta\tau}\right) |E_1|^2 E_2 + \frac{E_1^2 E_3^* e^{i\Delta k z}}{1+i\delta\tau} \right], \quad (3.7)$$

$$\frac{\partial E_3^*}{\partial z} = -i \frac{2\pi\omega_3}{n_3 c} (A+B/2) \left[\left(1 + \frac{1}{1+i\delta\tau}\right) |E_1|^2 E_3^* + \frac{(E_1^*)^2 E_2 e^{i\Delta k z}}{1+i\delta\tau} \right], \quad (3.8)$$

where $\omega_3 = \omega + \delta$ and the phase-mismatch is $\Delta k = 2k_1 - k_2 - k_3$. It is seen that the sidebands are coupled to each other through the last terms in Eqs. (3.7) and (3.8), and that the form of the coupling depends on the relative phase of the fields and on the propagation directions. This type of coupling is referred to as parametric forward four-wave mixing. Note that each probe field is driven by the complex conjugate of the other; hence if $E_3(z=0) = 0$, the anti-Stokes field generated by the nonlinear interaction will be the phase-conjugate of the Stokes field. This interaction is one possible method of generating the phase-conjugate of a wave by four-wave mixing. For $\Delta k L \gg 1$, where L is the length of the medium, the last terms in Eqs. (3.7) and (3.8) oscillate rapidly and hence give no significant contribution to the evolution of the probe fields. In the large phase-mismatch limit (a typical geometry of such an interaction is shown in Fig. 3.2b), the sidebands are essentially decoupled, and the results for the single side-band calculation presented above are valid. That is, the Stokes field E_2 experiences gain and the anti-Stokes field E_3 experiences loss.

Significantly different behavior for the evolution of the probe-fields is predicted when $\Delta k L \leq 1$. The solution to Eqs. (3.7) and (3.8) is tedious but straightforward. It is found that the evolution of the Stokes field is given by

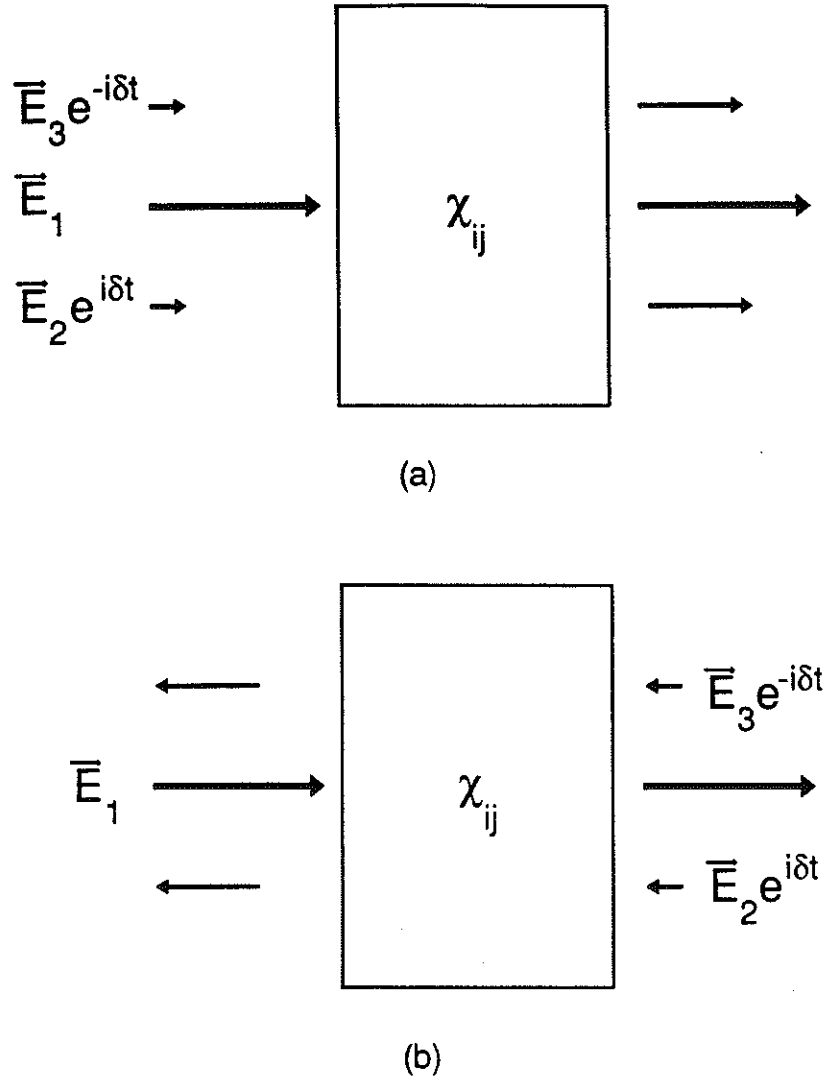


Figure 3.2 (a) Geometry of the interaction of a strong pump wave E_1 with two nearly copropagating sidebands in a Kerr medium characterized by a susceptibility tensor χ_{ij} and a response time τ . All wave have the same linear polarization. Strong parametric coupling between the pump wave and the two sidebands prevents the growth of the sidebands when the phasematching is perfect. (b) Geometry of the interaction of the strong wave with two nearly counterpropagating sidebands. The sidebands are essentially decoupled in this case because the phase-mismatch is large.

$$E_2(z) = c_1 e^{r_+ z} + c_2 e^{r_- z}, \quad (3.9)$$

where c_1 and c_2 are constants of integration which depend on the values of the probe fields at the input face of the nonlinear medium. The general expression for the eigenvalues $r_{\pm}$ is complicated; however, in the limit that the frequency shift δ is small in comparison to ω and in the limit of negligible dispersion ($n_1 = n_2 = n_3$), the eigenvalues are

$$r_{\pm} = \frac{|\alpha_x|}{2} \left\{ i(4 + \Delta k') \pm i \left[\Delta k'^2 - \frac{8\Delta k'}{1+i\delta\tau} \right]^{1/2} \right\} \quad (3.10)$$

where $\Delta k' = \Delta k / |\alpha_x|$. The coefficient $|\alpha_x|$ is equal to $2\pi\omega_2(A+B/2)I_1/n_2c$ which is the maximum value of the gain experienced by the Stokes wave in the absence of parametric coupling. The approximation that δ is much less than ω is equivalent to ignoring the frequency dependence of the coupling coefficients in the coupled-amplitude equations. The evolution of the anti-Stokes field $E_3(z)$ can be determined by substituting the expression for the Stokes field $E_2(z)$ into Eq. (3.7). Note that for $\Delta k = 0$, $r_{\pm}$ is purely imaginary and hence the waves do not experience exponential growth.

Figure 3.3a shows how the gain of the Stokes wave in the presence of parametric coupling (real part of $r_{\pm}$) depends on the phase-mismatch. It is seen that the exponential gain drops sharply to zero when $\Delta k = 0$, and that in the large phase-mismatch limit, the gain approaches that experienced by the wave in the absence of parametric coupling between the sidebands. Figure 3.3b shows the ratio $|E_3(z)/E_2(z)|$ for the case of no input anti-Stokes wave ($E_3(z=0) = 0$). For small values of the phase-mismatch, the two side-bands are strongly coupled and the anti-Stokes field is 'dragged' along by the Stokes field. For large phase-mismatch, the anti-Stokes field

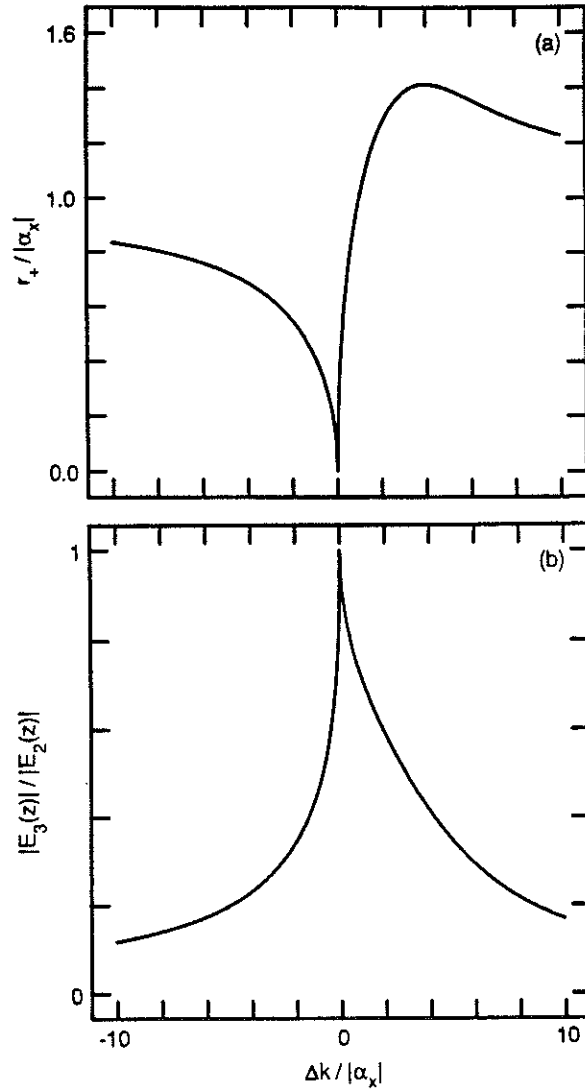


Figure 3.3 (a) Exponential gain experience by the Stokes wave E_2 in the presence of parametric coupling as a function of the phase-mismatch. The normalization coefficient $|\alpha_x|$ is equal to the maximum value of the gain experienced by the Stokes wave in the absence of parametric coupling. Gain suppression occurs at $\Delta k = 0$ due to interference between a process analogous to stimulated Rayleigh-wing scattering and parametric four-wave mixing. (b) Ratio of the anti-Stokes field E_3 to the Stokes field E_2 as a function of the phase-mismatch in the presence of parametric coupling. The Stokes field 'drags' the anti-Stokes field through the medium.

is a small fraction of the Stokes field-strength. However, this field can be significant because the Stokes wave experiences large exponential gain for large $\Delta k'$. This result is in contrast to the case when there is no parametric coupling for which the anti-Stokes wave experiences exponential absorption.

The exponential gain drops to zero for $\Delta k = 0$ because of competition between the gain process and the parametric four-wave mixing process. Since the parametric process depends on the relative phase of the fields, the contribution from the last terms in Eqs. (3.7) and (3.8) can destructively interfere with the other contributions. This interference effect is referred to as gain suppression and is known to exist in a variety of nonlinear optical processes in which side-bands interact with a strong pump wave, such as the processes of stimulated Raman scattering,⁸⁹ four-wave mixing enhance by two-photon atomic transitions,⁹⁰ four-wave mixing enhanced by Brillouin resonances,⁸⁷ and four-wave mixing enhance by a single-photon atomic transitions.⁹¹

It is of interest to determine what effects might inhibit the gain suppression since the strong pump wave is spatially stable for the case of zero phase-mismatch. It might be expected that any effect that disturbs the relative phases of the fields will disrupt the destructive interference because a well defined phase-relationship is required among the waves. Indeed, if the Kerr medium has dispersion in the refractive index, then the sidebands slip in phase with respect to each other and the interference is no longer perfect. In particular, Fig. 3.4a shows the exponential gain experienced by the Stokes wave as a function of the phase-mismatch for the case $n_2 = n_1 - \Delta n$, $n_3 = n_1 + \Delta n$, and $\Delta n/n_1 = 0.05$. It is seen that the Stokes wave now experiences exponential gain for all values of the phase-mismatch.

The strong nonlinear coupling can also be disrupted by the absorption of one of the sidebands, as is the case when the Kerr medium has a frequency dependent

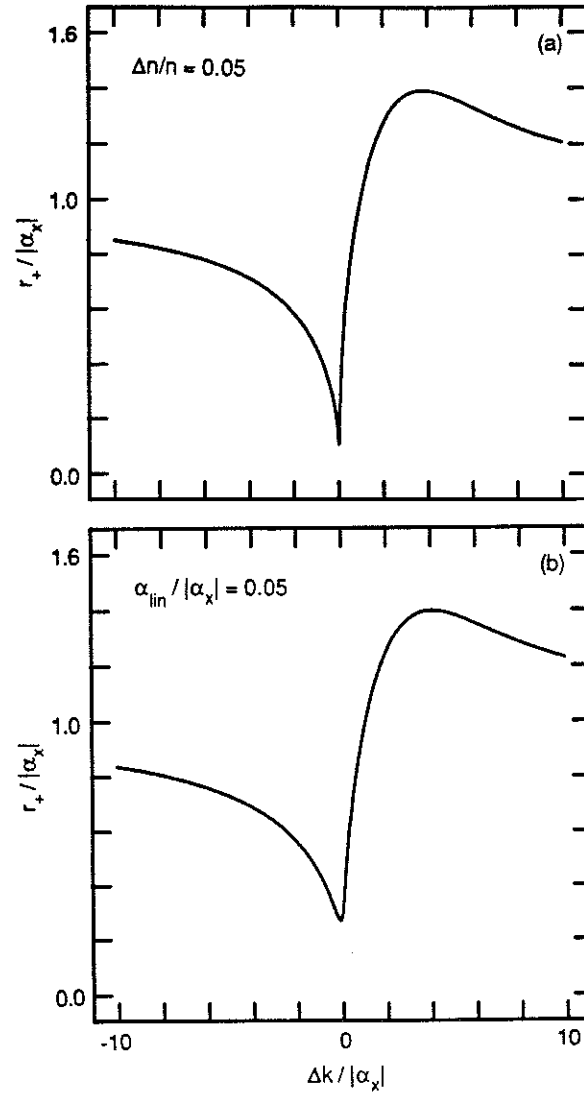


Figure 3.4 Exponential gain experienced by the Stokes wave as a function of the phase-mismatch in the presence of parametric coupling. The Stokes wave experiences exponential gain for all values of the phase-mismatch when (a) the dispersion in the medium is equal to $\Delta n/n = 0.05$ or when (b) the linear absorption at the anti-Stokes frequency is equal to $|\alpha_{lin}|/|\alpha_x| = 0.05$.

linear absorption. This effect is modeled by adding phenomenologically an additional contribution to the left-hand side of Eq. (3.8) of the form $-\alpha_{\text{lin}}^* E_3^*$. Figure 3.4b shows the exponential gain experienced by the Stokes wave assuming no linear dispersion and for a small amount of absorption at the anti-Stokes frequency ($\alpha_{\text{lin}}^*/\alpha_x = 0.05$). It is seen that the Stokes wave experiences exponential gain for all values of the phase-mismatch and hence the pump wave is unstable to the growth of new fields. Therefore, we expect that the threshold for instabilities in the amplitude of the pump field will be lowered by the presence of dispersion in the linear refractive index or by the presence of frequency dependent linear absorption.

B2. Stability of the Polarization of the Stationary Solution

--- The Tensor Interaction

In this section, the stability of the strong wave E_1 to the growth of polarization fluctuations will be investigated. It can be seen from expression for the susceptibility tensor (Eq. 3.1) that there will be coupling between different polarization components of the field only when $B \neq 0$. The ratio of B/A depends on the mechanism that is the origin of the optical nonlinearity. For the process of electrostriction B is equal to zero, and hence there will be no coupling between different polarization components. For most other nonlinearities B is nonzero. In particular, Saikan⁹² has shown that for the case of a single-photon atomic transition between a ground-level and an excited-level each having an angular momentum J equal to $1/2$, B/A is equal to -1 . This result was verified by Saikan for the case of pulsed laser light interacting with the sodium $3S_{1/2} \rightarrow 3P_{1/2}$ atomic transition. Only the case $B/A = -1$ will be studied in this section.

To investigate the stability of the polarization of the pump wave, the perturbing field is taken to have a state of polarization orthogonal to that of the pump wave and is of the form

$$E_{\text{pert}} = E_2 e^{-i[(\omega-\delta)t-k_2 z]} \hat{y}. \quad (3.11)$$

The coupled-amplitude equation for the probe wave is similar to Eq. (3.2) and is given by

$$\frac{dE_2}{dz} = i \frac{2\pi\omega_2}{n_2 c} \left[(A-B/2) + \frac{B}{2(1+i\delta\tau)} \right] |E_1|^2 E_2 \quad (3.12)$$

The solution to Eq. (3.12) is the similar to the solution to Eq. (3.5) and is $I_2(z) = I_2(0)e^{-\alpha' z}$, where the absorption coefficient in this case is $\alpha' = -2|\alpha_y| \delta\tau / [1+(\delta\tau)^2]$, where $|\alpha_y| = \pi\omega_2 B I_1 / n_2 c$. It is seen that the strong wave will be unstable to the growth of new polarizations components for $\delta > 0$ and $B > 0$ or *vice versa*. The E_2 wave will be referred to as the Stokes wave and it will be assumed that $\delta > 0$ and that $B > 0$.

To determine if the presence of a side-band at frequency $\omega+\delta$ alters the polarization stability of the pump wave as dramatically as for the case of the stability of the amplitude, a perturbing field of the form

$$E_{\text{pert}} = (E_2 e^{-i[(\omega-\delta)t+k_2 z]} + E_3 e^{-i[(\omega+\delta)t+k_3 z]}) \hat{y} \quad (3.13)$$

is considered. It is assumed that both waves are orthogonally polarized to the pump wave because the probe waves will be parametrically coupled for this case. The spatial evolution of the perturbing fields is then given by

$$\begin{aligned} \frac{dE_2}{dz} = i \frac{2\pi\omega_2}{n_2 c} \left\{ \left[(A-B/2) + \frac{B}{2(1+i\delta\tau)} \right] |E_1|^2 E_2 \right. \\ \left. + \frac{B E_1^2 E_3^* e^{i\Delta k z}}{2(1+i\delta\tau)} \right\}, \end{aligned} \quad (3.14)$$

and

$$\begin{aligned} \frac{dE_3^*}{dz} = -i \frac{2\pi\omega_3}{n_3 c} \left\{ \left[(A-B/2) + \frac{B}{2(1+i\delta\tau)} \right] |E_1|^2 E_3^* \right. \\ \left. + \frac{B (E_1^*)^2 E_2 e^{i\Delta k z}}{2(1+i\delta\tau)} \right\}. \end{aligned} \quad (3.15)$$

We see immediately that the case of a perturbation of the polarization of the pump wave is different than the case of a perturbation of the amplitude because the material coefficients enter into the equations in a different manner. The stimulated Rayleigh-wing scattering terms and the parametric four-wave mixing terms in this case are proportional only to B . The first terms in Eqs. (3.14) and (3.15) represent the nonlinear phase-shift of the probe waves induced by the pump wave. For the case when the amplitudes are perturbed, the phase-shift experienced by the probe waves is the same as the phase-shift of the pump beam induced by itself (in the limit $\delta/\omega \ll 1$ and in the limit of no dispersion). However, in the present case, the phase-shift of the probe waves is different than the phase-shift that the pump induces on itself. Therefore, we expect that the interference between the various nonlinear processes

will not be complete for $\Delta k = 0$ because of this difference in the relative phases of the fields.

The evolution of the Stokes wave in the presence of parametric coupling is given by

$$E_2(z) = c_1' e^{r_+^{\prime} z} + c_2' e^{r_-^{\prime} z}. \quad (3.16)$$

The eigenvalues $r_{\pm}'$ are equal to

$$r_{\pm}' = \frac{|\alpha_y|}{2} \left\{ i(4 + \Delta k'') \pm i \left[\left(8 + \frac{4}{1+i\delta\tau} - \Delta k'' \right)^2 - \frac{16}{(1+i\delta\tau)^2} \right]^{1/2} \right\}, \quad (3.17)$$

where $\Delta k'' = (|\alpha_x|/|\alpha_y|)\Delta k/|\alpha_x|$ and where it has been assumed that $\delta/\omega \ll 1$ and $n_1 = n_2 = n_3$. The coefficient $|\alpha_y|$ is equal to $\pi\omega_2 B I_1/n_2 c$ which is the maximum gain experienced by the Stokes wave in the absence of parametric coupling. The evolution of the anti-Stokes $E_3(z)$ wave is determined by substituting the expression for the Stokes wave $E_2(z)$ into Eq. (3.14). Figure 3.5a shows the gain experienced by the Stokes wave in the presence of parametric coupling as a function of the phase-mismatch. It is seen that there is a sharp drop in the gain at $\Delta k/|\alpha_x| = 8$, which is exactly the phase-shift required to compensate for the difference in the phase-shifts of the fields induced by the strong pump beam. Figure 3.5b shows the ratio $|E_3(z)/E_2(z)|$ as a function of the phase-mismatch. As in the previous case, it is seen that the coupling of the anti-Stokes wave to the Stokes wave causes the anti-Stokes wave to experience gain, not loss.

In this section and in the previous one, two nonlinear optical processes have been identified which are important when a single strong wave propagates in a Kerr

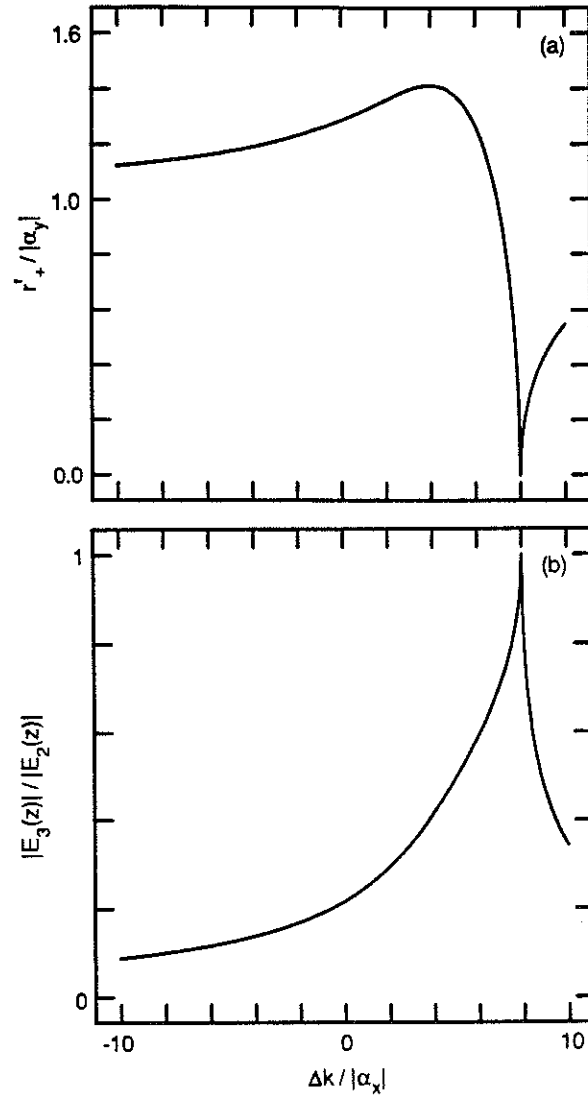


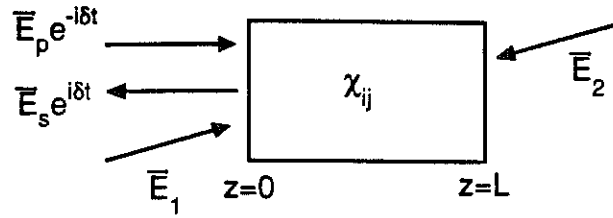
Figure 3.5 (a) Exponential gain experienced by the Stokes wave as a function of the phase-mismatch in the presence of parametric coupling. The probe waves are linearly polarized with a polarization orthogonal to that of the pump wave. Gain suppression occurs for a large value of the phase-mismatch. (b) Ratio of the anti-Stokes field E_3 to the Stokes field E_2 as a function of the phase-mismatch. The anti-Stokes wave experiences gain for all values of the phase-mismatch.

medium. One of these processes is stimulated Rayleigh-wing scattering. The gain experienced by a probe wave for this process is zero when the medium responds infinitely fast. The process does not depend on the relative phase of the fields. The second nonlinear optical process is parametric four-wave mixing. This process depends on the relative phases of the interacting fields and can occur in a medium that responds infinitely fast. Competition due to destructive interference between these two processes can dramatically alter the stability characteristics of the strong wave.

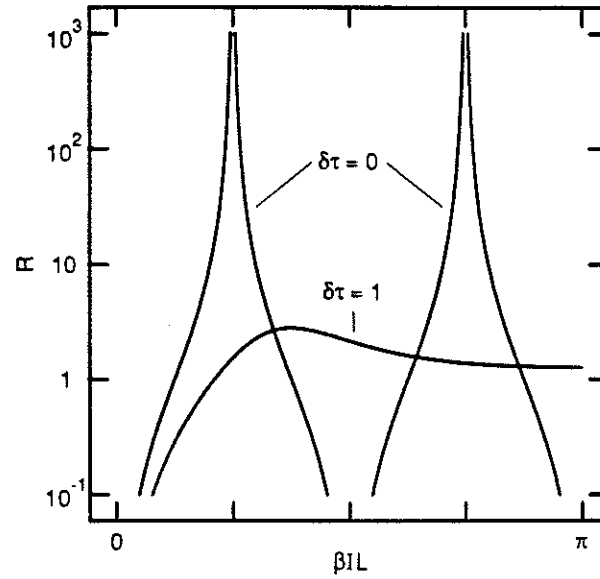
C. Phase-Conjugation by Nearly-Degenerate Four-Wave Mixing

In this section, the spatial stability of two strong counterpropagating waves interacting in a nonlinear Kerr medium will be considered. In particular, the geometry of Fig. 3.6a is analyzed where two pump waves E_1 and E_2 are shown interacting with a weak probe beam E_p to generate a signal beam E_s . It will be shown that the signal beam is a wavefront-reversed replica of the input probe beam and that the ratio of the generated field strength to the input probe-beam field strength can be infinite. This nonlinear optical interaction leads to phase-conjugation by nearly degenerate four-wave mixing. In this example, only linearly polarized fields will be considered.

The electric field of the (strong) pump waves is taken to be of the form $E = (E_1 e^{ikz'} + E_2 e^{-ikz'}) e^{-i\omega t}$ where the z' -axis is nearly along the z -axis. Steady-state solutions for the evolution of the two strong counterpropagating pump waves will be obtained. The stability of the stationary solutions will be determined by applying weak probe and signal fields as perturbations. The steady-state expression for the nonlinear susceptibility tensor for this case is given by



(a)



(b)

Figure 3.6 (a) Geometry of the interaction of two strong laser beams $\bar{E}_1$ and $\bar{E}_2$ counterpropagating in a Kerr medium of length L characterized by a susceptibility tensor χ_{ij} and a response time τ . The stability of the pump waves to the growth of a probe field $\bar{E}_p$ and a signal field $\bar{E}_s$ is determined. The signal field is the phase conjugate of the probe field. (b) Phase-conjugate reflectivity R as a function of the nonlinear phase-shift βIL for two different pump-probe detunings. The reflectivity becomes infinite for $\delta\tau = 0$ and $\beta IL = m\pi/4$, where m is an integer. Hence, the pump waves are unstable even when there are no input perturbing fields. The reflectivity is much larger than one when $\delta\tau = 1$ for a broad range of the nonlinear phase-shift; however, the reflectivity never becomes infinite.

$$\chi_{xx} = (A+B/2)(|E_1|^2 + |E_2|^2 + E_1 E_2^* e^{i2kz'} + E_1^* E_2 e^{-i2kz'}), \quad (3.18)$$

and all other elements equal to zero. We see from Eq. (3.18) that the susceptibility tensor contains interference terms, showing that the refractive index will be spatially modulated. It will be shown in the next section that the modulation of the refractive index can have a significant affect on the dynamical stability of the counterpropagating waves.

The coupled-amplitude equations governing the spatial evolution of the pump fields E_1 and E_2 are obtained by substituting Eq. (3.18) into the wave equation (Eq. (2.2)) and retaining only the nearly phase-matched terms. They are given by

$$\frac{dE_1}{dz'} = 2\pi i k (A+B/2)(|E_1|^2 + 2|E_2|^2)E_1, \quad (3.19)$$

and

$$-\frac{dE_2}{dz'} = 2\pi i k (A+B/2)(|E_2|^2 + 2|E_1|^2)E_2. \quad (3.20)$$

By inspection of these equations it is seen that the intensities do not vary spatially and hence that the waves do not exchange energy. The solutions to Eqs. (3.19) and (3.20) are obtained straightforwardly and, for the proper choice of the phases of the input fields, are given by

$$E_1(z') = I_1^{1/2} e^{i\beta(I_1 + 2I_2)z'}, \quad (3.21)$$

$$E_2(z') = I_2^{1/2} e^{-i\beta(I_2 + 2I_1)z'}, \quad (3.22)$$

where $I_1 = |E_1(0)|^2$, $I_2 = |E_2(L)|^2$ and $\beta = 2\pi k(A+B/2)$.

These steady-state solutions for the strong waves are perturbed by a field of the form

$$E_{\text{pert}} = (E_p e^{i(k_p z + \delta t)} + E_s e^{-i(k_s z + \delta t)}) e^{-i\omega t} \hat{x}, \quad (3.23)$$

where $k_p \approx k_s$. For simplicity, it will be assumed that the intensities of the pump waves are equal and are given by $I_1 = I_2 = I$. It is also assumed that the probe wave is nearly copropagating with the E_1 pump field so that $k \approx k_p$, but incident at a sufficiently large angle so that the contribution of the cross-coupled waves can be ignored.¹⁸ The coupled amplitude equations for the probe and signal field are given by

$$\frac{dE_p}{dz} = 2aE_p + bE_s^* e^{i\Delta kz}, \quad (3.24)$$

$$\frac{dE_s^*}{dz} = 2aE_s^* + bE_p e^{-i\Delta kz} \quad (3.25)$$

where the coupling coefficients are $a = i\beta I[(2+i\delta\tau)/(1+i\delta\tau)]$, $b = i2\beta I/(1+i\delta\tau)$, the phase-mismatch between the probe and signal wave is $\Delta k = k_s - k_p$, and where it is assumed that $\delta/\omega \ll 1$. It is seen that if no signal field is injected, the resulting signal field generated by the nonlinear interaction will be the phase-conjugate of the input probe field. The phase-conjugate reflectivity is defined as $R = |E_s(0)/E_p(0)|^2$ and is obtained from the solutions of Eqs. (3.24) and (3.25) for the boundary conditions $E_p(z=0) = E_p(0)$ and $E_s(L) = 0$ and is found to be

$$R = \left| \frac{b(e^{\alpha_+ L} - e^{\alpha_- L})}{(\alpha_+ - 2a)e^{\alpha_+ L} - (\alpha_- - 2a)e^{\alpha_- L}} \right|^2 \quad (3.26)$$

where $\alpha_{\pm} = [(4a + i\Delta k) \pm (4b^2 - \Delta k^2)^{1/2}]/2$. When $\delta\tau = 0$ and $\Delta kL = 0$, the expression for the reflectivity simplifies to

$$R = \tan^2(2\beta IL). \quad (3.27)$$

It is seen that the reflectivity becomes infinite whenever the nonlinear phase shift is $\beta IL = m\pi/4$ where m is an integer. The actual reflectivity at these values of the nonlinear phase-shift will be finite due to depletion of energy from the pump waves. An early analysis of phase conjugation by degenerate four-wave mixing⁸⁰ noted that the reflectivity can be infinite and that novel devices could be constructed using a phase-conjugate mirror operated in this regime. Figure 3.6b shows the dependence of the phase-conjugate reflectivity as a function of the nonlinear phase-shift as predicted by Eq. (3.26). When $\delta\tau = 0$, it is seen that the reflectivity becomes very large for multiple values of the nonlinear phase-shift, as expected from the simplified expression (Eq. (3.27)). Note that the value of the nonlinear phase-shift required for infinite phase-conjugate reflectivity is well below the threshold for the growth of sidebands from a single strong pump wave ($\beta IL = 30$), as discussed previously in section B of this chapter. For $\delta\tau = 1$ and $\Delta kL = 0$, it is seen that the reflectivity is greater than one for a wide range of values of the nonlinear phase-shift, but never becomes infinite.

To determine the frequency response of the phase-conjugate mirror, the phase-conjugate reflectivity as a function of the frequency difference of the pump and probe is shown in Fig. 3.7. It is seen that for a nonlinear phase-shift of $\beta IL = \pi/4$,

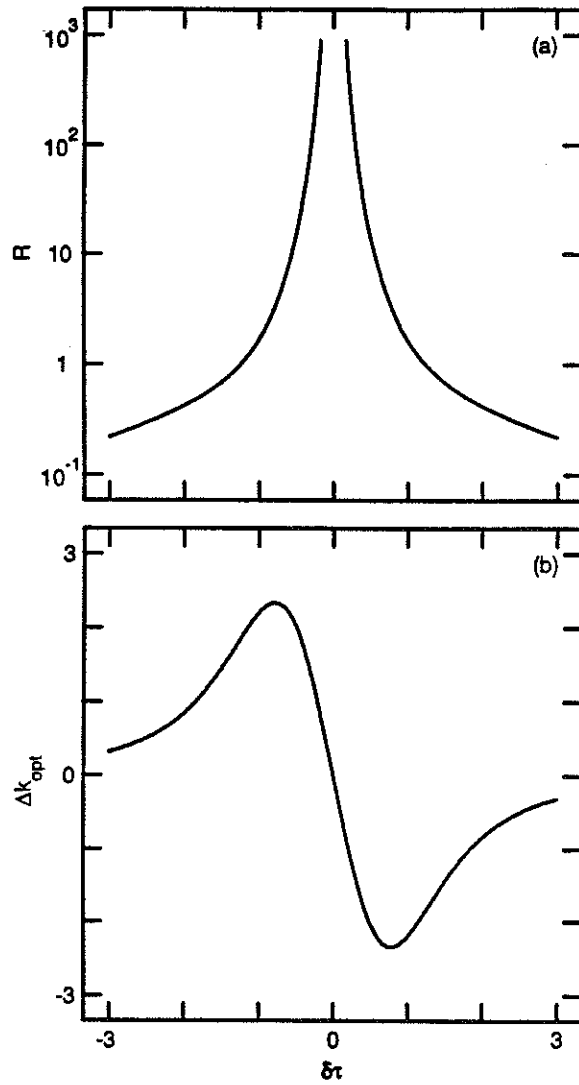


Figure 3.7 (a) Phase-conjugate reflectivity as a function of the pump-probe detuning for $\beta IL = \pi/4$. The phase-mismatch is varied to determine the largest value of the reflectivity and the optimum phase-mismatch $\Delta k_{\text{opt}}L$ is plotted in (b). The phase-conjugate reflectivity is large only for small pump-probe detunings.

the reflectivity is sharply peaked at $\delta\tau = 0$, and hence such a phase-conjugate mirror could be used as a narrow-band filter.⁹³ The reflectivity is dependent on the phase-mismatch of the probe and signal wave, therefore the value of the phase-mismatch which gives the largest reflectivity Δk_{opt} was used in calculating the curve in Fig. 3.7a and its dependence on the frequency difference is shown in Fig. 3.7b. This is a reasonable procedure since, in an experiment, the direction of the wave which experiences the largest gain will be the observed wave.

D. Dynamical Stability of Counterpropagating Laser Beams

In this section, the dynamical stability characteristics of counterpropagating laser beams in a Kerr medium will be reviewed. Silberberg and Bar-Joseph^{26,27} analyzed the case of a sluggish (τ finite), scalar ($B = 0$) Kerr medium. It was found that the amplitudes of the transmitted fields are dynamically unstable above a well defined value of the input intensities. Above this threshold, the intensities oscillate periodically and, for even larger values of the input intensities, the transmitted waves fluctuate chaotically.

For the case of counterpropagating waves, it was found in the last section that the susceptibility tensor is spatially modulated for the case of counterpropagating laser beams; hence, it is useful to make the *ansatz*

$$\chi_{ij} = \chi_{ij}^0 + \chi_{ij}^+ e^{i2kz} + \chi_{ij}^- e^{-i2kz}. \quad (3.28)$$

Kogelnik and Shank⁸⁴ have investigated the effects of a spatially modulated refractive index and find that such a structure displays interesting reflection properties. In particular, for a spatial modulation frequency of $2k/2\pi$, it is found that the reflectivity

of the structure is peaked at $mc/4L$ (m an integer) side-mode frequencies. This type of refractive-index structure is referred to as a distributed-feedback structure and can provide feedback between the counterpropagating laser beams.

Following the work of Silberberg and Bar-Joseph, the amplitude stability of linearly polarized pump waves of the form $\mathbf{E} = [E_1 \exp(ikz) + E_2 \exp(-ikz)] \exp(-i\omega t) \hat{x}$ will be studied. The temporal evolution of each of the spatial Fourier amplitudes of the susceptibility is determined by substituting the expression for the field and Eq. (3.28) into the differential equation governing the temporal evolution of the total nonlinear susceptibility (Eq. (3.1)). The coupled amplitude equations for the fields are then given by

$$\frac{\partial E_1}{\partial z} + \frac{n}{c} \frac{\partial E_1}{\partial t} = i \frac{2\pi\omega}{cn} (\chi_{xx}^0 E_1 + \chi_{xx}^+ E_2) \quad (3.29)$$

and

$$-\frac{\partial E_2}{\partial z} + \frac{n}{c} \frac{\partial E_2}{\partial t} = i \frac{2\pi\omega}{cn} (\chi_{xx}^0 E_2 + \chi_{xx}^- E_1). \quad (3.30)$$

The steady-state solution to Eqs. (3.29)-(3.30) and (3.1) for equal input intensities are the same as that found in the previous section (Eq. (3.21) and (3.22)) and are denoted by E_1^{ss} and E_2^{ss} , respectively.

Silberberg and Bar-Joseph investigate the dynamical stability of Eqs. (3.29) and (3.30) by applying a perturbing field of the form

$$\begin{aligned} E_{\text{pert}} = & [E_1^{ss}(f_1(z)e^{\lambda t} + f_2(z)e^{\lambda^* t})e^{ikz} \\ & + E_2^{ss}(b_1(z)e^{\lambda t} + b_2(z)e^{\lambda^* t})e^{-ikz}]e^{-i\omega t} \hat{x}. \end{aligned} \quad (3.31)$$

with the boundary conditions $f_m(0) = b_m(L) = 0$ ($m=1,2$). By inserting Eq. (3.31) into Eqs. (3.1), (3.29) and (3.30), and linearizing about the small, normalized field quantities f_m and b_m ($m=1,2$), coupled-amplitude equations for the perturbing fields are derived. From these coupled amplitude equations, the boundary for instability can be determined by setting $\text{Re}(\lambda) = 0$ and solving for the intensity corresponding to the instability threshold and for the frequency of oscillation $\text{Im}(\lambda)$. It was found that there are an infinite number of solutions for the instability boundary. Each solution corresponds to the excitation of a different mode of the distributed-feedback structure. It was found that the lowest instability threshold occurs when the response time of the medium is approximately equal to the transit time ($\tau \approx c/L$) and for the lowest order mode ($m = 1$) of the distributed-feedback structure. The nonlinear phase-shift corresponding to the minimum instability threshold was found to be $\beta IL = 2$. It was also found that the counterpropagating waves were dynamically stable if the medium had an instantaneous response time ($\tau \rightarrow 0$). The origin of this instability is attributed to stimulated Rayleigh-wing gain, parametric four-wave mixing and distributed feedback from the spatial modulation of the refractive index.

It is of interest to determine if counterpropagating waves are dynamically unstable to the growth of polarization fluctuations since a single strong pump wave is unstable to the growth of polarization fluctuations when the material parameter $B \neq 0$. Winful and Marburger⁹⁴ treated the case of counterpropagating waves interacting in a tensor Kerr medium and found that the states of polarization of the transmitted waves were multistable. Gaeta *et al.*²⁹ repeated the stability analysis outlined above, except that the perturbing field was polarized in the y-direction, rather than the x-direction. When the material parameters were such that $B/A \rightarrow 0$, the strong pump waves were stable to the onset of polarization fluctuations. For $B/A = 3$, it was found that the intensities of the waves were stable, but the polarizations of the

transmitted waves were rotated to a new, constant value for $0.79 < \beta_{IL} < 1.4$. For a nonlinear phase shift greater than 1.4, which is significantly lower than the value required to see intensity instabilities, both the intensities and polarizations of the waves were temporally unstable. It was also found that the instability threshold for polarization rotation was independent of the response time of the medium. For the case of a fast response time medium, the mechanism for the instability is parametric four-wave mixing and not stimulated Rayleigh-wing gain. For other ratios of the material parameters B/A , it is found in general that the threshold for instability is always lower than that predicted for intensity instabilities. The reason that the threshold for polarization instabilities is lower than that for amplitude instabilities is that gain suppression effects the dynamical instability in a similar manner as it effects the convective instability (section B of this chapter). From these results it is expected that it would be difficult, if not impossible, to observe intensity fluctuations unaccompanied by polarization fluctuations unless a scalar medium is used.

CHAPTER IV

INSTABILITIES OF COUNTERPROPAGATING LASER BEAMS IN SODIUM VAPOR

A. Introduction

Wu *et al.*⁹⁵ observed that a strong, near-resonant laser beam propagating through a collection of sodium atoms is unstable to the growth of a wave detuned with respect to the pump frequency by the generalized Rabi frequency. This experiment demonstrated that instabilities due solely to the effects of propagation can occur. In many instances instabilities can be useful for the efficient upconversion of infrared radiation,⁹⁶ for the transfer of energy from one beam of light into another beam⁹⁷ and for the generation of quantum states of the radiation field.⁹⁸ However, experimental results are presented in this chapter that demonstrate that laser beams counterpropagating through sodium vapor can fluctuate chaotically and that the instability can degrade the performance of nonlinear optical devices. These new results demonstrate that a passive, cavity-less, nonlinear optical system can be chaotic and that the dynamical evolution of the system is characterized by a strange attractor.

In particular, it will be shown that when the frequency of the laser is tuned near to the sodium $3S_{1/2} \rightarrow 3P_{1/2}$ atomic transition, the states of polarization and intensities of the counterpropagating fields are unstable. The attractors describing the evolution in phase space are constructed from measurements of the resulting fluctuations and the fractal dimensions and the entropies of the attractors are deter-

mined. The nonlinearity of sodium vapor cannot totally be explained by the Kerr-medium model; however, the experimental results are in good qualitative agreement with the theoretical predictions of Gaeta *et al.*²⁹ Theoretical models will be developed in the next chapter that explain some of the experimental results presented in this chapter which are not predicted by the Kerr-medium model.

The study of the stability characteristics of laser beams counterpropagating in nonlinear optical media is an active area of research. Recently, Khitrova, Valley and Gibbs⁹⁹ have observed an instability in the intensities and not the polarizations of laser beams counterpropagating in sodium vapor. The results presented in Ref. 99 only demonstrate the existence of an instability; no attempt was made to reconstruct the dynamical evolution of the system. Soon thereafter, Gauthier, Malcuit and Boyd²⁸ presented preliminary results of the experiments described in this chapter. It is difficult to determine whether or not the results of Gauthier *et al.* are consistent with those of Khitrova *et al.* because several experimental parameters required to make a quantitative comparison were not given in Ref. 99. Quite recently, Gaeta *et al.*⁸⁷ have observed an instability in the intensities of laser beams counterpropagating in a Brillouin active medium. The results for the case of a Brillouin medium are in good agreement with theoretical predictions; however, it was difficult to measure the time evolution of the fluctuations and hence it was difficult to construct the attractors describing the dynamical evolution.

B. Experimental Method

The experimental set-up used to observe unstable behavior of counterpropagating laser beams consists of two laser beams with a well-defined state of polarization that are input to a sodium vapor, is shown in Fig. 4.1a. Measurements

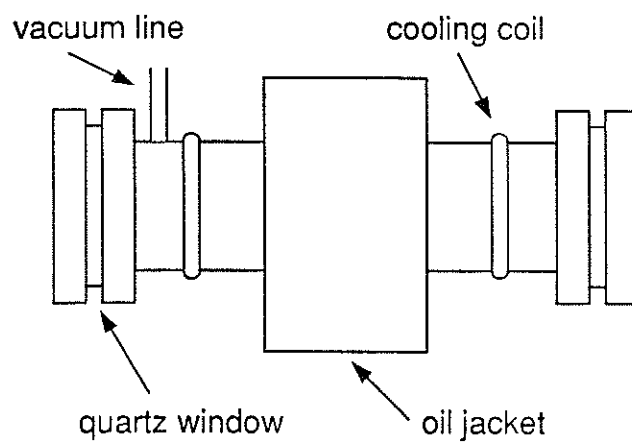
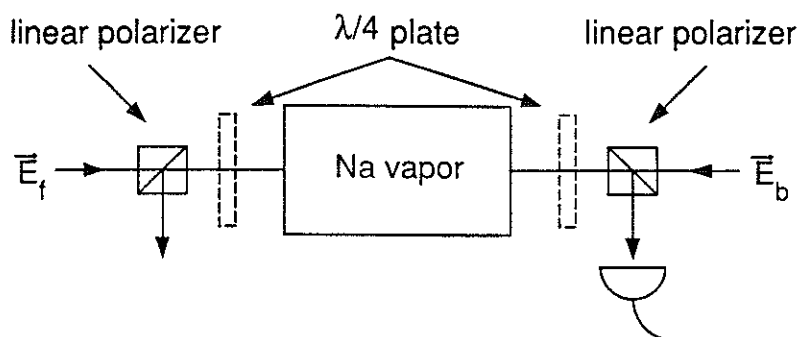


Figure 4.1 (a) Experimental set-up used to observe unstable behavior of counterpropagating laser beams in sodium vapor. The quarter-wave plate are inserted for the study of instabilities of circularly polarized waves. (b) Diagram of the sodium heat-pipe oven.

are performed to determine the states of the transmitted fields as functions of the intensities of the input beams, the detuning of the laser frequency from the atomic transition, the atomic number density, and the helium buffer gas pressure. The dependence of the instability threshold for various states of the input polarizations is also determined. The transmitted fields are characterized by measurements of the state of polarization, the optical spectrum, the temporal evolution of the intensities and the spatial distribution of the transmitted radiation.

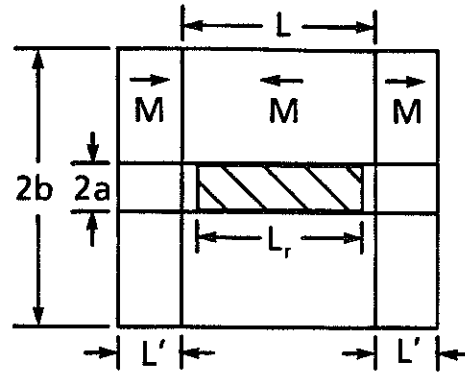
The counterpropagating laser beams are derived from the output of a single-mode, stabilized, continuous-wave ring dye-laser (Coherent model CR-699-21) tuned near to the atomic transition. The linewidth of the laser is ≤ 1 MHz, which is less than the ~ 10 MHz atomic linewidth, and the laser frequency can be scanned over a range of 30 GHz, which is much larger than the ~ 2 GHz Doppler width of the transition. Over 1 Watt of power can be obtained for frequencies near to the atomic sodium transition when Rhodamine 6G is used as the laser gain medium. Measurements of the input laser power are made with a calibrated power meter (United Detector Technology model 61). The laser beams are weakly focused at the center of the 5 cm long sodium vapor interaction region to a spot size of $\sim 725 \mu\text{m}$ and are carefully aligned to ensure that the beams are counterpropagating.

A sodium heat-pipe oven is used to produce the atomic sodium vapor because high optical quality windows which do not modify the state of polarization of the input and transmitted laser beams can be used. A schematic diagram of the heat-pipe oven is shown in Fig. 4.1b. A buffer gas is required for proper operation of the heat-pipe¹⁰⁰ and the effects of collisions between the buffer gas atoms and the sodium atoms complicates the nature of the nonlinear interaction. However, when a helium buffer gas is used, these complicating effects can be minimized because a helium atom colliding with a sodium atom essentially disrupts the phase of the coherence

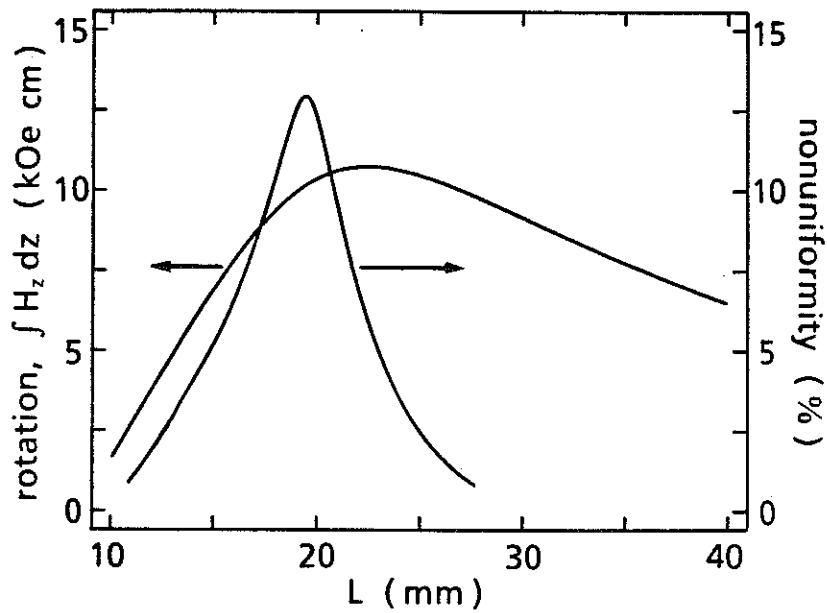
only and does not change the state of the sodium atom.¹⁰¹ For the number densities used in this experiment, the required buffer gas pressure was only ~15 mTorr resulting in a collisional linewidth determined predominantly by the self-broadened collisional width.

To ensure that the observed instabilities are not due to feedback into the laser, a Faraday isolator is used to decouple the laser from the experiment. Commercially available Faraday isolators have a low isolation ratio and are expensive. Therefore, Gauthier *et al.*¹⁰² developed a high-performance, permanent-magnet Faraday isolator. A Faraday isolator that provides high transmission and good isolation consists of two linear polarizers placed around a Faraday rotator that rotates the polarization direction of the incident light by 45 degrees and has uniform rotation across the clear aperture. The design procedure used in Ref. 102 for the Faraday rotator is the first to include the effects of uniformity of the rotation across the clear aperture.

The design criteria for the Faraday rotator are that the rotation angle be 45 degrees at 633 nm, that the clear aperture be 5 mm and that the isolation ratio be greater 43 dB for a Gaussian beam diameter of 2 mm. A schematic of the design is shown in Fig. 4.2a. The magnets¹⁰³ are right-circular cylinders of diameter $2b$ with a concentric hole of diameter $2a$ and a uniform magnetization $\mathbf{M}$ parallel to the axis of the cylinder and are $\text{Sm}_2\text{Co}_{17}$ permanent magnets with a residual magnet induction of 10^4 Gauss. The rotator rod of length L_r is placed in the center of the central magnet of length L nearly equal to L_r . The rotator glass¹⁰⁴ has a Verdet constant of $-4.12 \text{ deg kOe}^{-1} \text{ cm}^{-1}$ at a wavelength of 633 nm. For this value of the Verdet constant, an integrated magnetic intensity equal to 10.8 kOe is required to obtain 45 degrees rotation. Auxiliary magnets of length L' are placed on both ends of the central magnet with their magnetization vectors oriented anti-parallel to that of the central magnet.



(a)



(b)

Figure 4.2 (a) Cross-sectional view of the permanent-magnet Faraday rotator. (b) The on-axis rotation and fractional difference in the rotation angle between the axis and at a distance of 2.5 mm from the axis as a function of the length L of the central magnet. The length of the rotator rod is held fixed at 19.5 mm, and $a=2.8$ mm, $b=16.5$ mm, $L'=7.6$ mm, and the residual inductance is 10^4 Gauss.

The rotation angle θ of the incident linear polarization introduced in a single-pass through the Faraday rotator is given by

$$\theta = V(\lambda) \int H_z dz \quad (4.1)$$

where $V(\lambda)$ is the Verdet constant, H_z is the axial component of the magnetic intensity vector, and the integration is to be performed over the length of the rotator rod. The Verdet constant of the glass scales approximately as λ^{-2} ; therefore, a rotation of 45 degrees can be obtained for wavelengths shorter than 633 nm by displacing the rod from the center of the magnet stack.

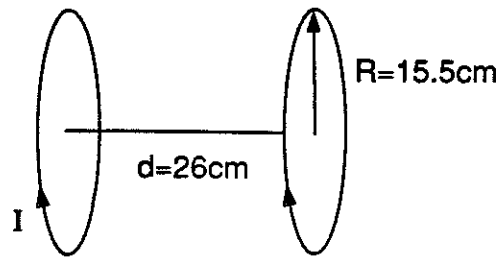
It is found that the rotation increases by increasing the length of the auxiliary magnets, by increasing the outer diameter of the magnets and by decreasing the inner diameter of the hole. It is also found that the rotation approximately increases quadratically with the radial distance from the axis of the rotator. A goal of the design is to minimize the change in rotation as a function of the radial distance from the axis of the rotator. An isolation ratio of 43 dB for a Gaussian beam diameter of 5mm corresponds to a maximum change of the rotation of 5%, where the maximum change of rotation occurs at the edge of the aperture. Figure 4.2b shows the integrated on-axis magnetic intensity and the nonuniformity of the rotation as a function of the central magnet length for $a = 2.8$ mm, $b = 16.5$ mm, $L' = 7.6$ mm. The length of the rod is 19.5 mm and it is assumed that the rod is in the center of the magnet stack. It is seen that the rotation is insensitive to the length of the central magnet for $L \approx L_T$. However, the nonuniformity is sharply peaked at $L = L_T$. Therefore, the performance of the rotator can be dramatically improved by making L slightly greater than L_T . An isolator constructed with $L = 22.8$ mm has a rotation of 45 degrees and an isolation ratio greater than 45 dB at a wavelength of 633 nm for a

Gaussian beam diameter of 2 mm. It is found that this Faraday rotator, when combined with high-quality calcite polarizer, provides sufficient isolation when the isolator is placed between the laser and the experiment. This was verified by observing that the frequency and amplitude stability of the laser is unaffected by direct feedback of a beam towards the laser.

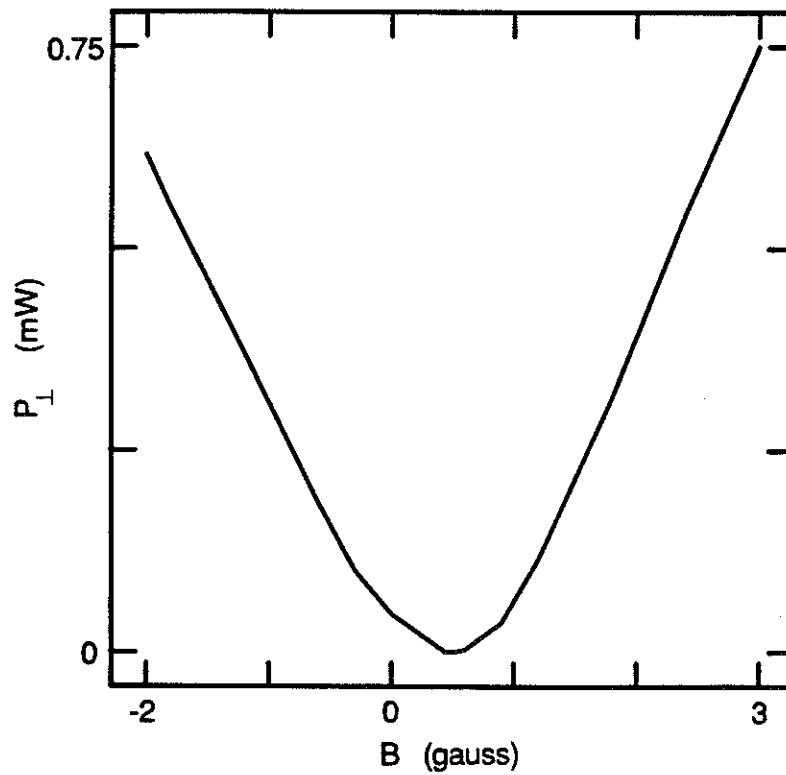
The observation of instabilities of laser beams propagating through sodium vapor are obscured by the rotation of the polarization of an incident linearly polarized wave by the Faraday effect. Magnetic fields passing through the sodium vapor of the order of 1 Gauss along the axis defined by the counterpropagating fields (axial-magnetic-field) were sufficient to rotate the incident polarization by a few degrees. Therefore, the heat-pipe oven is heated by oil capable of withstanding high temperatures rather than the common technique of electrical heating. A Helmholtz coil with a geometry shown in Fig. 4.3a is placed around the heat-pipe to cancel any stray magnetic fields. The field strength at the center of the coil is calculated to be 3 Gauss for 41 turns of wire on each loop and for a current of 1 A and the uniformity of the field along the axis of the coil over the 5 cm sodium interaction length is >98%.

Figure 4.3b shows the average power emitted in the polarization direction orthogonal to the input polarization direction for a single input laser beam as a function of the axial-magnetic-field strength produced by the Helmholtz coil. The input field is linearly polarized, has a power of 69 mW, the number density is 1.6×10^{13} atoms cm^{-3} , and the buffer gas pressure is ~ 15 mTorr. It is seen that for a field strength of ~ 0.5 Gauss, there is essentially no power emitted in the orthogonal polarization. In all experiments, the magnetic field is kept at this value.

Sodium displays the Faraday effect because the ground level and the excited level of the D_1 transition are comprised of several Zeeman sub-levels. The structure



(a)



(b)

Figure 4.3 (a) Diagram of the Helmholtz coil used to cancel stray magnetic fields. The field at the center of the coil is equal to 3 Gauss when the current I is 1 A. (b) Average power emitted in the orthogonal polarization direction for the transmitted forward wave as a function of the axial magnetic field strength. The input forward wave power is 69 mW. In the experiments described in the Chapter, the axial magnetic field is held fixed at 0.5 Gauss.

of the transition is also complicated due to the effects of the hyperfine interaction because the nuclear spin I , which is equal to $3/2$, couples with angular momentum J , which is equal to $1/2$ for both the ground and excited levels. Therefore, the ground and excited levels are each comprised of eight Zeeman sub-levels. The energy level diagram¹⁰⁵ for the transition is shown in Fig. 4.4.

The optical spectrum of the transmitted fields is measured by a confocal, scanning Fabry-Perot with a free spectral range of 7.5 GHz (Coherent model 240-1-B). The time evolution of the transmitted intensities is measured using a fast photodiode (EG&G model FND-100, back-biased with 90V) which has a response from dc to ≥ 1 GHz. In some experiments, the output of the detector is amplified with a low-noise, high-bandwidth (10 kHz-500 MHz) amplifier (Trontech model W500K) and is monitored with either a ten-bit transient digitizer with a sampling rate of up to 200 MHz and capable of collecting a long time series (Sony-Tektronix model RTD710), or with a spectrum analyzer having a bandwidth of 100 Hz to 1.9 GHz (Tektronix model 2753P). Measurements of the average power of the transmitted beams are made with a photodiode and a lockin amplifier (Stanford Research model SR510). All data is collected under computer control using either an IBM PC or a LSI-11/23-based computer. The spatial distribution of the transmitted radiation is recorded with photographic film.

The range of experimental parameters that affect the instability threshold is quite large. However, it is found that a change in the state of polarization of the field is the first indication of the existence of an instability. In no case is it found that the intensities of the transmitted beams are unstable unaccompanied by polarization fluctuations. Therefore, the boundary for instability is easily determined by measurements of the average power of the transmitted beams in the polarization direction orthogonal to the polarization direction of the input beams. One of the beams

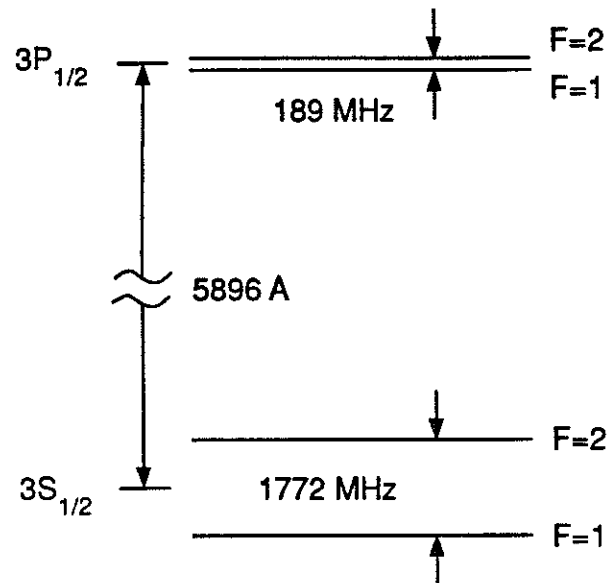


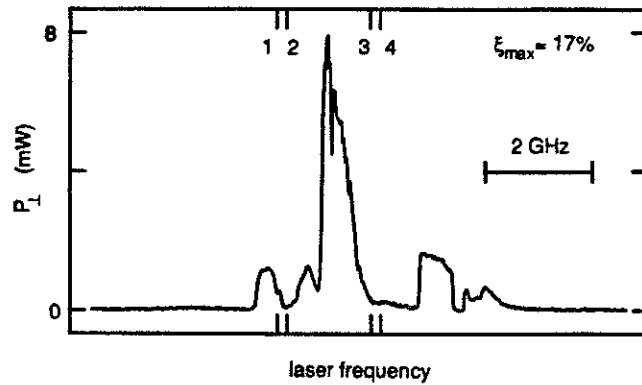
Figure 4.4 Energy level diagram of the sodium $3S_{1/2} \rightarrow 3P_{1/2}$ atomic transition. The hyperfine interaction is responsible for the splitting of the levels.

(denoted the forward beam) is held at a fixed power and measurements of the average power of the transmitted forward wave in the orthogonal polarization direction are made as the laser is tuned through the atomic resonance for various powers of the other beam (denoted the backward beam). By repeating these measurements for various forward beam intensities, the dependence of the instability threshold on the sum of the input intensities and on the ratio of the two input intensities can be determined for fixed values of the number density and buffer gas pressure. The dependence on atomic number density (buffer gas pressure) was similarly determined for fixed input intensities and buffer gas pressure (atomic number density). All of the parameter space is not explored in the experiments; however, general trends found from these measurements are representative of the behavior of the instability.

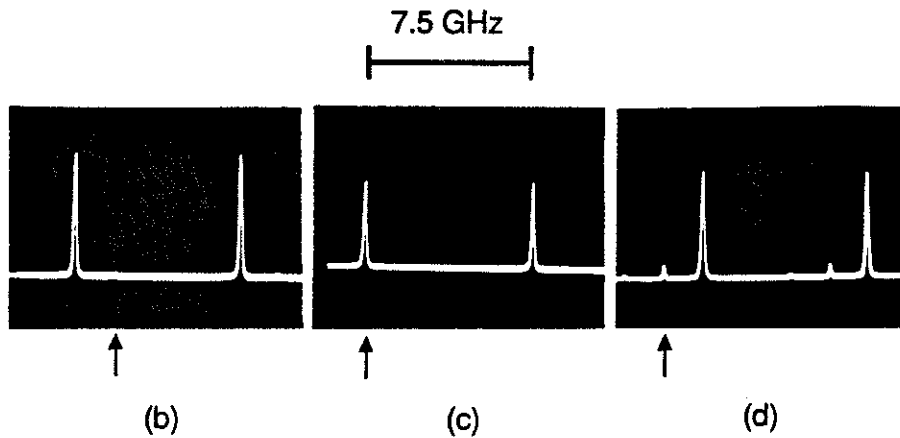
C. Observation of Instabilities using Linear and Parallel Pump Beam Polarizations

C1. Instability Boundary as Determined by Measurements of the Average Power Emitted in the Orthogonal Polarization

The results for linear-and-parallel polarized pump beams are presented in this section. High quality calcite polarizers are used as the polarizing elements and the radiation rejected by the polarizers (i.e. radiation orthogonal to that of the input polarization) of the transmitted beams is monitored. The instability is first characterized by its dependence on laser frequency. Figure 4.5a shows the average power generated in the orthogonal polarization component of the transmitted forward beam for equal input forward and backward beam powers (P_f and P_b , respectively) of 46



(a)



(b)

(c)

(d)

Figure 4.5 (a) Dependence of the instability in the polarization of counterpropagating waves on the laser frequency. The average power of the generated radiation as a function of the laser frequency is shown for a number equal to 1.6×10^{13} atoms cm^{-3} , a buffer gas pressure of 15 mTorr and $P_f = P_b = 46$ mW. The four tick marks on the horizontal axis, labeled 1-4, indicate the positions of the $(F=2) \rightarrow (F=1)$, $(F=2) \rightarrow (F=2)$, $(F=1) \rightarrow (F=1)$, and $(F=1) \rightarrow (F=2)$ hyperfine components of the $3S_{1/2} \rightarrow 3P_{1/2}$ sodium transition, respectively. Spectra of the generated radiation for a laser frequency tuned to the (b) low-frequency, (c) central, and (d) high-frequency feature of the excitation spectrum shown in (a). The arrows indicate the position of the laser frequency.

mW, an atomic number density equal to 1.6×10^{13} atoms cm^{-3} and a buffer gas pressure of ~ 15 mTorr. The tick marks indicate the positions of the four hyperfine components of the $3S_{1/2} \rightarrow 3P_{1/2}$ transition. It is seen that as much as 17% of the incident power can be converted into radiation in the orthogonal polarization direction; therefore, the instability dramatically alters the evolution of the interacting fields. The excitation spectrum is also seen to be very rich, containing spectral features that are much narrower than the ~ 2 GHz Doppler width. It is found that each of these features has a distinct optical spectrum. Figs. 4.5b-d displays the output of the confocal Fabry-Perot for each of the features for a fixed laser frequency.

The central feature (i.e. for a laser detuning of 1.01 GHz to the high-frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition), the generated light is found to be at the same frequency as the input light, as can be seen by Fig. 4.5c. For this value of the laser detuning and forward beam power, the instability threshold corresponds to $P_b \simeq 3$ mW. Slightly above the threshold, the power emitted in the orthogonal polarization is observed to be nearly constant in time, corresponding to the dc instability predicted by the Kerr-medium model.²⁹ For powers well above the instability threshold, weak oscillations in the transmitted intensity are observed with a frequency of oscillation equal to ~ 10 MHz which corresponds to the natural width of the atomic transition.

The emission spectrum of the spectral feature to the low-frequency side of the central feature contains three frequency components: a component at the input laser frequency and two sidebands symmetrically displaced by 1.8 GHz, which is the ground-state hyperfine splitting. As can be seen from Fig. 4.5b, the majority of the power is contained in the low-frequency or Stokes sideband. This observation suggests that for certain laser frequencies, stimulated hyperfine Raman scattering¹⁰⁶ is

the relevant nonlinear coupling mechanism. The spectrum of the strong feature (trapezoidal) to the high-frequency side of the central feature has similar frequency components as the low-frequency feature except that the high-frequency or anti-Stokes sideband contains the majority of the power, as can be seen from Fig. 4.5d. Both the high- and low-frequency features have interesting temporal behavior, as will be shown later in this section.

The spatial distribution of the radiation corresponding to each of the spectral features is unique. For the low powers and low atomic number densities used in the experiment, there is little self-focusing¹⁰⁷ of the incident laser beams. However, it is found that the generated radiation is emitted in a cone surrounding the transmitted laser beam. Figure 4.6a-c shows the radiation pattern of the generated light corresponding to the low-, central-, and high-frequency feature of Fig. 4.5a, respectively. The divergence angle (full angle) of the laser beam is ~ 2 mrad. It is seen that the divergence angle of the generated radiation is as large as ~ 12 mrad. The complicated radiation patterns are reproducible, but are very sensitive to the alignment of the counterpropagating pump beams. It is believed that the radiation is emitted in a cone because of phase-matching effects.¹⁰⁸ Note that the spatial distribution of the radiation generated in the orthogonal polarization component for the central- and high-frequency features is not cylindrically symmetric about the optic-axis. Similar spontaneous symmetry breaking in the spatial structure of electromagnetic waves has been predicted by Lugiato and Lefever;¹⁰⁹ however, the system that they analyzed was comprised of plane waves interacting in a passive nonlinear optical medium surrounded by an optical cavity.

The dependence of the instability threshold on laser power and laser frequency for $P_f = P_b$, for a fixed atomic number density equal to 1.6×10^{13} atoms cm^{-3} and for a buffer gas pressure equal to ~ 15 mTorr is shown in Figs. 4.7. For low

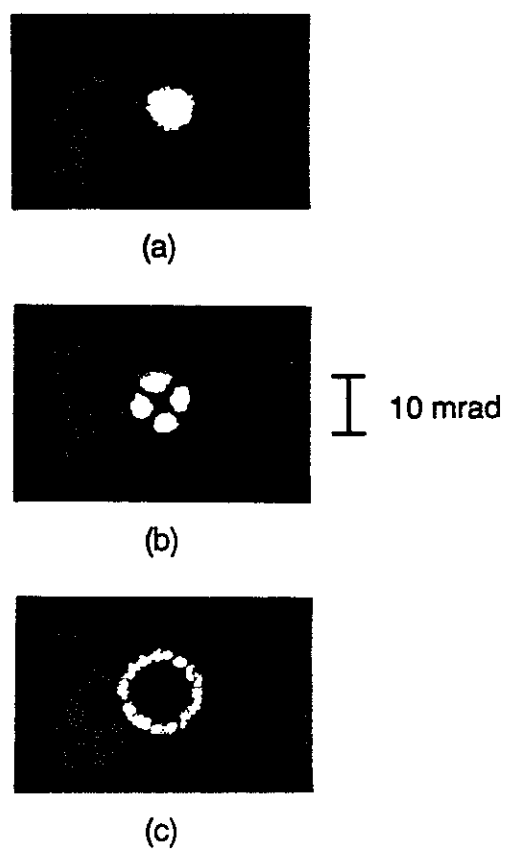


Figure 4.6 Radiation patterns of the light emitted in the orthogonal polarization component for the (a) low-frequency, (b) central, and (c) high-frequency feature of the excitation spectrum shown in Fig. 4.5a.

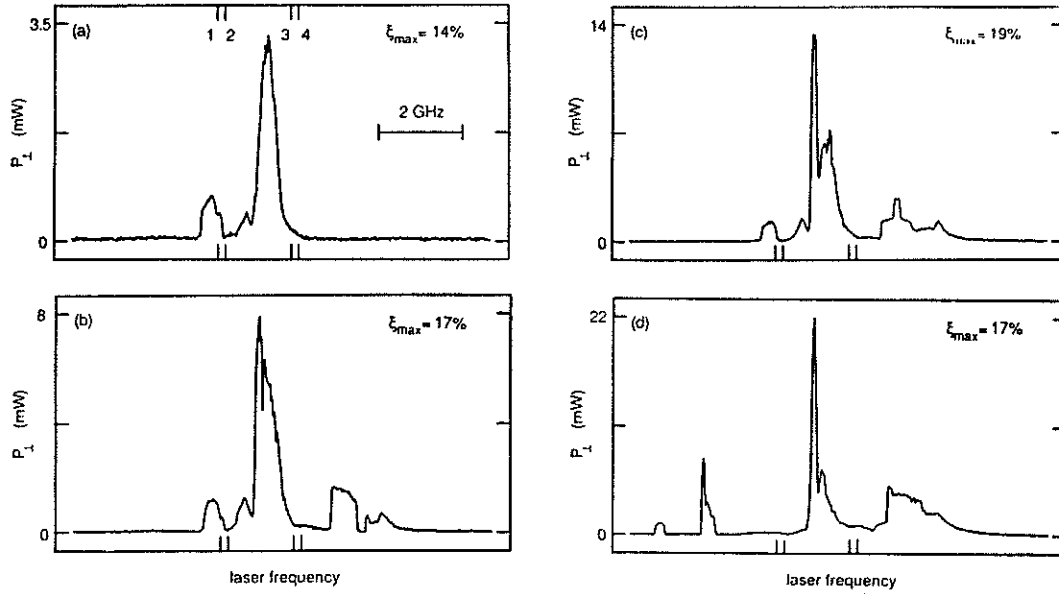


Figure 4.7 Dependence of the instability in the polarization of counterpropagating waves on the laser frequency for various input powers. The average power of the generated radiation as a function of the laser frequency is shown for a number density is 1.6×10^{13} atoms cm^{-3} , a buffer gas pressure of 15 mTorr and $P_f = P_b = P$ where (a) $P = 23$ mW, (b) $P = 47$ mW, (c) $P = 69$ mW, and (d) $P = 131$ mW.

powers only the central feature of Fig 4.5a is present in the excitation spectrum. For a power of 23 mW (Fig. 4.7a) new features appear and, as the power is increased, the features broaden as shown in Fig. 4.7b. The highest conversion efficiency into the orthogonal polarization component $\xi_{\max}$, which is determined by choosing the optimum laser detuning, is obtained for a power of 69 mW and is equal to 19% (Fig. 4.7c). It is also seen in Fig. 4.7c that the central feature is much narrower than at lower powers, and that at a power of 131 mW (Fig. 4.7d), the feature to the low-frequency side of the central feature vanishes due to the effects of strong saturation. Note that the new feature in Fig. 4.7d at the low-frequency side of resonance occurs at ~ 1.8 GHz below the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition.

Qualitatively different behavior for the instability boundary is found when the sum of the input powers is held fixed and the ratio of the input powers is varied. Figures 4.8 shows the average power emitted in the orthogonal polarization of the transmitted forward wave for $P_f + P_b \approx 90$ mW. Note, for comparison, that Fig. 4.8a is the same as Fig. 4.7b. As the ratio of the powers is increased, $\xi_{\max}$ is seen to decrease, as shown in Fig. 4.8b. For greater ratios (Fig. 4.8c), $\xi_{\max}$ is dramatically reduced and all the features, except for the central feature, are completely broadened. Therefore, we see that a single beam can saturate the nonlinearity even though the sum of the power of the two beams is held constant.

The dependence of the instability threshold on the atomic number density is investigated next. Figures 4.9 shows that for $P_f = P_b = 46$ mW the power converted into the orthogonal polarization initially increases with increasing number density, reaches a maximum value at a density of 2.6×10^{13} atoms cm^{-3} (Fig. 4.9b), and then decreases with increasing number density as is shown in Fig. 4.9c for the case of 4×10^{13} atoms cm^{-3} . It is also seen from Figs. 4.9 that the generation efficiency of the central spectral feature decreases with respect to the other features as the atomic

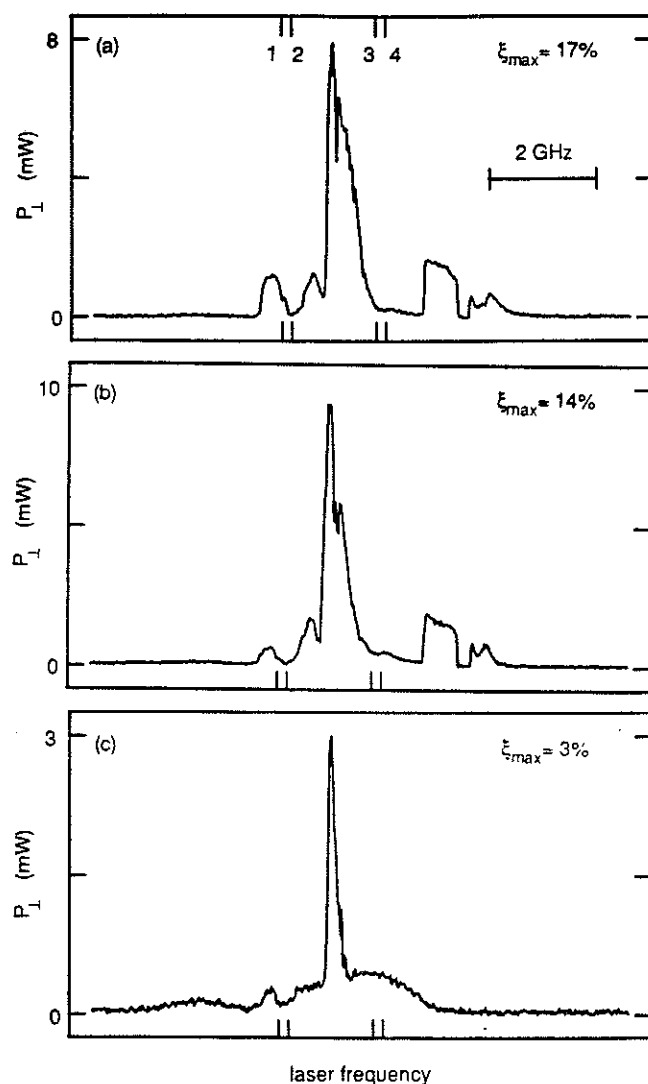


Figure 4.8 Dependence of the instability in the polarization of counterpropagating waves on the laser frequency for various ratios of the forward to backward wave input powers. The average power of the generated radiation as a function of the laser frequency is shown for a number density equal to 1.6×10^{13} atoms cm^{-3} , a buffer gas pressure of 15 mTorr, $P_f + P_b = 90$ mW, and (a) $P_f/P_b = 1$ ($P_f = 46$ mW), (b) $P_f/P_b = 2.8$ ($P_f = 69$ mW), and (c) $P_f/P_b = 5.7$ ($P_f = 92$ mW).

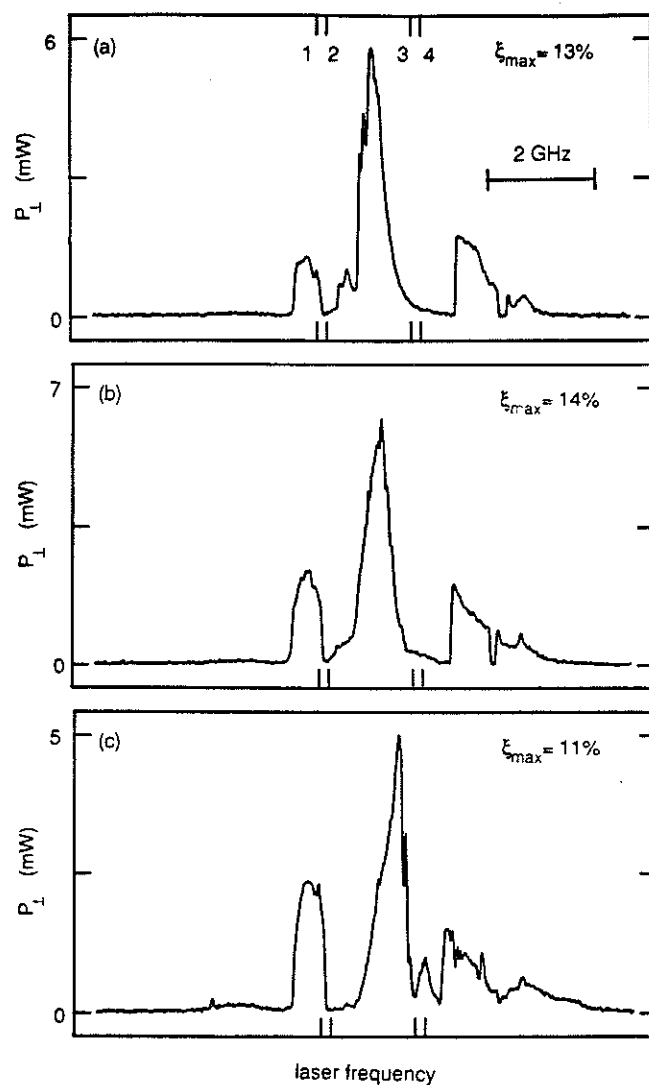


Figure 4.9 Dependence of the instability in the polarization of counterpropagating waves on the laser frequency for various values of the atomic number density. The average power of the generated radiation as a function of the laser frequency is shown for the same power in the two input waves is the same and is equal to 46 mW, a buffer gas pressure of 15 mTorr, and a number density equal to (a) 1.6×10^{13} atoms cm^{-3} , (b) 2.6×10^{13} atoms cm^{-3} , and (c) 4×10^{13} atoms cm^{-3} .

number density is increased and that all of the features are broadened for increasing density. At very high atomic number densities, the state of polarization is unstable only far from resonance. The waves are stable near to the resonance because the waves are absorbed entirely by the medium.

Figures 4.10 shows the dependence of the instability on the buffer gas pressure for an atomic number density of 1.6×10^{13} atoms cm^{-3} , $P_f = 46$ mW, and $P_b = 28$ mW. It is seen that the central-spectral-feature is very sensitive to the effects of collisions and that the highest conversion efficiency into the orthogonal polarization of $\xi_{\text{max}} = 19\%$ is obtained at a pressure of ~ 2 mTorr (Fig. 4.10a). For a buffer gas pressure of 100 mTorr, new features in the excitation spectrum appear (Fig. 4.10b) and broaden considerably at a pressure of 1 Torr (Fig. 4.10c). For pressures greater than 50 Torr, the central feature vanishes and the state of polarization of the field is unstable for laser detunings greater than 20 GHz from the atomic transition frequency.

From these measurement of the boundary for instability, it is found that the maximum efficiency of the light generated in the orthogonal polarization by the instability depends on several factors. The threshold can be lowered by using a low buffer gas pressure, by using input beams with equal power, and by choosing the optimum atomic number density. In the next sections, the buffer gas pressure is kept fixed at a low value of ~ 15 mTorr and the number density is varied to determine the optimum value.

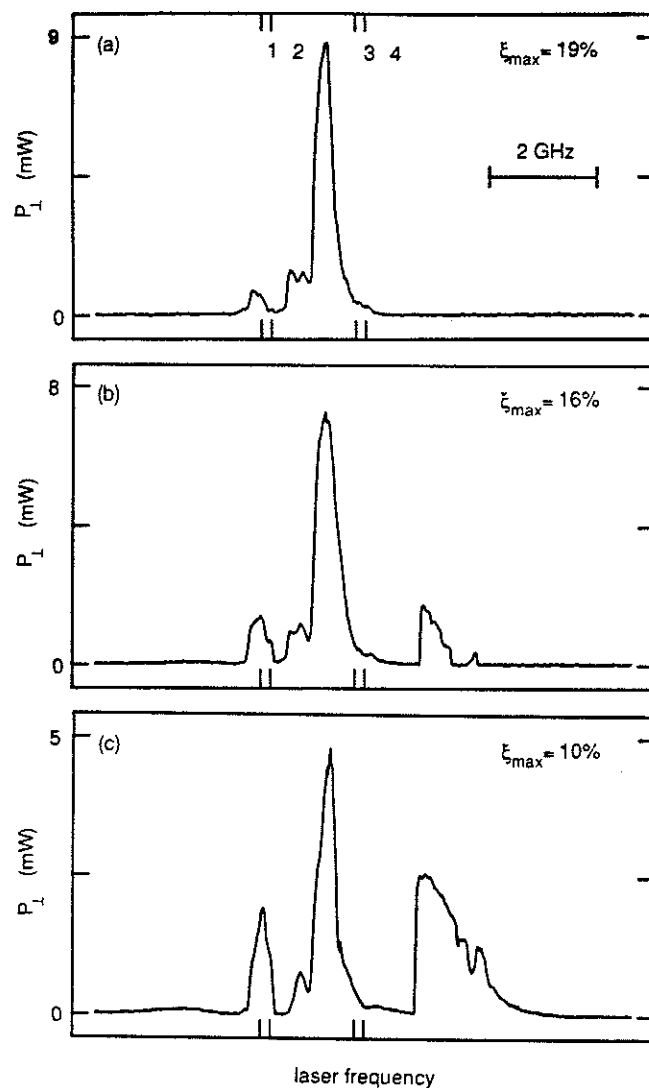


Figure 4.10 Dependence of the instability in the polarization of counterpropagating waves on the laser frequency for various values of the buffer gas pressure. The average power of the generated radiation as a function of the laser frequency is shown for $P_f = P_b = 46$ mW and a number density is equal to 1.6×10^{13} atoms cm^{-3} , and for a buffer gas pressure of (a) 2 mTorr, (b) 100 mTorr, and (c) 1 Torr.

C2. Temporal Evolution of the Light Generated in the Orthogonal Polarization Component

Interesting temporal behavior is observed for the low- and high-frequency features of the excitation spectrum shown in Fig. 4.3a. Figures 4.11a-o shows the temporal evolution of the radiation emitted in the orthogonal polarization in the forward direction when the laser is tuned to the low-frequency feature for various values of the backward-wave power. A time series containing 32,698 points is collected at a sample time of 5 ns and a small portion of these records is displayed in the figures. The number density is $3.2 \times 10^{13} \text{ atoms cm}^{-3}$, $P_f = 39 \text{ mW}$ and the laser is detuned 490 MHz to the low frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition. The threshold for instability corresponds to $P_b = 5.7 \text{ mW}$ for these values of the experimental parameters.

A dramatic change in the evolution is observed at the threshold for instabilities where a Hopf bifurcation occurs; that is, the evolution makes a transition from a fixed-point (no power emitted in the orthogonal polarization) to a limit cycle (periodic self-pulsing). As the backward wave power is increased, the frequency of oscillation increases, and for $P_b = 6.3 \text{ mW}$ (Fig. 4.11a) the oscillation frequency is $\sim 1.2 \text{ MHz}$. The frequency of oscillation increases to $\sim 5.2 \text{ MHz}$ at a power of 7.8 mW , as can be seen from Fig. 4.11b, and then changes to wildly fluctuating oscillations at a power of 8.1 mW (Fig. 4.11c). As the backward wave power is increased, the complexity and the characteristic frequency of oscillation of the fluctuations increases (Fig. 4.11d-g). However, at a backward wave power of 16.6 mW , regions of rather periodic behavior is evident, as can be seen from Fig. 4.11h, suggesting that the complexity of the system is decreasing. Indeed, for $P_b = 19.5 \text{ mW}$ (Fig. 4.11i), the oscillations are periodic with an oscillation frequency of $\sim 12 \text{ MHz}$. For increasing

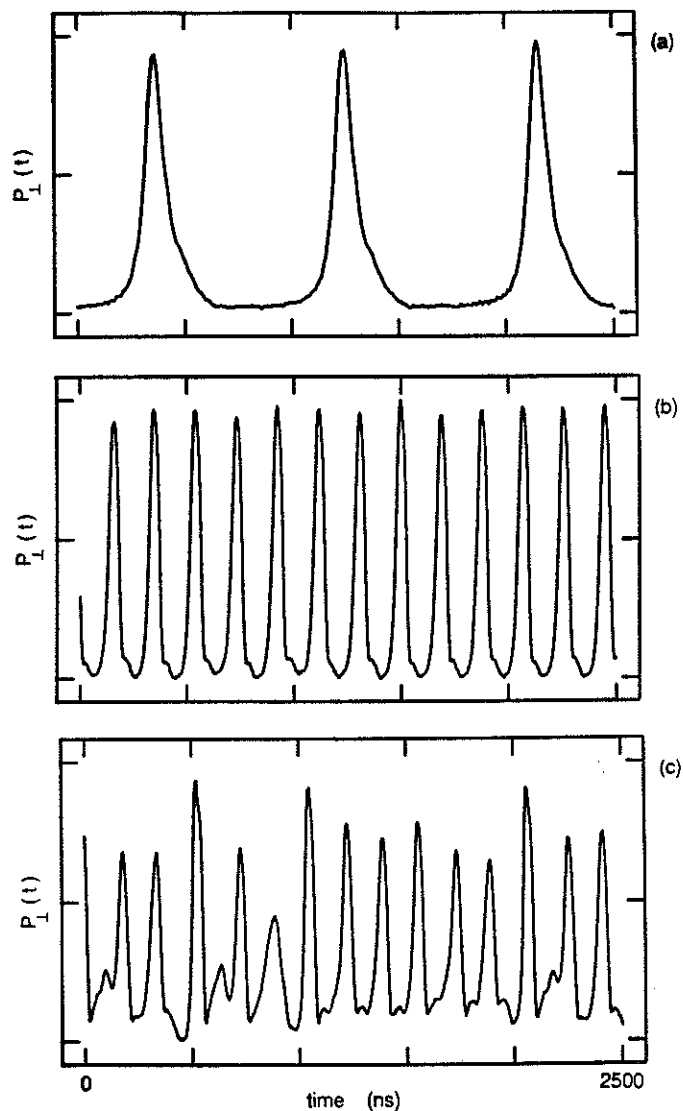


Figure 4.11 Temporal evolution of the light generated in the orthogonal polarization. The laser is tuned 490 MHz to the low-frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition, the buffer gas pressure is 15 mTorr, the number density is 3.2×10^{13} atoms cm^{-3} , the input power of the forward beam is 39 mW, and the backward wave power is (a) 6.3 mW, (b) 7.9 mW, and (c) 8.1 mW.

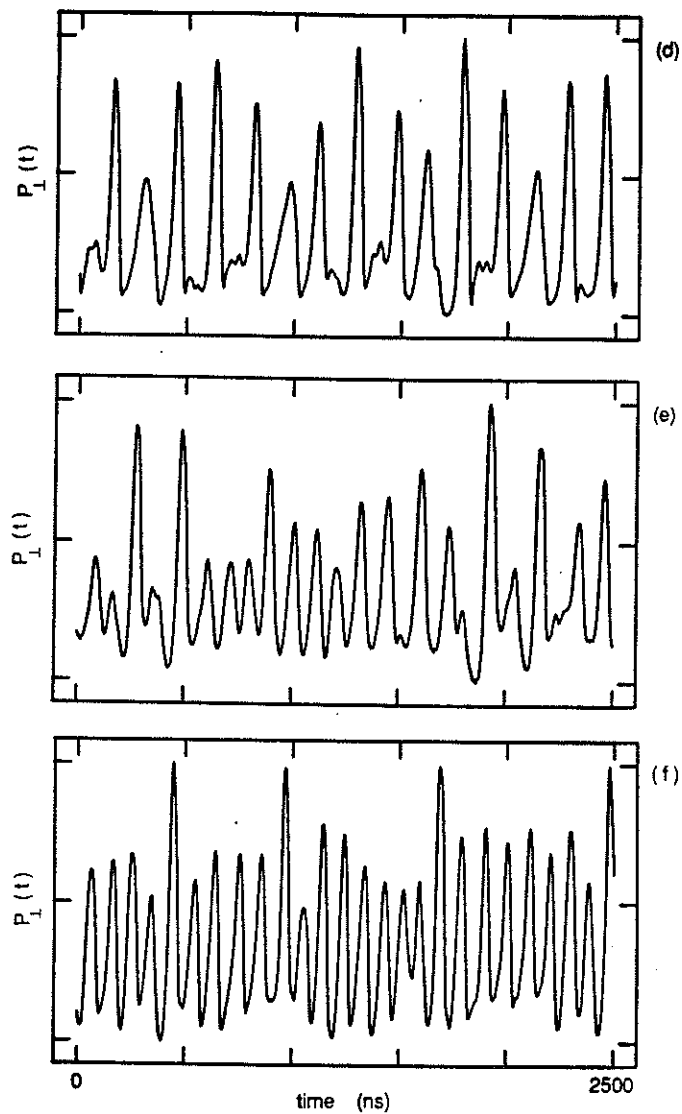


Figure 4.11 Temporal evolution of the light generated in the orthogonal polarization. The laser is tuned 490 MHz to the low-frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition, the buffer gas pressure is 15 mTorr, the number density is $3.2 \times 10^{13} \text{ atoms cm}^{-3}$, the input power of the forward beam is 39 mW, and the backward wave power is (d) 8.8 mW, (e) 9.8 mW, and (f) 12 mW.

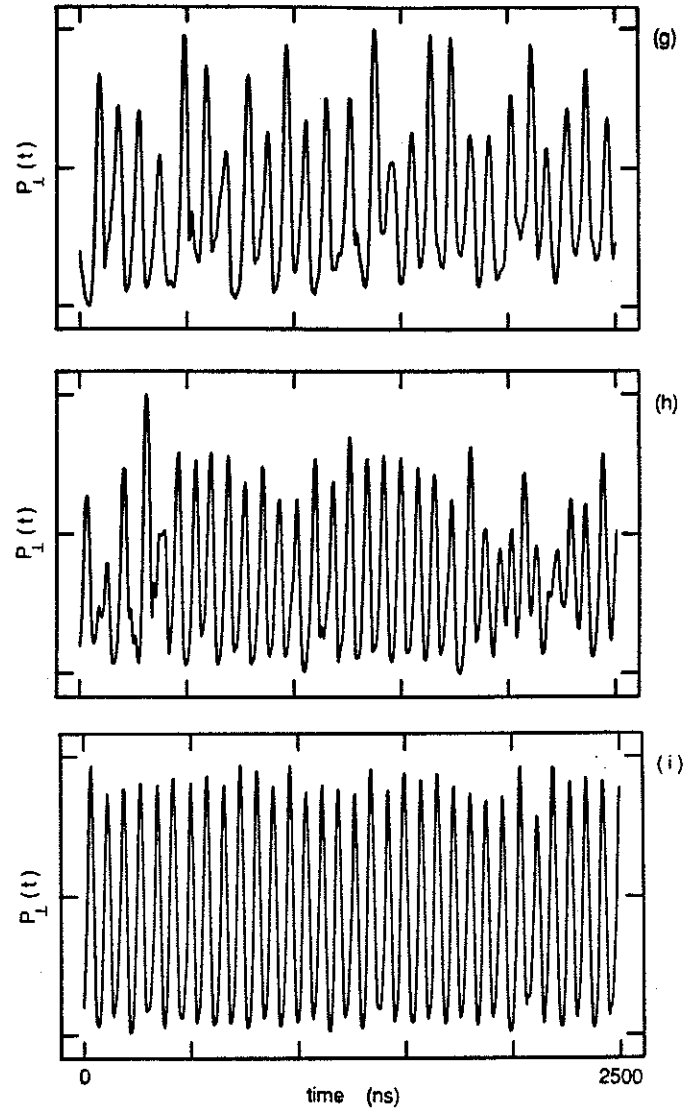


Figure 4.11 Temporal evolution of the light generated in the orthogonal polarization. The laser is tuned 490 MHz to the low-frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition, the buffer gas pressure is 15 mTorr, the number density is 3.2×10^{13} atoms cm^{-3} , the input power of the forward beam is 39 mW, and the backward wave power is (g) 16 mW, (h) 18 mW, and (i) 20 mW.

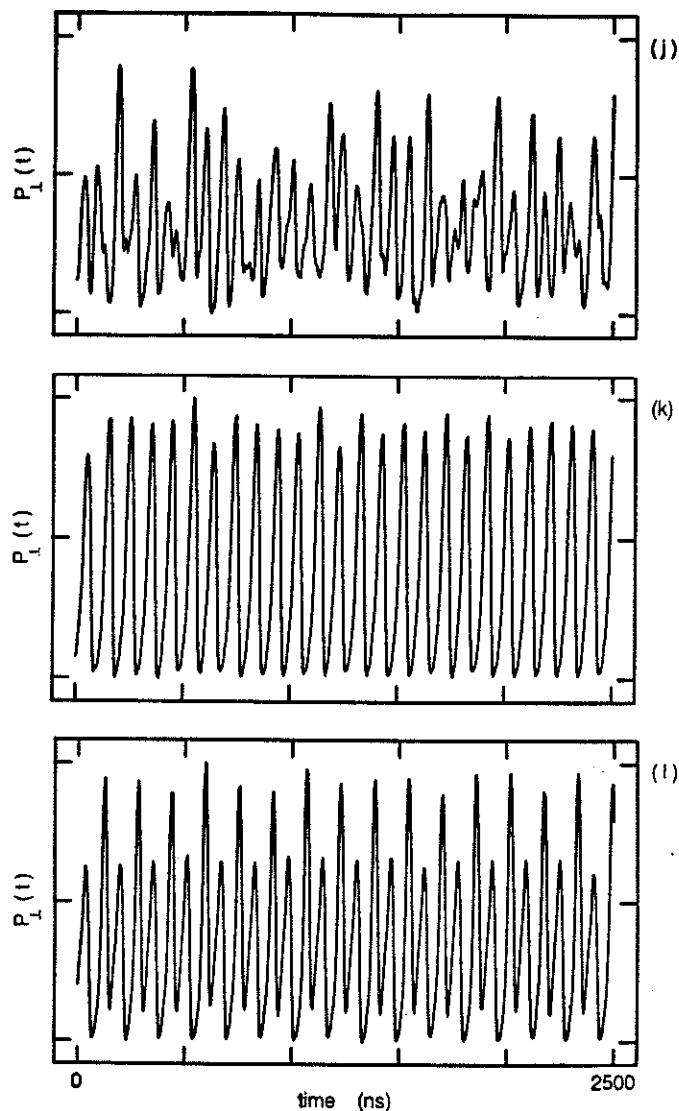


Figure 4.11 Temporal evolution of the light generated in the orthogonal polarization. The laser is tuned 490 MHz to the low-frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition, the buffer gas pressure is 15 mTorr, the number density is 3.2×10^{13} atoms cm^{-3} , the input power of the forward beam is 39 mW, and the backward wave power is (j) 21 mW, (k) 24 mW, and (l) 28 mW.

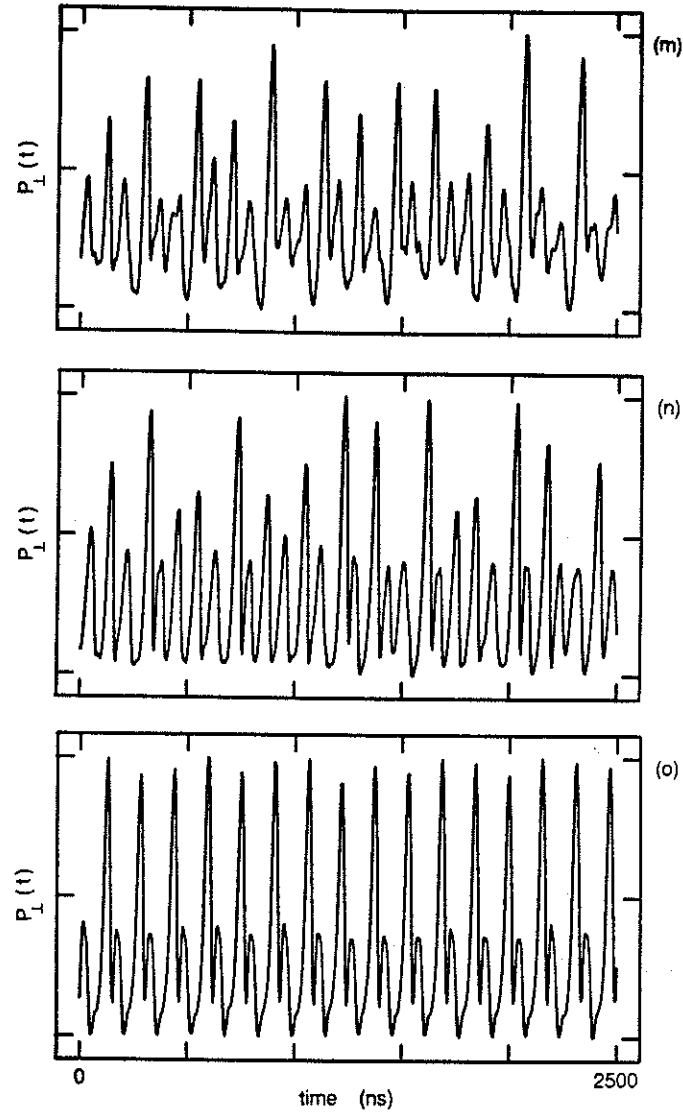


Figure 4.11 Temporal evolution of the light generated in the orthogonal polarization. The laser is tuned 490 MHz to the low-frequency side of the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition, the buffer gas pressure is 15 mTorr, the number density is 3.2×10^{13} atoms cm^{-3} , the input power of the forward beam is 39 mW, and the backward wave power is (m) 30 mW, (n) 31 mW, and (o) 34 mW.

power (Fig. 4.11j), the temporal evolution is again quite complex demonstrating that the periodic behavior at a power of $P_b = 19.5$ mW is a 'periodic window' embedded in a region of chaotic behavior. For $P_b = 24.1$ mW (Fig. 4.11k), the evolution switches back to periodic, then period-doubles for an increase in the power to 27.9 mW, as can be seen from Fig. 4.11l. For even higher powers (Fig. 4.11m-n), the time evolution is again erratic, but the fluctuations are not as complex as that shown for lower powers. At the highest powers used, the evolution is periodic, as shown in Fig. 4.11o.

It was shown in Chapter I that it is useful to study the power spectrum and the autocorrelation function of the resulting fluctuations, and the evolution of the system in a two-dimensional time-delay phase-space in order to characterize the nature of the time behavior. Plots of these quantities are shown in Figs. 4.12-4.15 for several different values of the backward-wave power. The corresponding time series is reproduced in each figure for ease of comparison. For the time series shown in Fig. 4.12a, the power spectrum (Fig. 4.12b) sharply peaks at ~ 5.2 MHz and at several harmonics of this fundamental frequency. The autocorrelation function is near one periodically (Fig. 4.12c), and the phase space trajectory (Fig. 4.12d) forms a closed loop. All of these results indicate that the evolution is periodic.

For an increase in the backward wave power, the time series is complex (Fig. 4.13a), the peaks in the power spectrum are broadened (Fig. 4.13b), the autocorrelation function decays with increasing delay time (Fig. 4.13c) and the phase space trajectory is complex. It is seen that the trajectory in phase space does not form closed loops and does not fill phase space. These results indicate that the temporal evolution shown in Fig. 4.13a is chaotic and that the attractor is a strange attractor. Note that the autocorrelation function looks periodic with an envelope that decays with increasing delay time. In this case, one might expect that there would be a large

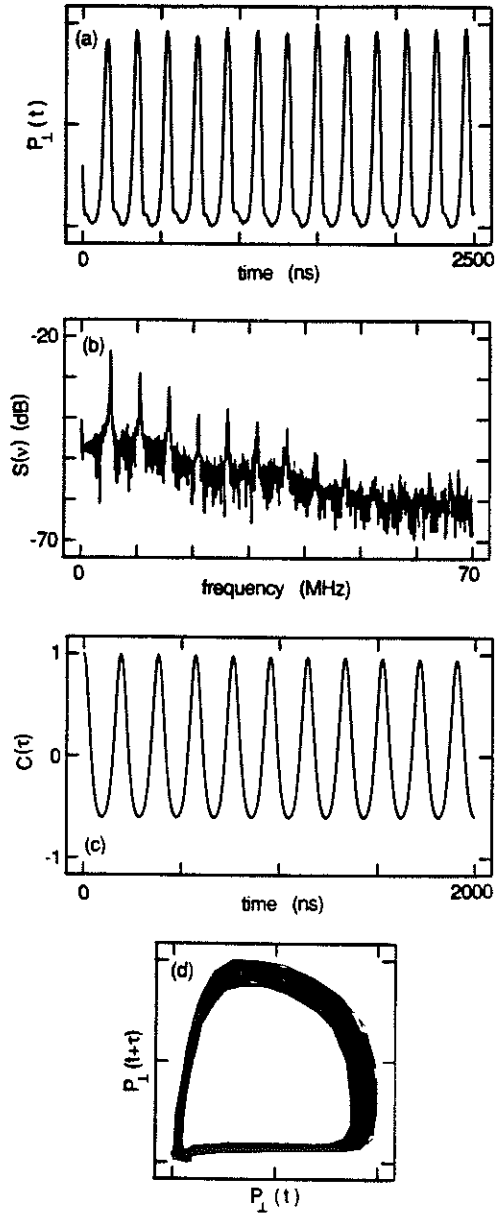


Figure 4.12 Temporal evolution (a) of the light generated by the instability and the associated power spectrum (b), autocorrelation function (c) and attractor for a backward wave power equal to 7.9 mW. The attractor is reconstructed by plotting the power emitted in the orthogonal polarization at time t vs. that emitted at time $t+\tau$ for a delay time equal to 25 ns.

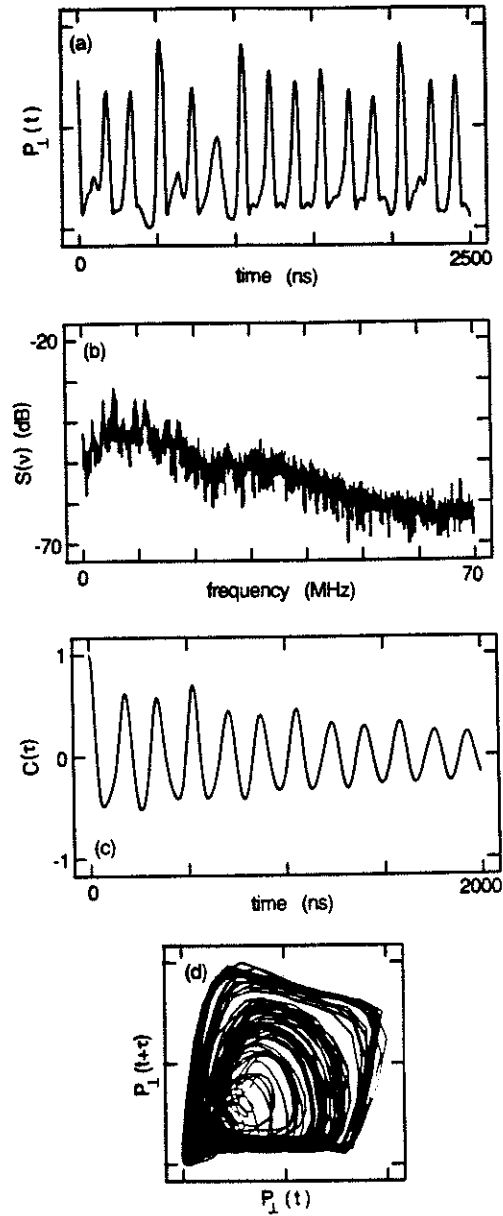


Figure 4.13 Temporal evolution (a) of the light generated by the instability and the associated power spectrum (b), autocorrelation function (c) and strange attractor for a backward wave power equal to 8.1 mW. The attractor is reconstructed by plotting the power emitted in the orthogonal polarization at time t vs. that emitted at time $t+\tau$ for a delay time equal to 20 ns. The correlation dimension D_2 and order-2 entropy K_2 characterizing the attractor are equal to 2.7 and 36 Mbits/sec, respectively.

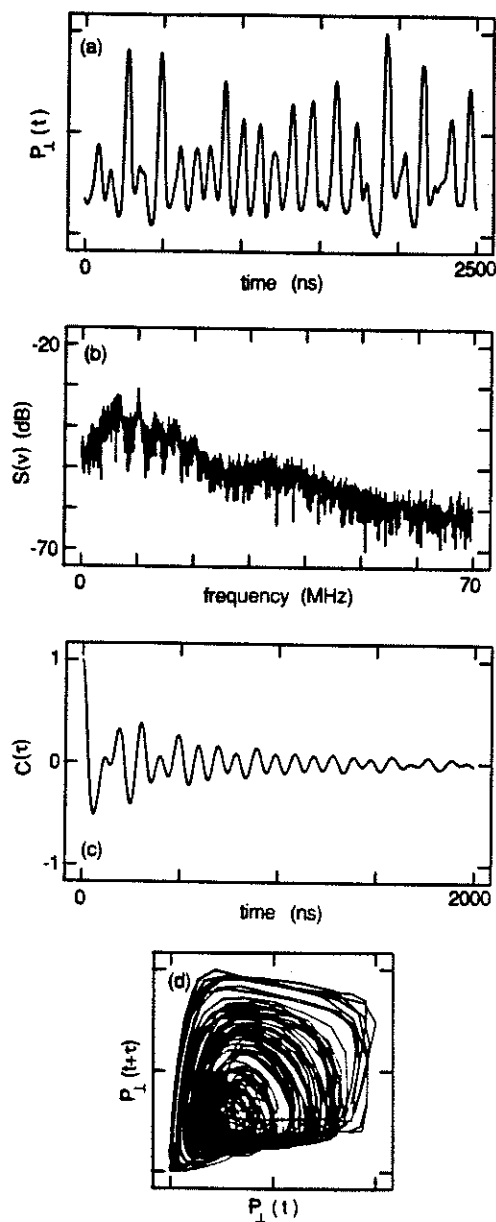


Figure 4.14 Temporal evolution (a) of the light generated by the instability and the associated power spectrum (b), autocorrelation function (c) and strange attractor for a backward wave power equal to 9.8 mW. The attractor is reconstructed by plotting the power emitted in the orthogonal polarization at time t vs. that emitted at time $t+\tau$ for a delay time equal to 20 ns. The correlation dimension D_2 and order-2 entropy K_2 characterizing the attractor are equal to 4.2 and 60 Mbits/sec, respectively.

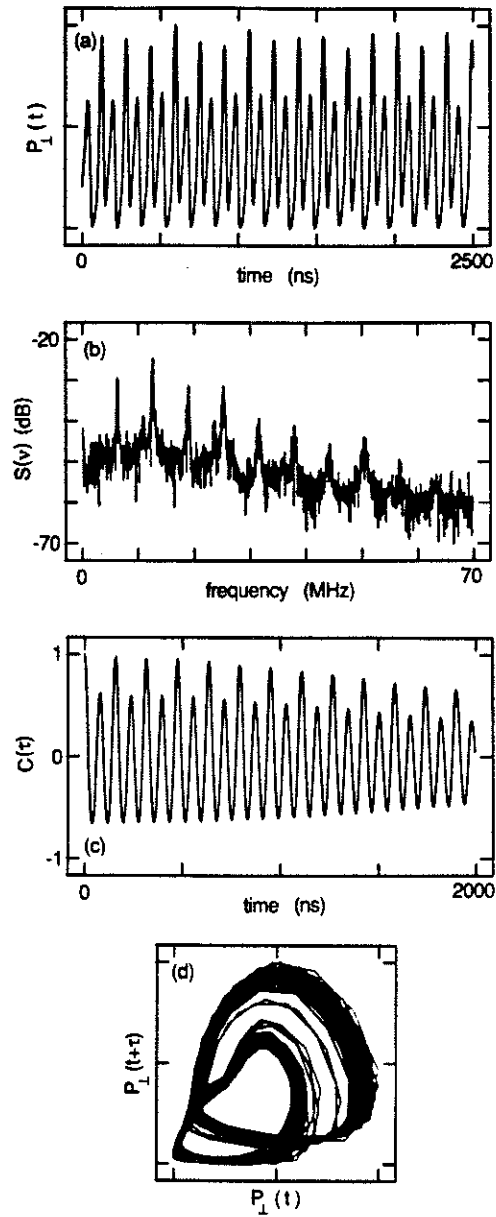


Figure 4.15 Temporal evolution (a) of the light generated by the instability and the associated power spectrum (b), autocorrelation function (c) and periodic attractor for a backward wave power equal to 28 mW. The attractor is reconstructed by plotting the power emitted in the orthogonal polarization at time t vs. that emitted at time $t+\tau$ for a delay time equal to 20 ns.

peak in the power spectrum around 6 MHz. However, a close inspection of the autocorrelation function reveals that the period of each oscillation has a different value, resulting in the observed small, broadened peak in the power spectrum at ~6 MHz. For a slight increase in power, the time series becomes even more complex (Fig. 4.14a), the power spectrum contains less structure (Fig. 4.14b), and the autocorrelation function decays rapidly (Fig. 4.14b). It is seen in Fig. 4.14d that the trajectory fills phase space in a highly nonuniform manner, indicating that it is also a strange attractor. At the highest powers, strong saturation of the atomic transition decreases the nonlinearity and the evolution is periodic, as seen in Fig. 4.15a. It is seen in this case that the power spectrum contains discrete frequency components with the strongest frequency occurring at ~12.5 MHz. The autocorrelation function decays slowly (Fig. 4.15c) and the phase space trajectory forms a closed loop.

To verify these conclusions quantitatively, the correlation dimension D_2 and the order-2 entropy K_2 are calculated using the fixed-volume method of Grassberger and Procaccia²⁴ and the fixed-mass method of Badii and Politi⁶⁹ for the time series shown in Figs. 4.11. The correlation dimension is a lower bound to the fractal dimension and usually is a good approximation to the fractal dimension. The order-2 entropy is a lower bound to Kolmogorov entropy which is the sum of the positive Liapunov exponents. Therefore, a necessary condition for a system to be chaotic is that the order-2 entropy be finite and greater than zero. Figure 4.16 shows D_2 and K_2 as a function of the laser power. It is seen that the dimension (Fig. 4.16a) undergoes a sudden change at the instability threshold (Hopf bifurcation), initially increases with increasing intensity, and then decreases with increasing intensity. It is also seen that very low-dimensional evolution occurs at windows embedded in regions of high-dimensional evolution. The entropy (Fig. 4.16b) is found to be zero below a power of 7.9 mW, indicating that the evolution is on a fixed point or limit cycle. For a

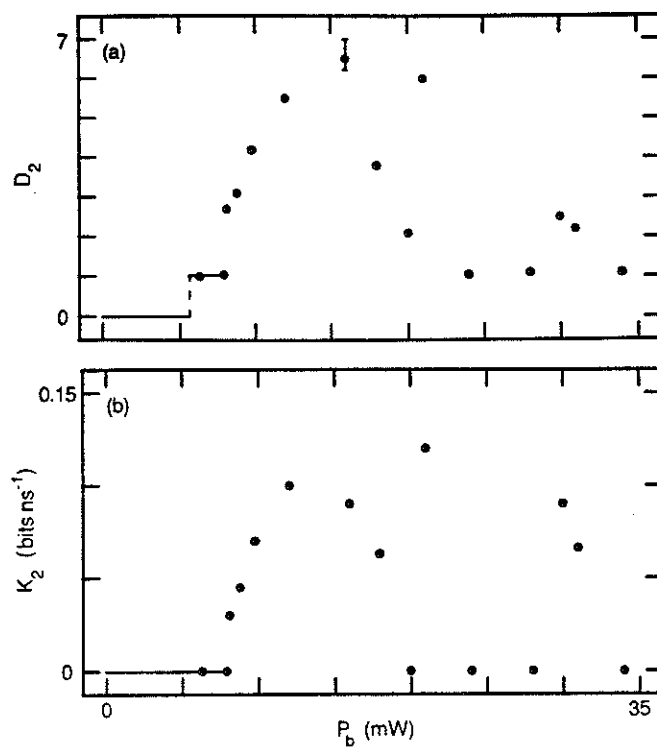


Figure 4.16 Correlation dimensions D_2 (a) and order-2 entropies K_2 (b) of the attractors associated with the time series shown in Figs. 4.11. It is seen that the complexity of the system increases with increasing laser power due to an increase in the nonlinear coupling strength. However, high laser powers saturate the atomic transition and the complexity of the system is seen to decrease.

power of 8.1 mW, the entropy is equal to 36 Mbits/sec showing that the evolution is chaotic and verifying our qualitative assessment of the data based on visual inspection of the attractors. The entropy drops to zero at intermediate laser powers at the positions of the windows of low-dimensional evolution, showing that the evolution is periodic in these cases. The error in the determination of the dimension becomes large for the data points above a dimension of 4 because the scaling region is small due to the limited size of the 32,689 data set and the limited ten-bit resolution of the digitizer. The eventual decrease of the dimension and entropy as the laser power is increased is due to saturation of the nonlinearity of atomic resonances. This point will be investigated in the next chapter.

Very different temporal behavior is observed for the spectral feature to the high-frequency side of the central feature of the excitation spectrum. The temporal behavior of the generated radiation is a constant value in time (except for very high intensities); however, this constant value of the power in the orthogonal polarization is multivalued. To illustrate this observation, Fig. 4.17 shows the power emitted in the orthogonal polarization for the transmitted forward beam as a function of the backward wave power for various atomic number densities. The backward-wave power is slowly ramped from a value of 5 mW to 19mW over 1.7 msec and then back to 5 mW using an acousto-optic modulator (Isomet model 1205C-2). Two fast photodiodes are used to monitor the input backward power and the power of the radiation generated in the orthogonal polarization. The forward-wave power is fixed at 74 mW and the laser frequency is 730 MHz to the high-frequency side of the $3S_{1/2}(F=1) \rightarrow 3P_{1/2}(F=2)$ transition. For an atomic number density below 1.3×10^{13} atoms cm^{-3} , no power is emitted in the orthogonal polarization. For a number density of 1.6×10^{13} atoms cm^{-3} it is seen from Fig. 4.17a that the power in the orthogonal polarization jumps abruptly from zero to ~ 0.6 mW at $P_b \approx 9$ mW. The amount of

generated radiation slowly increases for a backward wave power above this value. As the power in the backward wave is decreased, it is found that the behavior is hysteretic in that the value of the generated radiation is high even for $P_b < 9$ mW. As the nonlinear coupling is increased by increasing the number density (Fig. 4.17b), it is seen that a new stable states exist for high input power. At a density of 2.6×10^{13} atoms cm^{-3} , the upper stable state becomes temporally unstable, as seen in Fig. 4.17c. No significant change in the behavior of these states is found as the ramp time is increased to just a few tenths of a second or decreased to times of ~ 0.5 μsec . In fact, it is found that the state of the system could be forced to another stable state just by blocking the backward wave with a card and suddenly unblocking it and that the system remains in this state until another large perturbation is applied to the system.

Figure 4.18 shows similar data for a detuning of 1330 MHz to the high-frequency side of the $3S_{1/2}(F=1) \rightarrow 3P_{1/2}(F=2)$ transition. The results for a density of 2.2×10^{13} atoms cm^{-3} (Fig. 4.18a) are very similar to the results shown in Fig. 4.17a, except that a much higher value of the backward wave power is required for the system to switch to the 'high' state. However, rather different behavior is observed at the higher atomic number densities. It is seen that the output can be tristable for a density of 2.6×10^{13} atoms cm^{-3} and at high powers, the system is temporally unstable. At the highest densities, it is seen in Fig. 4.18c that there are several stable states of the system.

In this section, we have found that the dynamical evolution of the state of polarization of laser beams counterpropagating in sodium vapor is very rich. It is found that the evolution can be steady, periodic or chaotic. These conclusions have been based on visual inspections of phase space trajectories and quantitative measurements of the power spectrum, autocorrelation function, the correlation dimension and the order-2 entropy.

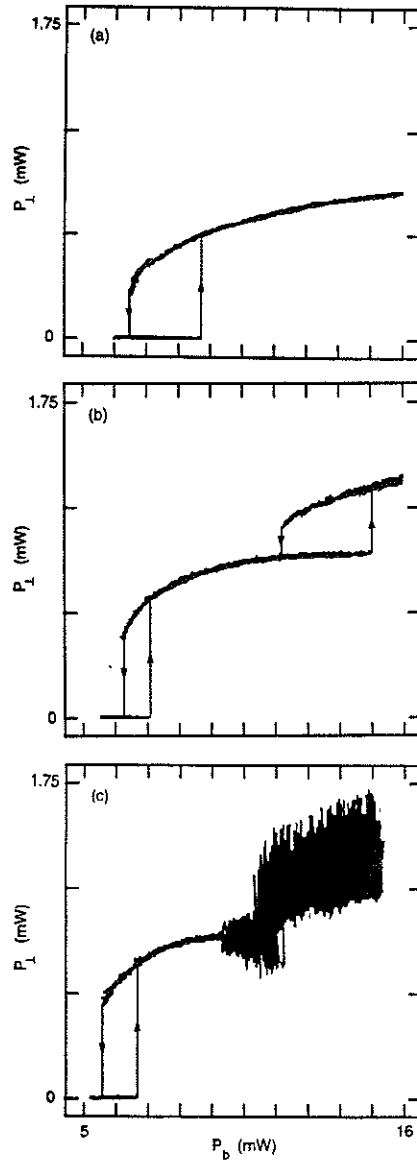


Figure 4.17 Power emitted in the forward direction in the polarization component orthogonal to that of the input waves as a function of the backward wave power. The power of the forward wave is fixed at 74 mW, the buffer gas pressure is 15 mTorr, the laser is detuned 730 MHz to the high-frequency side of the $3S_{1/2}(F=1) \rightarrow 3P_{1/2}(F=2)$ transition, for a density equal to (a) 1.6×10^{13} atoms cm^{-3} , (b) 1.9×10^{13} atoms cm^{-3} , and (c) 2.6×10^{13} atoms cm^{-3} .

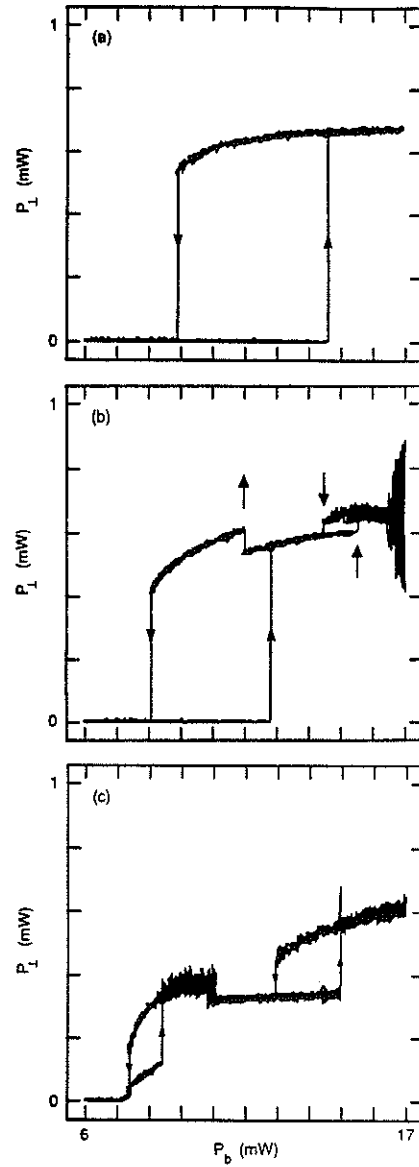


Figure 4.18 Power emitted in the forward direction in the polarization component orthogonal to that of the input waves as a function of the backward wave power. The power of the forward wave is fixed at 74 mW, the buffer gas pressure is 15 mTorr, the laser is detuned 1330 MHz to the high-frequency side of the $3S_{1/2}(F=1) \rightarrow 3P_{1/2}(F=2)$ transition, for a density equal to (a) 1.6×10^{13} atoms cm^{-3} , (b) 2.6×10^{13} atoms cm^{-3} , and (c) 3.2×10^{13} atoms cm^{-3} .

D. Results for other Pump-Wave Polarizations

It is found that the results for different pump-wave polarizations are very similar to the results presented in the previous section. Therefore, only the dependence of the instability boundary on the laser frequency and the laser power is investigated as determined by measurements of the average power emitted in the orthogonal polarization. All data presented below is for a number density of $3.2 \times 10^{13} \text{ atoms cm}^{-3}$, a buffer gas pressure of $\sim 15 \text{ mTorr}$, and $P_f = P_b = 64 \text{ mW}$.

Figure 4.19a shows the dependence of the generated radiation on the laser frequency for the case of linear-and-parallel pump-wave polarizations which is the same polarization states used in the previous section. Figure 4.19b shows the power emitted in the orthogonal polarization for the case of linear-and-orthogonal pump-wave polarizations. This case is of importance because, for a scalar medium, no instabilities can exist for orthogonally polarized pump waves. It is seen that the central feature which is present in Fig. 4.19a is greatly reduced and that there is a sharp dip near to the $3S_{1/2}(F=2) \rightarrow 3P_{1/2}(F=2)$ transition. By placing quarter-wave plates in the incident laser beams, the dependence of the instability boundary for circular and counterrotating pump beam polarizations can be investigated. It is seen from Fig. 4.19c that this case is very similar to the case of linear and parallel polarizations except that the low- and high-frequency features are generated with higher efficiency with respect to the central feature. Note that in all cases, the maximum efficiency is approximately the same. Therefore, these results suggest that the tensor properties of the nonlinearity are such that any choice of pump wave polarization will lead to dynamical instabilities.

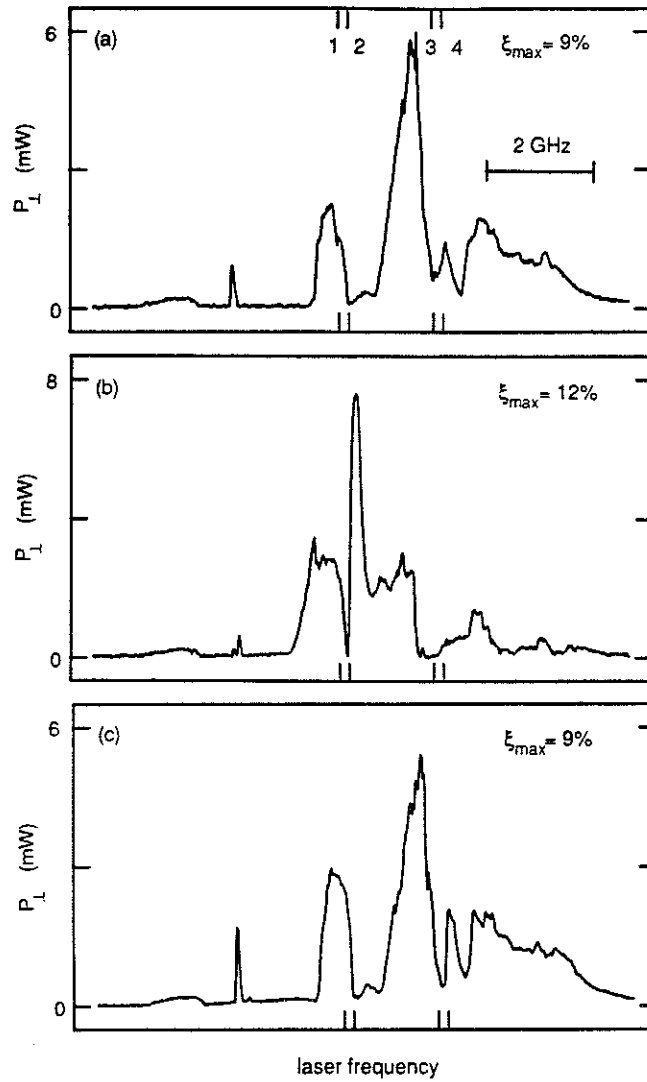


Figure 4.19 Dependence of the instability in the polarization of counterpropagating waves on the laser frequency for various states of polarization of the input waves. The average power of the generated radiation as a function of the laser frequency is shown for $P_f = P_b = 64$ mW, a number density equal to 3.2×10^{13} atoms cm^{-3} , and a buffer gas pressure of 15 mTorr. The states of polarization are: (a) linear-and-parallel polarizations, (b) linear-and-orthogonal polarizations, and (c) circular and counterrotating polarizations.

E. Phase Conjugation by Degenerate Four-Wave Mixing

To determine the effects of instabilities on degenerate four-wave mixing, a third beam derived from the output of the laser is input to the sodium heat-pipe oven in addition to the two pump waves. In this study, the beams all have same linear state of polarization. The total phase conjugate reflectivity and the reflectivity into the orthogonal polarization component is measured for various backward-wave powers as a function of the laser detuning from the atomic resonance. The probe wave is nearly copropagating with the forward wave with an incident angle greater than the cone angle of the radiation generated by the instability. The atomic number density is 1.0×10^{13} atoms cm^{-3} , the buffer gas pressure is ~ 15 mTorr, $P_f = 12$ mW, and the probe wave power is 6.4 mW. Figure 4.20a shows the total power in the phase conjugate wave as a function of the laser frequency for $P_b = 19$ mW. The phase conjugate reflectivity is only appreciable in the wings of the atomic transition and all the radiation is emitted in the same linear polarization. In this case, the polarization state of the pump waves is stable. When the backward wave power equal to 27 mW a sharp feature appears near to line center as seen in Fig. 4.20b. However, Fig. 4.20c shows that the polarization of the radiation corresponding to this new feature is orthogonal to that of all the input waves. In this case, it is found that the pump wave are sufficiently intense to be above the threshold for instability. Therefore, it is seen that the instabilities in the pump waves dramatically modifies the behavior of the four-wave mixing process. In some sense, instabilities in the pump waves are advantageous in that high phase-conjugate reflectivity can be obtained over spectral regions for which the reflectivity is low below the threshold for instability. However, in some instances, it is required that the phase-conjugate wave be of a particular polarization state which might not be possible when the pump waves are unstable.

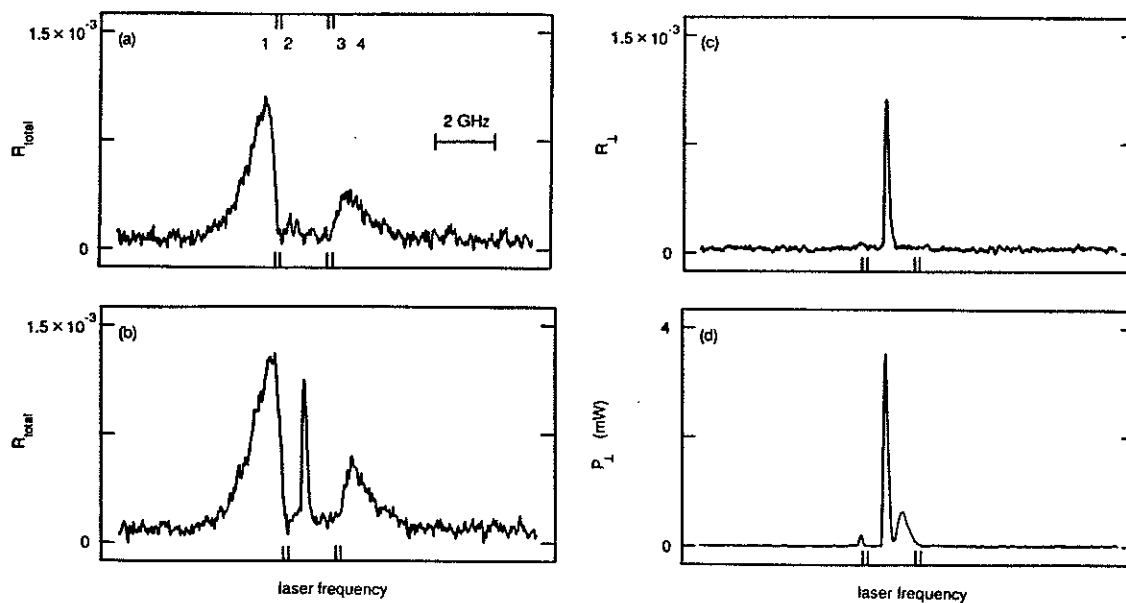


Figure 4.20 Phase conjugation by degenerate four-wave mixing for the case when all input waves have the same linear polarization. (a) Total phase-conjugate reflectivity as a function of the laser frequency for a backward wave intensity below the threshold for instability. The phase-conjugate signal is the same linear polarization as the input waves. At a backward wave power above the threshold for instability, the total reflectivity (b) contains a sharp peak at a frequency between the hyperfine components of the transition. (c) Phase-conjugate reflectivity into the polarization direction orthogonal to that of the input polarizations for the same backward wave power as in (b). Power of the radiation in the orthogonal polarization (d) of the transmitted forward wave. Instabilities in the pump wave polarization affect the polarization of the phase conjugate wave.

CHAPTER V

THEORY OF LASER BEAMS PROPAGATING THROUGH A SATURABLE ABSORBER

A. Introduction

In the previous chapter, an experiment is described that shows that laser beams counterpropagating through a sodium vapor are dynamically unstable. Many of the experimental results are in good qualitative agreement with the Kerr-medium model; however, it is found that absorption, saturation and optical pumping affect the nature of the instability. A simple two-level model will be analyzed in this chapter to investigate the effects of absorption and saturation on the threshold for instabilities of laser beams propagating through sodium vapor.

Mollow¹¹⁰ predicted that a strong, near-resonant optical radiation field dramatically modifies the spontaneous emission spectrum of an atom. Soon thereafter, Schuda *et al.*¹¹¹ confirmed these results by showing that the spectrum of the light scattered from sodium atoms interacting with a resonant laser field contains new frequencies peaked at the Rabi frequency. Harter *et al.*¹¹² have shown that the spontaneous spectrum is modified by the effects of propagation and that intense sidebands detuned symmetrically around the pump frequency by the Rabi frequency can be generated when a laser beam propagates through a dense atomic vapor. When the effects of damping are important and when the effects of spontaneous emission can be ignored, it has been found⁸⁹ that the semi-classical, density-matrix formalism is useful to predict the experimental observations for the case of laser beams propagating in a dense media.

It will be assumed that the atomic response of an atomic vapor is governed by the density-matrix equations of motion, which are taken to be of the form

$$\begin{aligned} \frac{d\rho_{ll}}{dt} = & -2 \sum_{l>i} \gamma_{li} \rho_{ll} + 2 \sum_{i<l} \gamma_{il} \rho_{ii} \\ & + \frac{i}{\hbar} \sum_i (\vec{\mu}_{li} \rho_{il} - \rho_{li} \vec{\mu}_{il}) \cdot \mathbf{E}(t) \end{aligned} \quad (5.1)$$

and

$$\frac{d\rho_{lm}}{dt} = - (i\omega_{lm} + \Gamma_{lm}) \rho_{lm} + \frac{i}{\hbar} \sum_i (\vec{\mu}_{li} \rho_{im} - \rho_{li} \vec{\mu}_{im}) \cdot \mathbf{E}(t), \quad (l \neq m) \quad (5.2)$$

where the sum on $l>i$ ($i>l$) is to be taken over those states with energy $W_l > W_i$ ($W_l < W_i$) and $\vec{\mu}_{li}$ is the li matrix-element of the dipole moment. The ll element of the density-matrix is the probability that the state l is excited. The lm density-matrix element is the coherence between the states l and m . It will be assumed that population is conserved in the system and that the field $\mathbf{E}(t)$ is a collection of plane-waves. In some cases, the density-matrix elements will be expanded in a perturbation series given by

$$\rho_{lm} = \rho_{lm}^{(0)} + \rho_{lm}^{(1)} + \rho_{lm}^{(2)} + \dots \quad (5.3)$$

in order to obtain the third-order contributions to the nonlinear polarization so that comparisons can be made to the Kerr-medium model. The polarization is obtained from the density-matrix and the dipole moment by the relationship

$$\mathbf{P} = N \text{Tr}(\vec{\mu} \rho), \quad (5.4)$$

where N is the atomic number density. The resulting expression is then used as the source term in the wave equation.

B. Two-Level Atomic Transition

A two-level atomic system is studied because it is a simple system and because, in many cases, it can adequately model the sodium D_1 transition. Figure 5.1 shows the atomic energy levels consisting of the ground-state $|a\rangle$ and the excited-state $|b\rangle$ separated by an energy $\hbar\omega_{ba}$. It is assumed that all field frequencies are near to the transition frequency so that the rotating wave approximation is a valid approximation. This interaction is scalar; hence the vector notation can be dropped and the dipole moment is assumed to be given by $\mu_{aa} = \mu_{bb} = 0$ and $\mu_{ba} = \mu_{ab}^*$. For a field of the form $E = \tilde{E}\exp(-i\omega t) + \text{c.c.}$, the density matrix equations of motion for this system are, in the rotating wave approximation, given by

$$\frac{d\sigma}{dt} = -(i\Delta + \Gamma)\sigma - i\frac{w}{2}\tilde{\Omega} \quad (5.5)$$

and

$$\frac{dw}{dt} = -2\gamma(w+1) - i(\tilde{\Omega}^* \sigma - \sigma^* \tilde{\Omega}) \quad (5.6)$$

where the slowly varying coherence is defined by $\sigma = \rho_{ba}\exp(i\omega t)$, the Rabi frequency is $\tilde{\Omega} = (2\mu_{ba}\tilde{E})/\hbar$, the atomic inversion is $w = (\rho_{bb} - \rho_{aa})$, and the atom-field detuning is $\Delta = (\omega_{ba} - \omega)$. The coherence decays at a rate $\Gamma = 1/T_2$, where T_2 is the transverse relaxation time, and the population decays from level b to level a at a rate $2\gamma = 1/T_1$, where T_1 is the longitudinal relaxation time.¹¹³ It is assumed that the atomic density is low enough so that local field effects can be ignored (i.e., there is only a density-dependent increase in the transverse relaxation rate due to sudden shifts in the phase

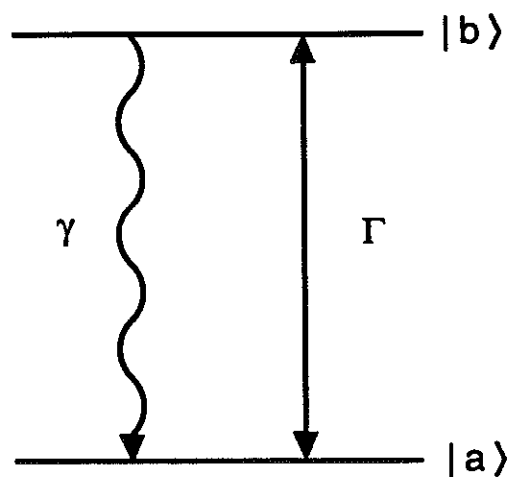


Figure 5.1 Energy-level diagram of the two-level atomic system. Population decays from state b to state a at a rate γ and the coherence connecting the two states decays at a rate Γ .

of the atomic coherence and there is no density-dependent shift in the atomic resonance frequency).

B1. Perturbation Theory

The stability of a wave $\tilde{E}_1 \exp(-i\omega_1 t)$ (pump wave) propagating through a collection of stationary two-level atoms will be investigated by applying a perturbing wave (probe wave) of the form $\tilde{E}_2 \exp(-i\omega_2 t)$. This stability analysis is similar to section B1 in Chapter II which was for the case of a Kerr medium. The density-matrix equations are solved in steady-state to lowest-order in the perturbing field and to third-order in the pump field. It is assumed that the unperturbed density matrix elements are $\sigma^{(0)} = 0$ and $w^{(0)} = -1$. For this case, the only nonzero, first-order correction to the density matrix is given by

$$\sigma^{(1)} = \frac{1}{2} \left[\frac{\tilde{\Omega}_1}{\Delta - i\Gamma} + \frac{\tilde{\Omega}_2 e^{i\delta t}}{\Delta + \delta - i\Gamma} \right] \quad (5.7)$$

where $\tilde{\Omega}_j = \tilde{E}_j$ and the pump-probe detuning is $\delta = (\omega_1 - \omega_2)$.

The lowest-order contribution to the polarization is given in terms of the first-order coherence by the relationship $P^{(1)} = N\mu_{ab}\sigma^{(1)}$. The linear susceptibility is then found from Eq. (2.1) and is given by

$$\chi^{(1)} = \frac{N|\mu_{ba}|^2}{\hbar(\Delta - i\Gamma)} \quad (5.8)$$

which is related to the dielectric constant ϵ by the relationship

$$\epsilon = 1 + 4\pi\chi^{(1)}. \quad (5.9)$$

The linear refractive index is equal to the square root of the dielectric constant and is given by

$$n' = 1 + 2\pi \operatorname{Re}(\chi^{(1)}) = 1 + \frac{2\pi N |\mu_{ba}|^2 \Delta}{\hbar(\Delta^2 + \Gamma^2)} \quad (5.10)$$

and

$$n'' = 2\pi \operatorname{Im}(\chi^{(1)}) = \frac{2\pi N |\mu_{ba}|^2 \Gamma}{\hbar(\Delta^2 + \Gamma^2)}, \quad (5.11)$$

where $n = n' + i n''$. The imaginary part of the refractive index is a Lorentzian with a width of Γ/π and is related to the ^{5 p} ~~absorption~~ coefficient by $\alpha = 2n''\omega/c$. The line-center absorption coefficient is then

$$\alpha_0 = \frac{4\pi N |\mu_{ba}|^2 \omega}{\hbar c \Gamma} \quad (5.12)$$

and the off-resonance absorption coefficient is

$$\alpha(\Delta) = \alpha_0 \frac{\Gamma^2}{\Delta^2 + \Gamma^2} = \alpha_0 \frac{1}{\Delta^2 \Gamma^2 + 1}. \quad (5.13)$$

The real part of the refractive index is equal to one at $\Delta = 0$ and is maximum (minimum) at $\Delta = -\Gamma$ ($\Delta = \Gamma$).

The second-order correction to the density-matrix is given by

$$w^{(2)} = \frac{1}{2} \left[\frac{\Gamma |\tilde{\Omega}_1|^2}{\gamma(\Delta^2 + \Gamma^2)} + \frac{(\delta - i2\Gamma) \tilde{\Omega}_1^* \tilde{\Omega}_2 e^{i\delta t}}{(\delta - i2\gamma)(\Delta + i\Gamma)(\Delta + \delta - i\Gamma)} + \frac{(\delta + i2\Gamma) \tilde{\Omega}_1^* \tilde{\Omega}_2 e^{-i\delta t}}{(\delta + i2\gamma)(\Delta - i\Gamma)(\Delta + \delta + i\Gamma)} \right]. \quad (5.14)$$

It is seen that the population oscillates¹¹⁴ at the beat-note of the applied field. The intensity of the applied field is given by $I = |\tilde{E}|^2$ and can be expressed as $I = I_0[1 + I \cos(\delta t)]$, where it has been assumed that the slowly varying field amplitudes are real. The inversion can also be expressed in a similar manner as $w^{(2)} = w_0^{(2)}[1 + w \cos(\delta t + \phi)]$. The phase-angle ϕ is the amount that the population oscillations lag or lead the oscillations in the intensity and is given by

$$\phi = -\tan^{-1} \left[\frac{2\delta(\Gamma - \gamma)}{\delta^2 + 4\Gamma\gamma} \right] \quad (\Delta \gg \delta, \Gamma) \quad (5.15)$$

in the limit that the pump detuning is large in comparison to the atomic linewidth and the pump-probe detuning. For the case of natural broadening, Γ is equal to γ , and it is seen from Eq. (5.15) that the population oscillations are always in phase with the intensity oscillations. For the case of large collisional broadening, the phase-angle is

$$\phi = -\tan^{-1} \left[\frac{\delta}{2\gamma} \right]. \quad (\Gamma \gg \gamma) \quad (5.16)$$

When the pump-probe detuning is equal to 2γ , the population oscillations lead the intensity oscillations by $\pi/4$. Population oscillations are the source of the third-order optical nonlinearity, as will be shown below.

The third-order correction to the density-matrix is given by

$$\begin{aligned}
\sigma^{(3)} = & -\frac{1}{4} \left[\frac{\Gamma |\tilde{\Omega}_1|^2 \tilde{\Omega}_1}{\gamma(\Delta^2 + \Gamma^2)(\Delta - i\Gamma)} \right. \\
& + \frac{\Gamma |\tilde{\Omega}_1|^2 \tilde{\Omega}_2 e^{i\delta t}}{\gamma(\Delta^2 + \Gamma^2)(\Delta + \delta - i\Gamma)} + \frac{(\delta - i2\Gamma) |\tilde{\Omega}_1|^2 \tilde{\Omega}_2 e^{i\delta t}}{(\delta - i2\gamma)(\Delta + \delta - i\Gamma)^2(\Delta - i\Gamma)} \\
& \left. + \frac{(\delta + i\Gamma) \tilde{\Omega}_1^2 \tilde{\Omega}_2^* e^{-i\delta t}}{(\delta + i2\gamma)(\Delta - \delta - i\Gamma)(\Delta - i\Gamma)(\Delta - \delta + i\Gamma)} \right]. \quad (5.17)
\end{aligned}$$

The first term in Eq. (5.17) is the change in the refractive index at frequency ω due to the pump wave and is responsible for self-focusing.¹¹⁵ The second term is the change in the refractive index at frequency $\omega - \delta$ due to the pump wave and is responsible for cross-phase modulation.¹¹⁶ The third term can be traced to the population oscillations (see the second term of Eq. (5.14)) and gives rise to coupling between the pump and probe when the pump is tuned off-resonance and gives rise to the homogeneous hole¹¹⁷ when the pump is tuned to exact resonance. The fourth term in Eq. (5.17) gives rise to a response at the other side band at frequency $\omega + \delta$ and depends on the relative phase of the fields and is a parametric four-wave mixing term.

The absorptive response of the atomic system at the probe wave frequency is given by $2\text{Im}\{[\sigma^{(1)}(\delta) + \sigma^{(3)}(\delta)]/(\tilde{\Omega}_2 T_2)\}$ and is equal to

$$\begin{aligned}
2\text{Im}\left\{\frac{\sigma^{(1)}(\delta) + \sigma^{(3)}(\delta)}{\tilde{\Omega}_2 T_2}\right\} = & \text{Im}\left\{\frac{(\Delta + \delta) T_2 + i}{(\Delta + \delta)^2 T_2^2 + 1} [1 - |\mathcal{E}_1|^2] \right. \\
& \left. - \frac{1}{2} \frac{T_2}{T_1} \frac{(\delta T_2 - i2)(\Delta T_2 + i)}{[\delta T_2 - i(T_2/T_1)] [(\Delta + \delta) T_2 - i]} |\mathcal{E}_1|^2 \right\}. \quad (5.18)
\end{aligned}$$

The pump field has been normalized by the square-root of the off-resonance saturation intensity so that $E_1' = E_1/I_{\text{sat}}(\Delta)$, where the off-resonance saturation intensity is defined as

$$I_{\text{sat}}(\Delta) = \frac{\hbar^2 (\Delta^2 T_2^2 + 1)}{4 |\mu_{ba}|^2 T_1 T_2}. \quad (5.19)$$

For the perturbation expansion of the density-matrix (Eq. (5.3)) to be valid, the normalized intensity $|E_1'|^2$ must be much less than one. In this case, the sum of first two terms in Eq. (5.18) is greater than zero and real. Therefore, these terms contribute to absorption of the probe wave. However, the last term can give rise to gain for the proper choice of the pump-probe detuning and for large collisional broadening. This contribution to the absorptive response is a pressure induced resonance¹¹⁸ in that it only exists for a collisionally broadened vapor ($\Gamma \neq \gamma$) and, as will be shown below, gives rise to an absorptive response reminiscent of that of stimulated Rayleigh-wing scattering. It is also seen that the gain can always overcome the loss for sufficiently large detunings because the linear absorption (the first term in Eq. (5.18)) scales as $(\Delta T_2)^{-2}$ whereas the gain scales as $(\Delta T_2)^{-1}$. The minimum value for the threshold for the pump wave to be unstable in the presence of the probe wave is for a pump-probe detuning of $\delta = -(\Delta/|\Delta|)(1/T_1)$ and for large collisional dephasing. Under these conditions, the pump wave is unstable when the inequality

$$|E_1'|^2 |\Delta| T_2 > 2 \quad (|\Delta| T_2 \gg 1) \quad (5.20)$$

is fulfilled.

The spatial evolution of the two wave is determined by substituting Eqs. (5.17) and (5.7) into Eq. (5.4) to obtain the polarization which is then substituted into the wave equation (Eq. (2.2)). Assuming that the pump wave is detuned far from resonance so that absorption can be ignored and that the fields are given by $\tilde{\Omega}_1 = \Omega_1 \exp[k_1 n'(\Delta)z]$ and $\tilde{\Omega}_2 = \Omega_2 \exp[k_2 n'(\Delta+\delta)z]$, the coupled amplitude equations are given by

$$\frac{dE_1'}{dz} = -i \frac{\alpha(\Delta)}{2n'(\Delta)} \Delta T_2 |E_1'|^2 E_1' \quad (5.21)$$

and

$$\begin{aligned} \frac{dE_2'}{dz} = & -i \frac{\alpha(\Delta+\delta)}{2n'(\Delta+\delta)} (\Delta+\delta) T_2 \left[1 + \right. \\ & \left. \frac{1}{2} \frac{2+i\delta T_1(T_2/T_1)}{1+i\delta T_1} \frac{\Delta T_2}{(\Delta+\delta) T_2} \right] |E_1'|^2 E_2', \end{aligned} \quad (5.22)$$

where the linear refractive index is given by Eq. (5.11) and the fields have been normalized by the square-root of the off-resonance saturation intensity. It is seen that there is no exponential gain or loss experienced by the probe wave if: (1) $T_2/T_1 = 2$ (the case of natural broadening), or (2) $\delta = 0$. Therefore, as can be seen from Eq. (5.15), there is only coupling between the pump and probe waves when there is a phase difference between the intensity oscillations and the population oscillations. Equation (5.22) has been written in a suggestive manner for comparison to Eq. (3.5) which describes the spatial evolution of a probe wave in the presence of a pump wave propagating through a Kerr medium. In the limit of large collisional dephasing ($T_2/T_1 \rightarrow 0$) and for $|\Delta| \gg |\delta|$, it is seen that Eqs. (5.22) and (3.5) are identical if the following relationships are made:

$$-\frac{\alpha(\Delta+\delta)}{2n'(\Delta+\delta)}(\Delta+\delta)T_2 = \frac{2\pi\omega_2}{n_2c}A \quad \text{and} \quad T_1 = \tau. \quad (5.23)$$

The Kerr-medium-model material constant B is equal to zero because the interaction of light with a two-level atomic system is a scalar interaction. The relationships (5.23) and the limits required to obtain them are important in that they specify those conditions under which a two-level atom mimics a Kerr medium and gives the material constant A and the response time τ .

Following the approach used in section B1 of Chapter II, the amplitude stability of the field $\tilde{E}_1 \exp(-i\omega t)$ in the presence of a perturbing field of the form $\tilde{E}_2 \exp[-i(\omega-\delta)t] + \tilde{E}_3 \exp[-i(\omega+\delta)t]$ is next investigated. It is assumed that the slowly-varying field amplitudes are given by $\tilde{E}_j = E_j \exp[ik_j n'(\Delta_j)z]$ ($\Delta_1 = \Delta$, $\Delta_2 = \Delta + \delta$, and $\Delta_3 = \Delta - \delta$), that the pump frequency is far from the atomic resonance and that there is large collisional dephasing. The nonlinear polarization is derived by solving Eqs. (5.5) and (5.6) to third-order in the the pump field and to lowest order in the probe fields and substituting the resulting coherences into Eq. (5.4). Coupled amplitude equations that govern the spatial evolution of the probe waves are then found by substituting the nonlinear polarization into the wave equation and are given by

$$\begin{aligned} \frac{dE_2'}{dz} = & -i \frac{\alpha(\Delta+\delta)}{2n'(\Delta+\delta)} (\Delta+\delta)T_2 \left[\left(1 + \frac{1}{1+i\delta T_1}\right) |E_1'|^2 E_2' \right. \\ & \left. + \frac{E_1'^2 E_3'^* e^{i\Delta kz}}{1+i\delta T_1} \right] \end{aligned} \quad (5.24)$$

and

$$\begin{aligned} \frac{dE_3'^*}{dz} = & i \frac{\alpha(\Delta - \delta)}{2n'(\Delta - \delta)} (\Delta - \delta) T_2 \left[\left(1 + \frac{1}{1+i\delta T_1}\right) |E_1'|^2 E_3'^* \right. \\ & \left. + \frac{(E_1')^*{}^2 E_2 e^{i\Delta k z}}{1+i\delta T_1} \right], \end{aligned} \quad (5.25)$$

where $\Delta k = 2k_1 n'(\Delta) - k_2 n'(\Delta + \delta) - k_3 n'(\Delta - \delta)$ and the fields have been normalized by the square-root of the saturation intensity. These equations for forward four-wave mixing in a two-level atomic medium are different than the third-order equations derived by Boyd *et al.*¹¹⁷ (Eq. 28) in that the $1/T_1$ resonance was ignored in Ref. 117. This is also the case for the third-order susceptibilities derived by Nilsen and Yariv¹¹⁹ for four-wave mixing with two counterpropagating pump waves. However, the main results of Ref. 117, which are valid to all orders of the pump field, properly take into account the $1/T_1$ resonance.

By inspection of Eqs. (5.24) and (5.25) and Eqs. (3.7) and (3.8), it is seen that the two set of equations are identical using relations (5.23). Therefore, all results obtained for the case of a Kerr medium in Chapter III are directly applicable to the present case. In particular, complete gain suppression occurs (recall Fig 3.3) when there is perfect phase-matching ($\Delta k = 0$). Therefore, the pump wave is stable in the presence of the two probe fields for $\Delta k = 0$ and unstable for all other values of the phase-mismatch.

A similar analysis can be performed for the case of laser beams counterpropagating through a collection of two-level atoms as was performed for the case laser beams propagating through a Kerr medium. Referring to the results derived in Chapter III, which assumed that the two pump waves have equal intensity, the lowest threshold for infinite phase conjugate reflectivity occurs for

$$\frac{\alpha(\Delta)L}{2n'(\Delta)} |\Delta|T_2 |E'|^2 = \frac{\pi}{4}, \quad (\delta=0) \quad (5.26)$$

where $|E'|^2$ is the intensity of the pump wave and L is the length of the medium. The lowest threshold for dynamical instabilities of the type predicted by Silberberg and Bar-Joseph²⁴ for the case of a Kerr medium occurs for

$$\frac{\alpha(\Delta)L}{2n'(\Delta)} |\Delta|T_2 |E'|^2 = 2. \quad \left(\frac{c}{4L} = \frac{1}{2\pi T_1}\right) \quad (5.27)$$

It will be shown in the next section how saturation and absorption affect the stability characteristics of the waves.

B2. The interaction of strong laser beams with a two-level atomic system

Boyd *et al.*¹¹⁷ showed that for the case of waves nearly copropagating in a collection of two-level atoms, strong resonances occur in the gain experienced by weak sidebands in the presence of a strong wave for pump-probe detunings equal to the generalized Rabi frequency and equal to $\sim 1/T_1$. Gruneisen *et al.*⁹⁷ used the resonance at $\sim 1/T_1$ to couple the energy from one beam of light into another beam. The Rabi resonance was not used in the experiment of Gruneisen *et al.* because the Rabi resonance depends sensitively on the intensity of the strong wave and on the detuning of the pump frequency from the atomic resonance. Hence pump depletion, absorption and the Doppler effect dramatically reduce the effective gain of the Rabi resonance. Gaeta *et al.*¹²⁰ treated the case of degenerate four-wave mixing with two strong counterpropagating pump waves and found that pump-propagation effects significantly alter the predictions for the phase-conjugate reflectivity.

The effects of saturation can be understood by solving Eqs. (5.5) and (5.6) in steady-state to all orders of a single applied field $\tilde{E}_1$. The coherence and inversion is found to be

$$\sigma_{ss} = -\frac{\Delta T_2 + i}{2} \left[\frac{T_2/T_1}{\Delta^2 T_2^2 + 1} \right]^{1/2} w_{ss} \tilde{E}_1 \quad (5.28)$$

and

$$w_{ss} = -\frac{1}{1 + |\tilde{E}_1|^2}, \quad (5.29)$$

respectively. It is seen that for $|\tilde{E}_1|$ much greater than one, the inversion and hence the coherence tend to zero. Therefore, the nonlinearity increases with increasing field strength for $|\tilde{E}_1| \ll 1$ and then decreases for $|\tilde{E}_1| > 1$.

The steady-state response of the medium to an additional weak field $\tilde{E}_2 \exp(i\delta t)$ can be determined by making the ansatz

$$\sigma = \sigma_{ss} + \sigma(\delta)e^{i\delta t} + \sigma(-\delta)e^{-i\delta t} \quad (5.30)$$

and

$$w = w_{ss} + w(\delta)e^{i\delta t} + w(-\delta)e^{-i\delta t}$$

where $w(\delta) = w(-\delta)^*$ as required for the inversion to be a real quantity. The perturbed solutions are determined by substituting Eq. (30) into Eq. (5.5) and (5.6) and are given by

$$\sigma(\delta) = -\frac{1}{2D} \left[(\delta - i2\gamma)(\Delta - \delta + i\Gamma) + \frac{1}{2} \frac{|\tilde{\Omega}_1|^2 \delta}{\Delta + i\Gamma} \right] w_{ss} \tilde{\Omega}_2, \quad (5.31)$$

$$\sigma(-\delta) = \frac{(\delta + i2\Gamma)}{4D^* (\Delta - i\Gamma)} w_{ss} \tilde{\Omega}_1^2 \tilde{\Omega}_2^*, \quad (5.32)$$

and

$$w(\delta) = - \frac{(\delta - i2\Gamma) (\Delta - \delta + i\Gamma)}{2D (\Delta + i\Gamma)} w_{ss} \tilde{\Omega}_1^* \tilde{\Omega}_2, \quad (5.33)$$

where

$$D = (\delta - i2\gamma)(\Delta + \delta - i\Gamma)(\Delta - \delta + i\Gamma) + |\tilde{\Omega}_1|^2 (\delta - i\Gamma). \quad (5.34)$$

The factor D in the denominator gives rise to a resonance in the absorptive response which is related to the Rabi frequency of the strong pump wave.

It was shown in the previous section that there is only coupling from the pump wave to the probe wave when there is a phase difference between the intensity oscillations and the population oscillations. For the case of a saturating pump wave, the phase difference is given by

$$\begin{aligned} \phi &= \tan^{-1} \left[\frac{\text{Im}[w(\delta)]}{\text{Re}[w(\delta)]} \right] \\ &= - \tan^{-1} \left[\frac{2\delta(\Gamma - \gamma) + \delta\Gamma |E_1'|^2}{(\delta^2 + 4\Gamma\gamma) + (\delta^2 + 2\Gamma^2) |E_1'|^2} \right], \end{aligned} \quad (5.35)$$

where the field has been normalized by the square-root of the saturation intensity and where it has been assumed that $|\Delta|T_2 \gg 1, \delta$ and that the fields are real. It is seen that even for the case of natural broadening, the population oscillations are out-of-phase with the intensity oscillations for $\delta \neq 0$. These results show that the simulated Rayleigh-wing resonance will appear in the absorptive response for probe beam for the case of a saturating pump wave in the limit of no collisional dephasing.

The absorptive response at the probe wave frequency is $2\text{Im}[\sigma(\delta)/(\tilde{\Omega}_2 T_2)]$ and is shown in Figs. 5.2 for $|\tilde{E}_1'|^2 = 5$ and two different values of the dipole dephasing rate.

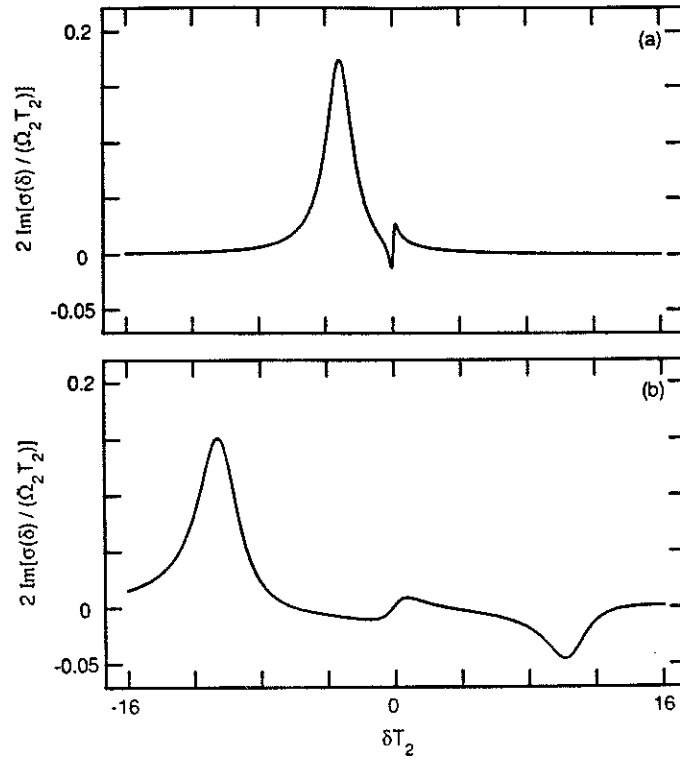


Figure 5.2 Absorptive response of a two-level atom interacting with a strong wave as a function of the pump-probe detuning. The pump frequency ΔT_2 is equal to 3 and the normalized pump wave intensity $|E_1'|^2$ is equal to 5. (a) The case of large collisional broadening ($T_2/T_1 = 0.02$) and (b) the case of natural broadening ($T_2/T_1 = 2$).

Figure 5.2a shows the absorptive response for the case of large collisional dephasing. It is seen that the peak absorption occurs at the atomic resonance and that the absorptive response for small pump-probe detunings is dispersive shaped and peaks near $1/T_1$. It is seen that for the case of natural broadening (Fig. 5.2b), the peak of the absorption is shifted significantly from the atomic resonance frequency and is at a pump-probe detuning equal to the generalized Rabi frequency $\tilde{\Omega}_g = (\Delta^2 + \tilde{\Omega}_1^2)^{1/2}$. For a probe detuning symmetrically displaced about the pump frequency by $\tilde{\Omega}_g$, a gain feature is evident and this feature is the Rabi gain feature. It is also seen that the Rayleigh feature is present for the case of natural broadening. Therefore, the Rayleigh feature exists even for the case of no collisions at high field strengths. These results demonstrate that the pump wave is unstable to the growth of a new field for a broad range of frequencies.

B3. Stability of Saturating Counterpropagating Waves in a Two-Level Atomic medium

In section A.1 of this chapter, it is shown that counterpropagating laser beams are unstable to the growth of temporal fluctuations when the frequency of the waves is far from the atomic resonance so that absorption effects can be ignored. Bar-Joseph and Silberberg⁸⁶ have studied the stability of counterpropagating waves for the case of saturating pump waves; however, they also assume that the effects of absorption can be ignored. In their analysis, it is found that for the case of natural broadening, the pump wave are unstable for sufficiently high intensities and that the frequency of oscillation is at the generalized Rabi frequency for a medium of a length given by $L \sim \pi c / (2\Omega_g)$. It was also shown that for the case of large collisional dephasing, the pump waves are unstable to the growth of oscillations at a frequency of

$\sim 1/T_1$ for a medium of a length given by $L \sim \pi c T_1/2$. The instability boundary is only determined and no attempt was made to investigate the behavior of the wave in the unstable regime.

It is difficult to perform the stability analysis when the intensities of the pump waves is spatially varying (as is the case when the waves experience absorption). Therefore, I have performed a numerical analysis of the stability characteristics of laser beams counterpropagating through a two-level atomic system by integrating the wave equation and the material equations in space and time. The temporal behavior of the waves above the threshold for instabilities can also be investigate through use of this numerical technique.

It is shown in the previous sections that instabilities most readily occur when the frequency of the waves is tuned far from the atomic resonance. In this case it is difficult to numerically integrate Eq. (5.5) because a very small integration step size must be used in order to follow the rapid phase variations due to the dispersive term in the equation. Therefore, I have assumed that there is a large amount of collisional dephasing so that the coherence σ decays rapidly (the rate equation limit). This limit is performed formally by setting the time derivative in Eq. (5.5) equal to zero. It believed that the effects of saturation and absorption on the threshold for instabilities is not overly sensitive to the amount of collisional dephasing. As will be seen below, this statement is supported by the fact that the results of the numerical simulations are very similar to the general trends in the experimental data, even though there was little collisional dephasing in the experiment.

The stability of an electric field of the form

$$\tilde{E} = E_f(z,t)e^{ikz} + E_b(z,t)e^{-ikz} \quad (5.36)$$

is investigated by numerically integrating the wave equation (Eq. (2.2)) and the material equations (Eqs. (5.5) and (5.6)) using the method of characteristics. The formulation of the problem is complicated because of the standing wave interference pattern of the saturating pump waves. For two strong pump waves, the atomic response contains all spatial Fourier harmonics of the fundamental spatial frequency. This effect can be taken into account by the *ansatz*¹²¹

$$\sigma = \left[\sum_{n=-\infty}^{\infty} \sigma_n e^{inkz} \right] e^{ikz} \quad (5.37)$$

and

$$w = \sum_{n=-\infty}^{\infty} w_n e^{inkz}, \quad (5.38)$$

where $w_n = w_{-n}^*$. Equations (5.37) and (5.38) are substituted into the material equations (Eqs. (5.5) and (5.6)) and into the wave equation (Eq. (2.2)), and terms with similar spatial Fourier dependence are equated.

The equations of motion governing the evolution of the field and the inversion are then given by

$$\begin{aligned} T_1 \frac{dw_n}{dt} = & -(w_n + \delta_{n,0})w_n - \left[w_n (|E_f'|^2 + |E_b'|^2) \right. \\ & \left. + w_{n-1} E_f' E_b'^* + w_{n+1} E_f'^* E_b' \right], \quad (n=-\infty, \dots, \infty) \end{aligned} \quad (5.39)$$

$$\frac{\partial E_f'}{\partial z} + \frac{1}{c} \frac{\partial E_b'}{\partial z} = -i \frac{\alpha(\Delta)}{2} (\Delta T_2 + i) [w_0 E_f' + w_1 E_b'], \quad (5.40)$$

and

$$- \frac{\partial E_b'}{\partial z} + \frac{1}{c} \frac{\partial E_b'}{\partial z} = -i \frac{\alpha(\Delta)}{2} (\Delta T_2 + i) [w_1^* E_f' + w_0 E_b'], \quad (5.41)$$

where the fields have been normalized to the square-root of the off-resonance saturation intensity. Equations (5.39)-(5.41) are integrated by assuming that the medium of length L is divided in slices sufficiently thin so that the variable are nearly constant throughout the slice. The derivatives are approximated by finite differences and a predictor-corrector scheme is used to integrate the wave equations whereas a simple Euler scheme was adequate for the integration of the equation for the spatial amplitudes of the inversion. The infinite sum in Eq. (5.39) is truncated at a sufficiently large value of n such that the results of the simulation are unchanged (typically $n = 5$).

Figures 5.3 shows the predicted temporal evolution of the transmitted forward intensity $|E_f'|^2$ for the case when $\Delta T_2 = 100$, $\alpha(\Delta)L = 1$, $E_f'(0) = E_b'(L)$, and $t_T/T_1 = 1$ where $t_T = L/c$ is the transit time of the medium. The threshold for instability is predicted by the perturbative calculation to be at a value of the nonlinear phase shift $\alpha(\Delta)L \Delta T_2 |E_f'|^2/2$ equal to 2. Therefore, it would be expected that the transmitted intensities would be unstable for $|E_f'|^2 = 0.04$ given the values of the other pertinent parameters listed above. However, the temporal evolution is steady-state after an initial transient for $|E_f'|^2 = 0.1$, as can be seen in Fig. 5.3a. It is found that the instability threshold corresponds to $|E_f'|^2 \sim 0.12$ ($\alpha(\Delta)L \Delta T_2 |E_f'|^2/2 = 6$). If the numerical integration was repeated for a larger detuning from resonance ($\Delta T_2 = 10,000$) and a smaller value of the absorption coefficient ($\alpha(\Delta)L = 0.01$), it is found that the perturbative calculation accurately predicts the threshold for instability. Therefore, these results show that absorption raises significantly the threshold for instabilities.

Periodic temporal evolution for the transmitted forward intensity is predicted for the case $\alpha(\Delta)L \Delta T_2 |E_f'|^2/2 = 10$ which shown in Fig. 5.3b. The system is seen to period triple (Fig. 5.3c) for an increase in the nonlinear phase shift equal to 15.

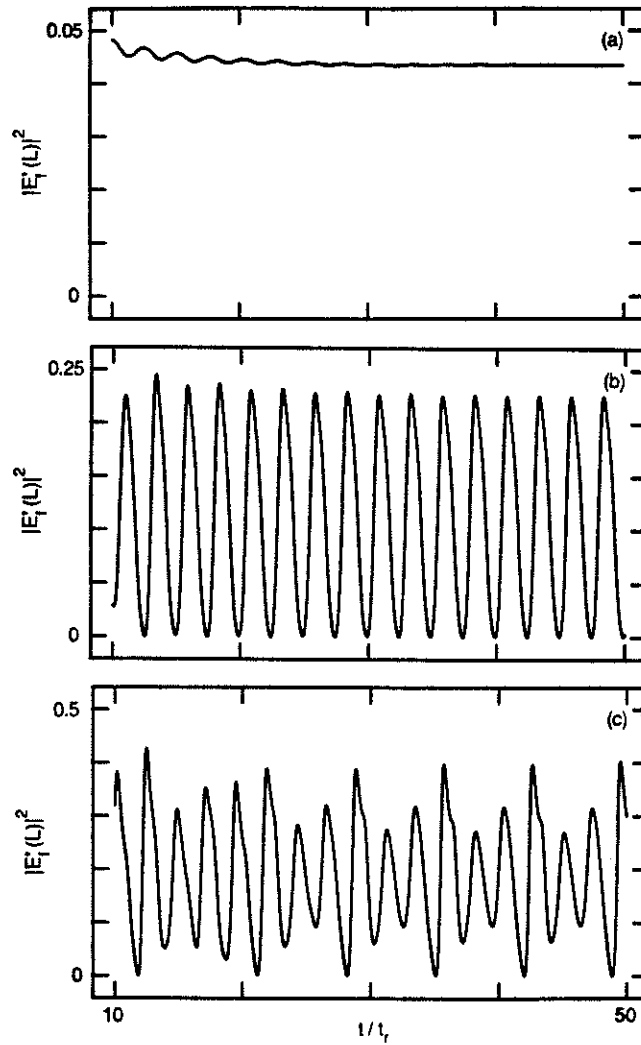


Figure 5.3 Theoretical predictions for the temporal evolution of the intensity of the transmitted forward wave for the case of laser beams counterpropagating in a collection of two-level atoms. The detuning of the laser frequency is $\Delta T_2 = 100$, the absorption coefficient is $\alpha(\Delta)L = 1$, and the input intensities are equal. The nonlinear phase shift $\alpha(\Delta)L \Delta T_2 |E'|^2/2$ is equal to (a) 6, (b) 10, and (c) 15.

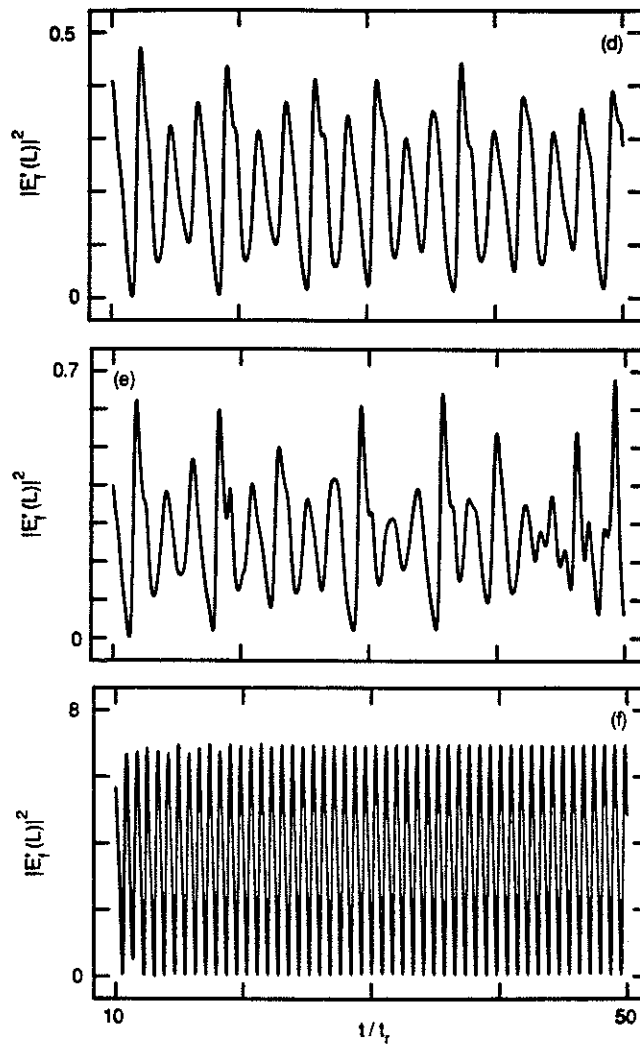


Figure 5.3 Theoretical predictions for the temporal evolution of the intensity of the transmitted forward wave for the case of laser beams counterpropagating in a collection of two-level atoms. The detuning of the laser frequency is $\Delta T_2 = 100$, the absorption coefficient is $\alpha(\Delta)L = 1$, and the input intensities are equal. The nonlinear phase shift $\alpha(\Delta)L \Delta T_2 |E_f|^2/2$ is equal to (d) 16.25, (e) 20, and (f) 200.

Chaotic temporal behavior is seen to occur for a value of the phase shift equal to 16.25 and for slightly higher values of the phase shift ($\alpha(\Delta)L \Delta T_2 |E_f|^2/2 = 20$) the evolution is very complex, as can be seen in Fig. 5.3e. The correlation dimension D_2 of the attractor corresponding to the time series shown in Fig. 5.3d is equal to 3.8 and the order-2 Reyni entropy K_2 is equal to 4.2 bits T_1^{-1} . The dynamics of the system are periodic for high intensities due strong saturation of the atomic transition as can be seen in Fig. 5.3e which is for the case $\alpha(\Delta)L \Delta T_2 |E_f|^2/2 = 200$.

It was shown in the previous section that there must be a phase lag between the population oscillations and the oscillations in the intensity for there to be coupling among the interacting waves. To show that there is indeed a lag between the population oscillations and intensity oscillations when the transmitted intensities are unstable, plots of the spatially uniform inversion $w_0(L)$ vs. the transmitted forward intensity $|E_f(L)|^2$ are shown in Figs. 5.4. Figure 5.4a is the phase space plot for the case when the transmitted forward intensity oscillates periodically. It is seen that the inversion also oscillates periodically and that there is a lag between the two since the trajectory does not fall on a straight line. Figure 5.4b shows the case when the evolution is periodic with period-3 oscillations and Fig. 5.4c shows the case when the evolution is chaotic. These results indicate that the population oscillations are crucial for the development of instabilities of laser beams counterpropagating in a two-level atomic media.

The experimental results shown in Chapter IV are in close qualitative agreement with the predictions of the two-level atom model. Periodic oscillations are predicted and observed for intensities just above the threshold for instabilities. Chaotic evolution is predicted and observed for small increases in the intensity, and for very high intensities the dynamics are periodic with a high frequency of oscillation.

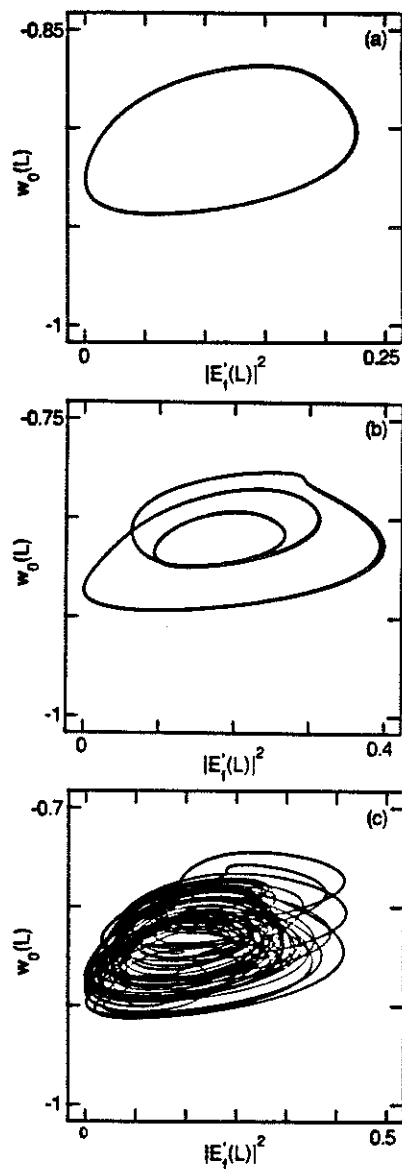


Figure 5.4 Attractors of characterizing the unstable behavior of the counterpropagating laser beams. The attractors are constructed by plotting the inversion as a function

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